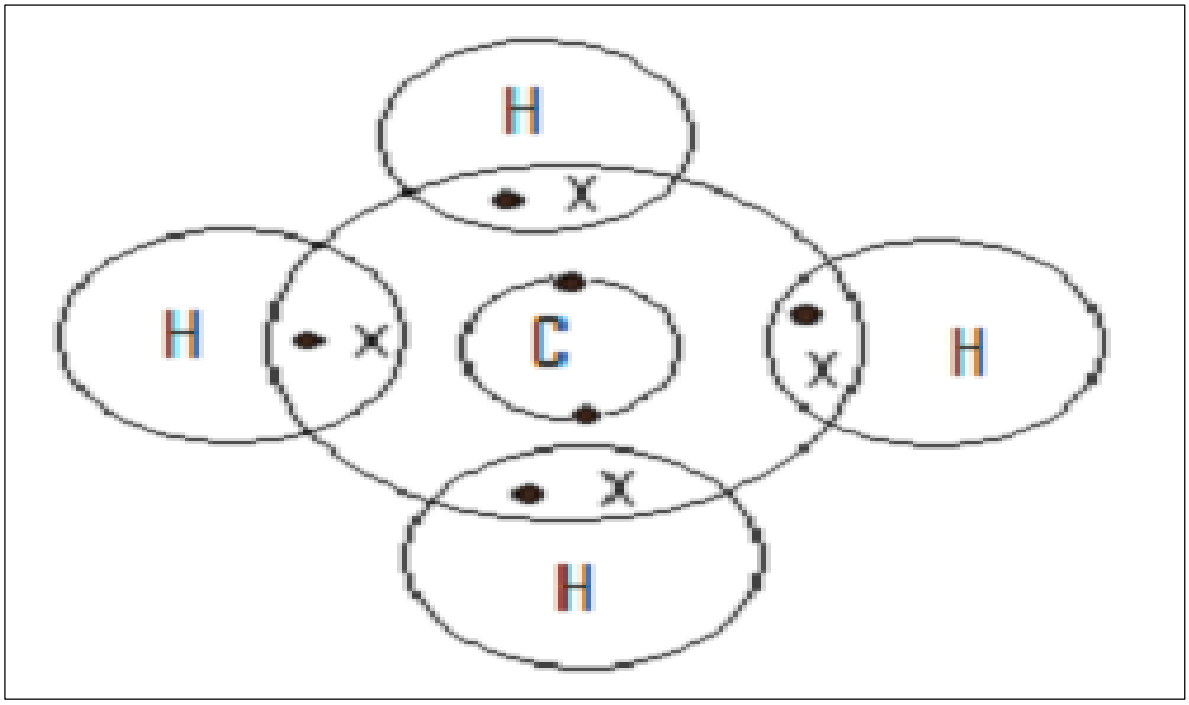
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GRADE 10 -12 2009 – 2019



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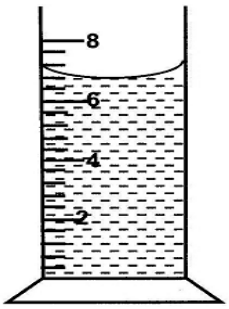
GRADE 10 E.C.Z QUESTIONS

2009 QUESTION PAPER 1

1. Which state(s) of mater exist(s) at the freezing point of a substance?

A. Solid only B. Solid and liquid C. Liquid only D. liquid and gas

2. A measuring cylinder below is used to measure the volume of a liquid. What is the volume of the liquid contained in the cylinder?



A. 6.4cm^3 B. 6.6cm^3 C. 6.8cm^3 D. 7.2cm^3

3. The best and suitable method of collecting pure water from a solution of ink is …..

A. chromatography B. Distillation C. Crystallization D. Filtration

4. Which of the following is true about isotopes?

A. Two or more elements belonging to the same Group of the periodic table.

B. They have the same chemical properties

C. They have the same number of nucleons.

D. They have the same physical properties

5. Which of the following sets contain particles with the same number of electrons?

A. Sodium, potassium and lithium ion C. sodium ion, neon and oxide ion

B. Helium, neon and argon D. Magnesium, calcium and beryllium

6. Below is a chemical equation.

a C_2H_6 + **b** O_2 $\longrightarrow$ **c** CO_2 + **d** H_2O

What are the correct values of **a**, **b**, **c** and **d.**

	a	**b**	**c**	**d**
A	2	7	4	6
B	1	7	2	3
C	1	5	4	6
D	2	5	4	6

7. Which of the following is true about an exothermic reaction?

A. Temperature of the surrounding decreases

B. The enthalpy change, ΔH is positive

C. Bonds formed are relative stronger than bonds broken.

D. Heat is absorbed from the surroundings.

2010 QUESTION PAPER 1

8. Diffusion occurs more quickly in gas than in liquid because ….

A. Molecules in a gas have more frequent collisions than those in liquid.

B. Gas molecules are larger.

C. Gas molecules move randomly

D. On average molecules in a gas are further apart than those in a liquid.

9. Air is used to inflate tyres because ……

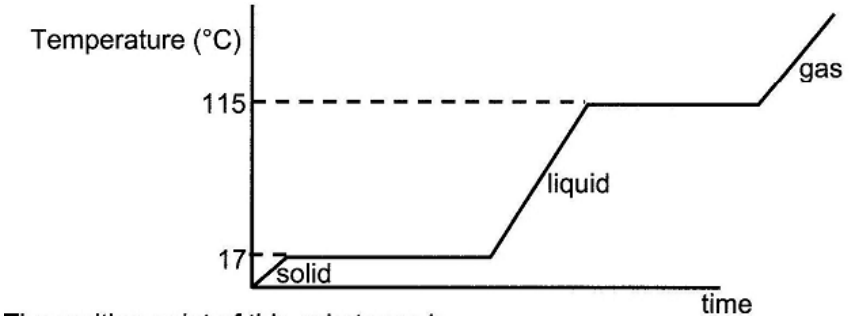
A. it is readily compressed. B. its molecules move randomly.

C. it serves as a coolant D. it get into tyres faster than other substances

10. If two liquids are miscible, they must be separated by …..

A. Separating funnel. B. Filtration C. Crystallization D. Fractional distillation

11. The graph below is a heating curve for a pure substance. It shows how the temperature rises with time, when the solid is heated until it melts, and then the liquid is heated until it boils.



The melting point of this substance is …

A. 0 ºC B. 17 ºC C. 100 ºC D. 115 ºC

12. The below shows the structure of several particles. Which particle is a negative ion?

Particle	Electrons	protons	Neutrons
A	12	12	12
B	12	12	14
C	10	12	12
D	10	8	8

13. When two atoms shares electrons, they form …

A. an ionic compound B. A molecule C. A lattice D. An allotrope

2011 QUESTION PAPER 1

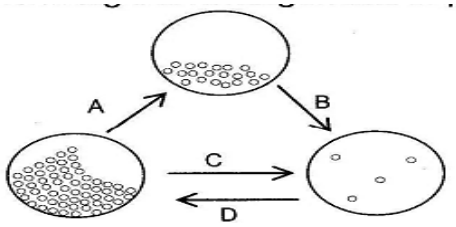
14. Which of the following is not one of the three classes of substances?

A. Compound B. Neutron C. Element D. Mixture

15. The formula for Copper (i) Oxide is ……

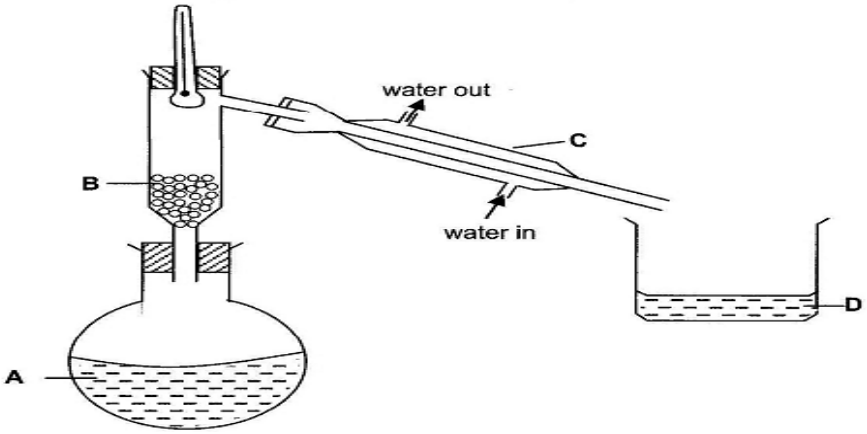
A. CuO B. CuO_2. C. Cu_2O . D. 2CuO

16. Study the diagram below showing the arrangement of particles during change of state of matter.

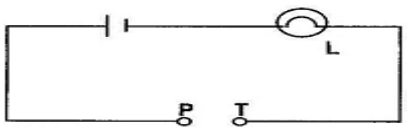


Which process **A**, **B**, **C** or **D** shown in the diagram does ammonium chloride undergo when heated?

17. The diagram below shows apparatus used to carry out fractional distillation. Which part labeled **A**, **B**, **C** or **D** represent a '**fraction**'?



18. The diagram below shows an incomplete electrical circuit. Pieces of elements are placed in turn between **P** and **T**.



Which of the following elements would not cause the light bulb, L, to light?

A. Boron B. Beryllium C. Lithium D. Scandium

2012 QUESTION PAPER 1

19. Which of the following is not one of the basic units of matter?

A. Atom B. Ion. C. Nucleus. D. Molecules.

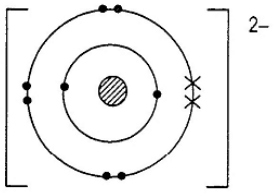
20. Which of the following can be used to measure a volume of 15.6 cm accurately?

A. Burette B. Pipette C. Measuring cylinder D. Volumetric flask

21. A condenser is used in distillation experiments to turn ….

A. liquid into gas B. Solid into liquid. C. Vapour into gas D. vapour into liquid

22. The diagram below represents the structure of a certain elementary particle, P.



Which of the following is not true about P?

A. P is an oxide ion C. P is an isotope of magnesium.

B. P has a noble gas electronic structure. D. P has 8 protons in its nucleus.

23. Element Q has atomic number 11 whereas element R has atomic number 8. The compound formed when atoms of Q and atoms R react together ……

A. will be a volatile liquid at r.t.p C. Will be a crystalline solid at r.t.p

B. contains diatomic molecules. D. is soluble in organic solvent but insoluble in water.

24. The formula of lithium phosphate is Li_3PO_4. What is the formula of magnesium phosphate?

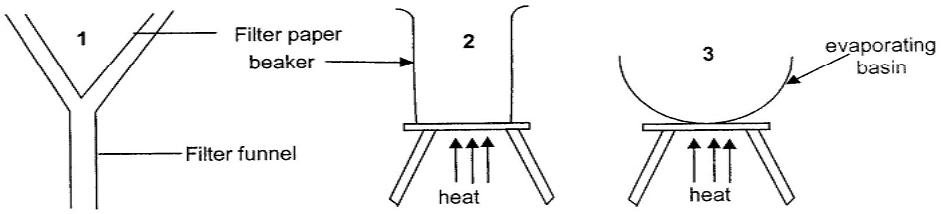
A. Mg_3PO_4 B. $MgPO_4$ C. $Mg_2 (PO_4)_3$ D. $Mg_3(PO_4)_2$

2013 QUESTION PAPER 1

25. Which of the following processes show that matter is composed of tiny particles?

A. Crystallisation B. Expansion C. Freezing D. Melting

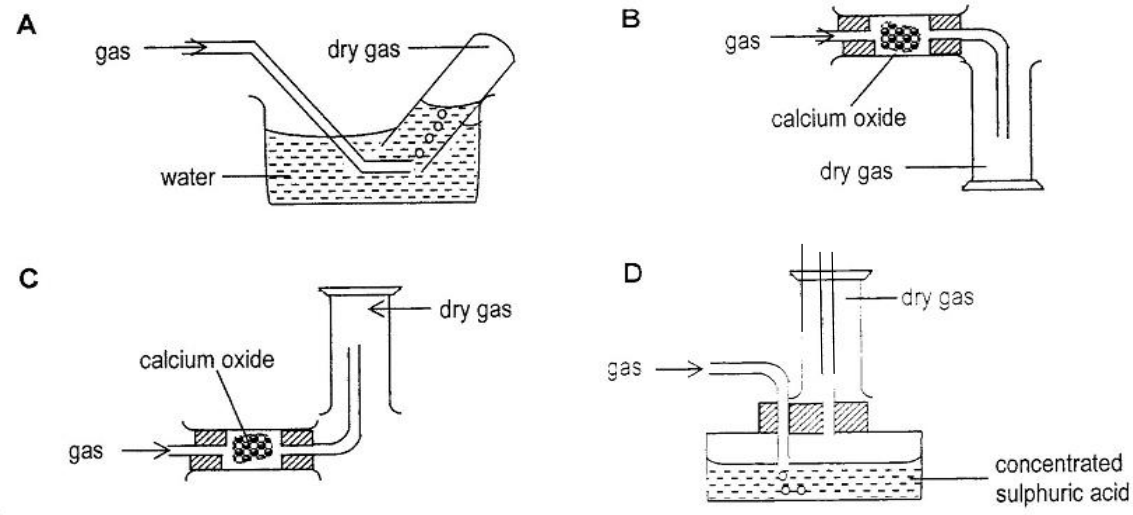
26. The diagrams below show three sets of apparatus.



What apparatus would be used to obtain separate samples of sand and salt from a mixture of sand and sea water?

A. 1 only B. 1 and 3 C. 2 and 3 D. 3 only

27. A gas is less dense than air, very soluble in water and is alkaline. Which method is used to collect a dry sample of the gas?



28. The symbol for an atom of phosphorus is $^{31}_{15}P$. What does the number 31 represent for an atom of phosphorus?

A. the number of nucleons. B. the number of electrons and neutrons

C. the number of protons. D. the position of an atom in the periodic table

29. Which pairs of elements form a compound by sharing electrons?

A. Carbon and chlorine B. Lithium and iodine C. Neon and oxygen D. potassium and bromine

30. How many atoms are present in one molecule of urea, $CO(NH_2)_2$?

A. 4 B. 6 C. 7 D. 8

2015 QUESTION PAPER

31. Which changes occur when a liquid at 50 ^{O}C becomes gas at 120 ^{O}C?

	Separation of particles	**Energy of particles**	**Attractive force between particles**
A	Decrease	increase	Decrease
B	Decrease	Decrease	Increase
C	Increase	Increase	Decrease
D	increase	Decrease	increase

32. Of the techniques below, which one can be used to separate cellular components of blood from blood plasma?

A. Centrifugation B. Chromatography C. Distillation D. filtration

33. Element **X** has an electronic configuration 2, 8, 8, 1 while that of **Y** is 2, 8, 6. Which one of the following is true about the compound formed between **X** and **Y**?

A. Covalent compound of formula X_2Y
C. Covalent compound of formula XY_2
B. Ionic compound of formula XY_2
D. Ionic compound of formula X_2Y

34. Which of the following contains a set of three elements?

A. Argon, lime, water
C. potassium, graphite, nitrogen
B. Silica, oxygen, hydrogen
D. copper, petrol, alcohol

35. Ethane C_2H_6 burns in oxygen completely according to the balanced equation below.

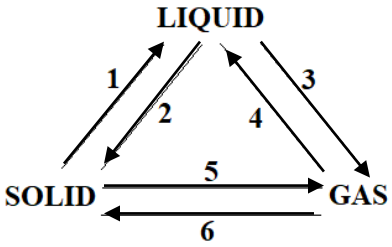
a C_2H_6 + **b** O_2 ⟶ **c** CO_2 + **d** H_2O

Which of the following sets of coefficients balances the equation correctly?

	a	**b**	**c**	**d**
A	1	3.5	2	3
B	1	7	4	5
C	2	7	4	6
D	2	3.5	4	6

2016 QUESTION PAPER 2

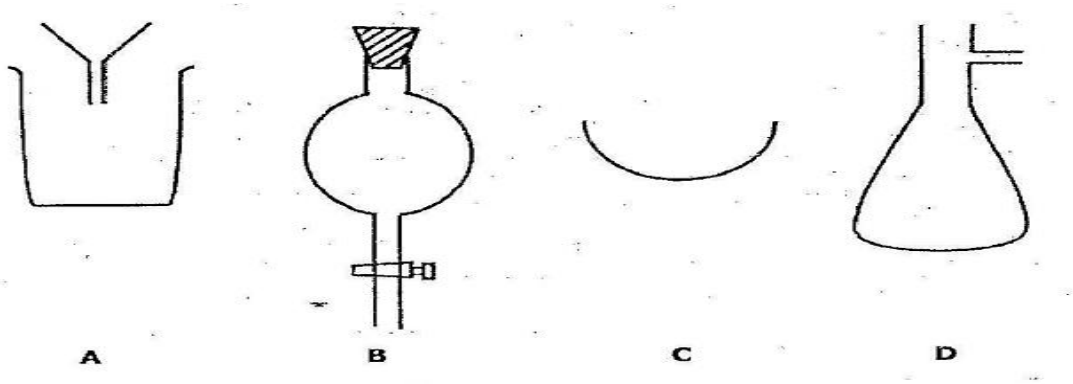
36. The diagram below shows the changes of state.



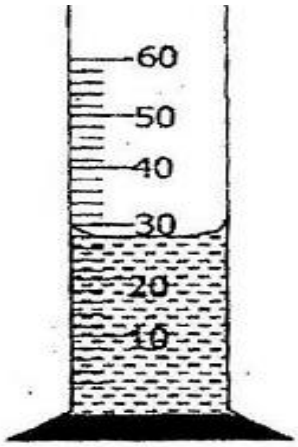
Which of the following sets of changes are exothemetic and endothermic?

	Exothermic	**Endothermic**
A	1, 2, 3	4, 5, 6
B	1, 2, 5	3, 4, 6
C	2, 4, 6	1, 3, 5
D	2, 3, 5	2, 4, 6

37. Which of the following apparatus can be used for separating immiscible liquids?



38. What is the volume of the liquid in the measuring cylinder below?



A. 30 cm^3 B. 29.5 cm^3 C. 29.0 cm^3 D. 28 cm^3

39. The element X and Y have the following nuclides:

$^{28}_{14}X \quad and \quad ^{16}_{8}Y$

What is the correct chemical formula of the compound formed when element X reacts with element Y?

A. XY_2 B. X_2Y C. X_3Y_2 D. X_2Y_3

40. All isotopes of an element contain ….

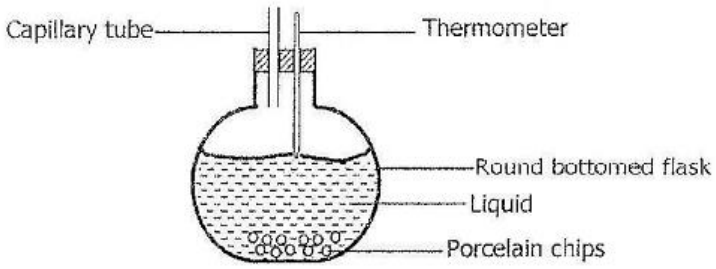
A. Different numbers of electrons
B. The same number of protons
C. different numbers of protons
D. the same number of neutrons

2017 QUESTION PAPER

41. A teacher asked Mutinta to explain what happens to the particles in a stone when it is heated. The correct explanation given by Mutinta was particles in a stone …

A. Will not move
B. Move randomly
C. vibrate more in their fixed positions
D. vibrate and begin to move randomly

42. The diagram below shows the experiment set up for the determination of the boiling point of a liquid.



Which statement explains the purpose of adding porcelain chips?

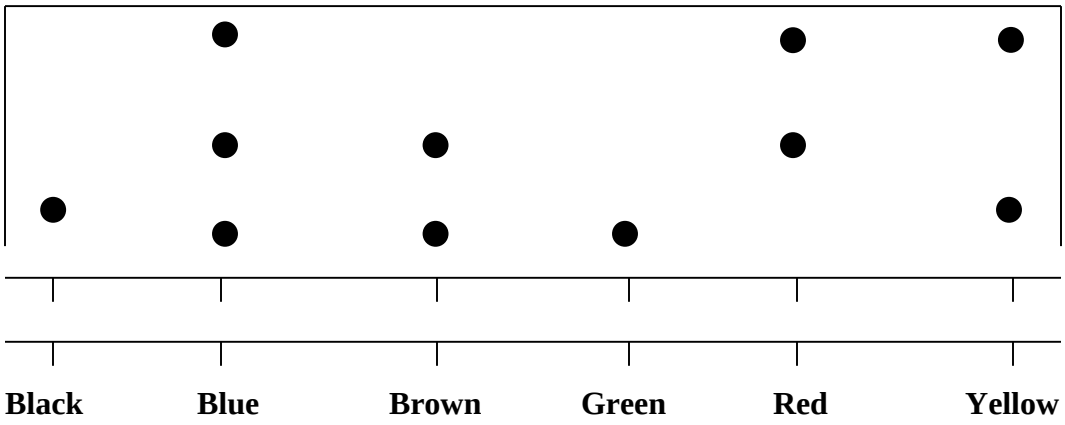
A. To ensure smooth boiling of the liquid.

B. to colour the liquid as it starts to boil

C. To enable the thermometer record the temperature of the boiling liquid easily

D. To make the liquid boil faster

43. The diagram below is a chromatogram for various types of ink.



Which statement is correct about the chromatogram?

A. Red ink contains black ink

B. Black ink and green ink are pure ink .

C. Blue ink can be made by mixing brown and green inks.

D. green ink contains red ink

44. A phosphorus ion contains ….

	Protons	Neutrons	Electrons
A.	15	15	13
B.	15	16	18
C.	16	15	16
D.	16	16	18

45. Hydrogen can form both ionic and covalent compounds. Which element will hydrogen form an ionic compound?

A. Zinc B. Sodium C. Nitrogen D. Sulphur

46. An endothermic reaction is one that …..

A. Evolves heat B. Produces light C. Absorbs energy D. produces sound

47. Consider the following chemical reaction.

X $Hg(NO_3)_2$ ⟶ **Y** Hg + **Z** NO_2 + O_2

The letters X, Y and Z represent …

	X	**Y**	**Z**
A.	2	2	2
B.	1	1	2
C.	3	3	2
D.	3	3	3

2017 G.C.E QUESTION PAPER 2

48. Which of the following is **not** true about evaporation?

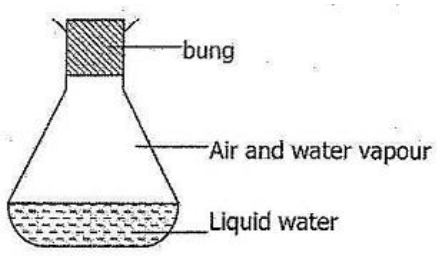
A. It involves a physical change of state C. The particles gain kinetic energy

B. It is non-reversible change D. It weakens the intermolecular forces of attraction

49. The nuclide of aluminium ion is written as $^{27}_{13}Al^{3+}$, state the numbers of neutrons and electrons in the nuclide of the ion.

	Neutrons	**Electrons**
A.	27	13
B.	14	13
C.	14	10
D.	27	10.

50. Some cold water is poured into a conical flask and a bung inserted. The diagram shows the flask after being left in open air for some time.

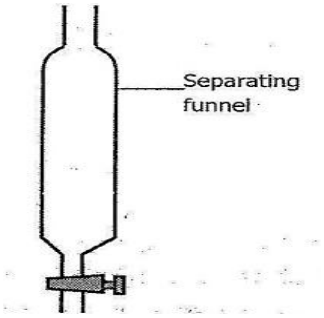


What is occuring in the flask?

A. Boiling and condensation C. Evaporation and condensation

B. Evaporation and freezing D. Freezing and melting

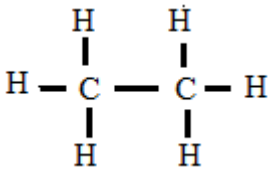
51. Identify a mixture of substance that can be separated using the apparatus below.



A mixture of …

A. Paraffin and water
C. common salt and iodine solution
B. Sugar solution and alcohol
D. alcohol and water

52. Ethane has the structure shown below.



How many of the electrons in a molecule of ethane are **not** involved in bonding?

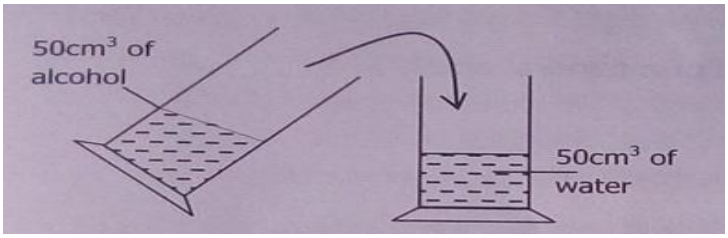
A. 4 B. 3 C. 2 D. 0

53. Which of the following is an exothermic reaction?

A. The reaction between hydrogen and iodine
C. photosynthesis
B. Development of photographs
D. Rusting

2019 G.C.E QUESTION PAPER

54. 50 cm^3 sample of alcohol is mixed with 50 cm^3 of water.



The volume of the mixture is found to be 97 cm^3. Which of the following is the best explanation for this observation?

A. Some alcohol molecules evaporated.
B. The alcohol molecules fit into the gaps between water molecules
C. Water and alcohol react to form a gas which escapes
D. Water and alcohol react to produce a salt which then dissolves

55. A mixture **Q** contains three compounds **T**, **U** and **V** whose solubility in ethanol and water are as shown in the table below.

Compounds	Solubility in	
	Ethanol	Water
T	Very soluble	Sparingly soluble
U	Insoluble	Very soluble
V	Insoluble	Insoluble

How would you separate the mixture Q to obtain pure crystals of compound **U**?

A. Add ethanol ⟶ filter ⟶ dry the residue

B. Add water ⟶ filter ⟶ crystallize

C. Add ethanol ⟶ filter ⟶ add water ⟶ filter ⟶ crystallize

D. Add water ⟶ filter ⟶ add ethanol ⟶ filter ⟶ crystallize

56. The chemical formula of ammonium sulphate is …

A. NH_4SO_4 B. $(NH_4)_2SO_3$ C. $(NH_4)_2SO_4$ D. NH_4SO_3

57. The atoms $^{31}_{15}P$ and $^{32}_{16}S$ have the same …

A. Nucleon number. B. number of electrons.

C. Number of neutrons D. Number of protons

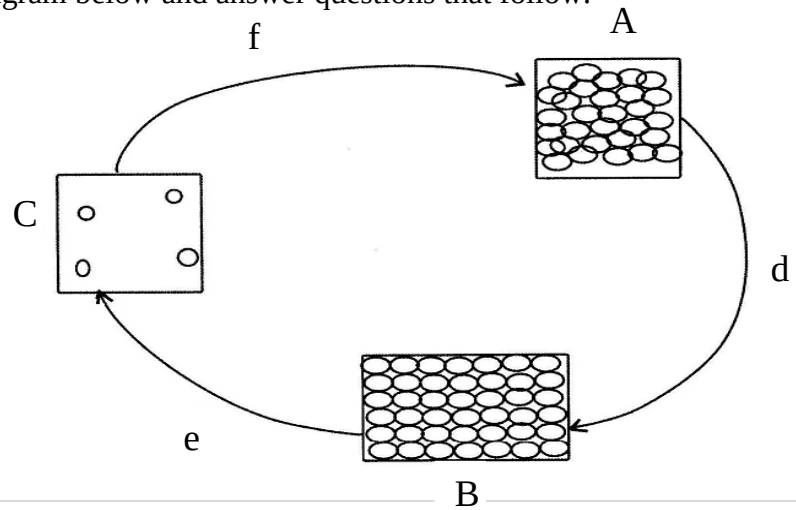
58. Which pair of elements can combine chemically to form single covalent bonds?

A. Hydrogen and nitrogen C. Oxygen and carbon

B. Potassium and nitrogen D. Sodium and chlorine

2009 QUESTION PAPER 3

59. Study the diagram below and answer questions that follow.



NB: **d**, **e**, **f** are processes

(a) Name the processes **d**, **e** and **f** [3]

(b) Describe what you would do to convert substance A to B. [1]

(c) State one of the basic units of matter. [1]

2010 QUESTION PAPER 3

60. Use your knowledge of the kinetic theory of matter to suggest a reason for each of the following.

(a) Wet clothes dry more quickly on warm days than cold days. [1]

(b) Solid ice loses its shape when it melts. [1]

(c) Salt dissolves faster in hot water than cold water. [1]

(d) When sugar is dissolved in a glass of water without stirring, all of the water soon tastes sweet. [1]

61. Below are some processes which are used in a laboratory, industry and at home:

- **Crystallisation, combustion, Distillation, esterification, Filtration, galvanizing, Neutralization and polymerization.**

(a) Which one of the processes listed above can be used to separate the following from a sample of sea water?

(i) Salt …… [1]

(ii) Water ……. [1]

(iii) Sand …….. [1]

(b) State the process from the list of processes above that can be used to:

(i) Manufacture plastic for making buckets …… [1]

(ii) Produce carbon dioxide from carbon …… [1]

62. An element has an isotope with the nucleon number of 7. Each neutral atom of this isotope has three electrons and a nucleus containing two different types of particles.

(a) (i) Give the name of these particles and the number of each particle present in each nucleus.

Name of particle	Number of particle

(ii) Compare the masses and the electrical charges of these particles. [2]

(b) What is the difference in the nuclei of this isotope and its isotope whose nucleon number is 6? [1]

2011 QUESTION PAPER P3

63. Use the information in the table to answer the following questions.

Substance	**Conduct electricity when solid**	**Melting point /^{O}C**	**Dissolves in water**
Sodium chloride	No	808	Yes
Sulphur	No	113	No
Tungsten	Yes	3317	No
Wax	No	35 – 50	No
Aluminium	Yes	660	No

(a) (i) Name one metal from the table [1]

(ii) How can you tell from the table that the substance you have chosen in (a) (i) above is a metal? [1]

(b) How can you tell from the table that wax is a mixture? [1]

(c) (i) Name a compound from the table. [1]

(ii) Explain the meaning of the word compound. [1]

64. Use the list of separation techniques below to answer the questions that follow below.

Fractional Distillation, Simple Distillation, Use of Separating Funnel, Magnetism, Chromatography, Evaporation and Filtration.

Choose one method from the list above which can be used to separate:

(a) Sand from water ……...……[1]

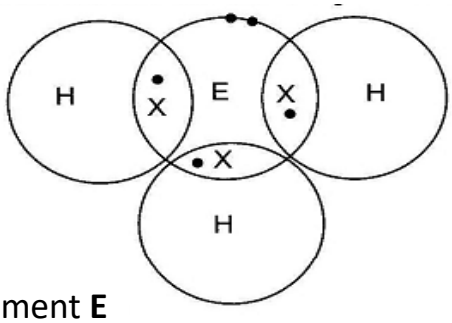
(b) Oil from water ……..……[1]

(c) Water from ink …… [1]

(d) Sulphur powder from iron filings ……...……..[1]

(e) Salt from paraffin ……...……..[1]

65. Element E whose proton number is 7 combines with hydrogen to form a gas. The diagram below shows the bonding in one molecule of this gas.



Element **E**

(a) Identify element E ……..……..[1]

(b) What is the name of the gas? ……...…….[1]

(c) Write the chemical formula of the gas…….…… [1]

(d) What type of bonding holds the atoms together in this compound? ……...…….[1]

(e) State one physical property of the gas that is due to the type of bonding it has …..……[1]

(f) Name another compound which has the same type of bonding……[1]

2012 QUESTION PAPER 3

66. Matter is classified as solid, liquid or gas. State two physical properties of each of the following:

(a) Solid ……..……[2]

(b) Liquid ……..……..[2]

(c) Gas ……[2]

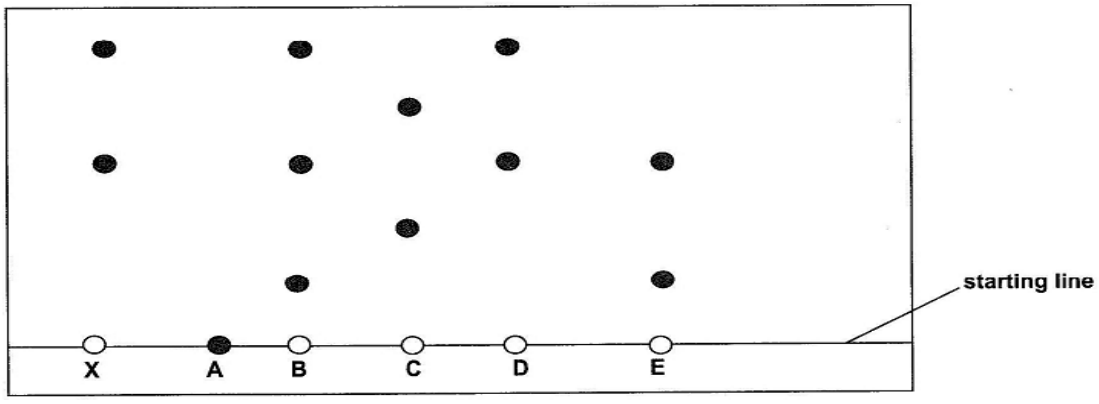
67. Two miscible liquids with boiling points of 78 ^{O}C and 100 ^{O}C were accidentally mixed.

(a) Name the process which can be used to separate the mixture…….[1]

(b) Draw a labeled diagram showing the arrangement of the apparatus used to separate the mixture. [3]

(c) Explain how you can obtain hydrated sodium sulphate crystals from an aqueous solution of sodium sulphate……...…….[2]

68. Paper chromatography was used to catch a forger. A sample of ink, X from a forged signature was compared with ink from the pens of five suspects. The diagram below shows the chromatogram obtained:

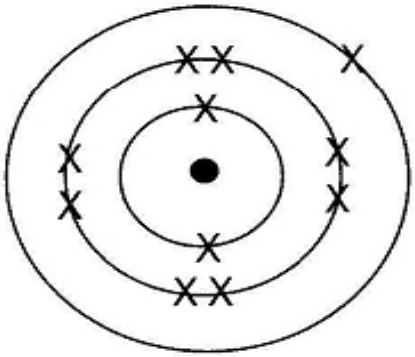


(a) Draw the apparatus you would use to produce this chromatogram [3]

(b) Which of the inks A, B, C, D or E could have been used to write the forged signature? [1]

(c) Which of the inks is insoluble in water? [1]

69. The diagram below represents electronic arrangement of a particular atom. Study this diagram and answer the questions that follow.



(a) The relative atomic mass of the atom represented is 23.

(i) What is its proton number? ……....[1]

(ii) What is its neutron number? ……..…….[1]

(b) (i) In which group of the periodic table is the element found? [1]

(ii) Explain your answer in **(b) (i).** [1]

70. (a) Nuclides of magnesium and calcium are shown below. What do the following numbers tell you about these atoms:

(i) 24 in the nuclide for magnesium? [1]

(ii) 20 in the nuclide for calcium? [1]

(b) Draw the electronic structure of the atom of magnesium. [1]

(c) Describe how the electronic structure of magnesium and calcium indicate that they are both in the same group of the periodic table. [1]

(d) An atom A (atomic number 11) burns in chlorine to produce a white solid chloride B. What is the charge of the atom:

(i) Before the reaction? [1]

(ii) After the reaction? [1]

(e) Write a balanced chemical equation for the reaction in (d) above. [2]

(f) State the type of bonding found in chloride B and discuss one of its properties. [2]

2013 QUESTION PAPER 3

71. Chlorine has two isotopes, $^{35}_{17}Cl$ *and* $^{37}_{17}Cl$.

(a) Define the term isotopes [1]

(b) State the number of neutrons in each of the following isotopes of chlorine.

(i) Chlorine – 35: ……...……[1]

(ii) Chlorine – 37: ……...……[1]

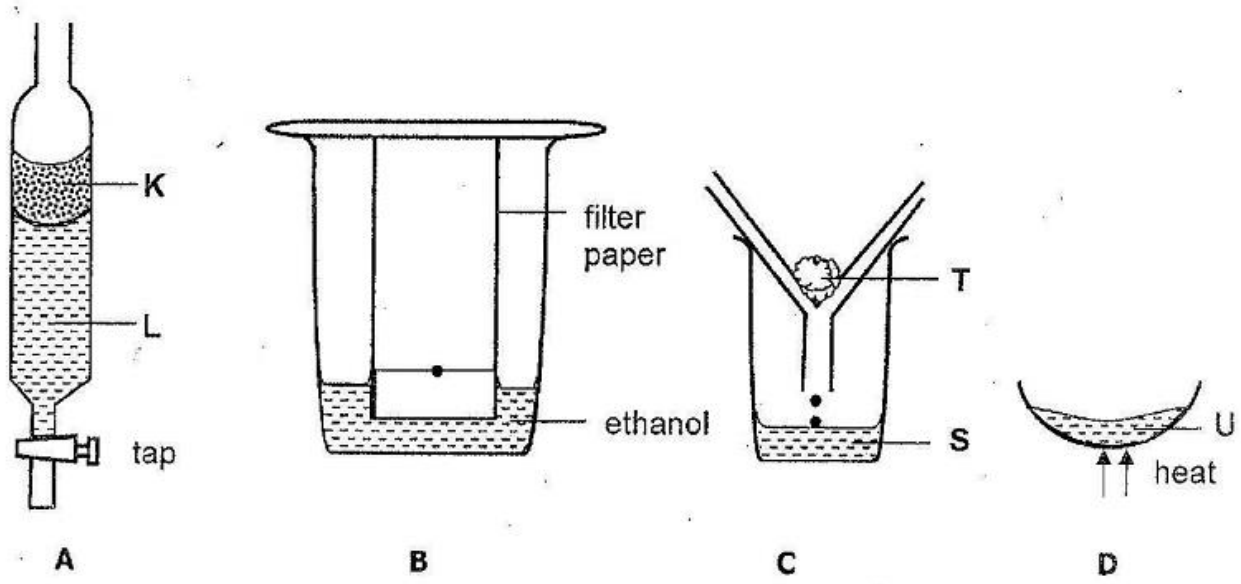
(c) (i) Draw the atomic structure of an atom of chlorine. [1]

(ii) Give the formula of a chlorine ion. [1]

(d) Explain why the relative atomic mass of chlorine on the periodic table is not a whole number. [1]

2014 QUESTION PAPER 3

72. The diagrams below show four different methods of separation.



(a) Using the letters **A**, **B**, **C** or **D** identify the method used to:

(i) Separate insoluble particles from water. [1]

(ii) Separate a mixture of different dyes. [1]

(iii) Make crystals of copper (ii) sulphate from copper (ii) sulphate solution. [1]

(iv) Separate water and cooking oil. [1]

(b) State **one** industrial application of chromatography. [1]

73. The table below shows the ions present in one litre of mineral water bottled for drinking by an industry.

Name of ion	**Formula of the ion**	**Mass of ion present in one litre/milligram**

Calcium	Ca^{2+}	5.0
Chloride	Cl^{-}	4.0
Bicarbonate	HCO_3^{-}	32
Sodium	Na^{+}	4.0
Sulphate	SO_4^{2-}	3.5

(a) Define the term ion. [1]

(b) Identify the cation with the lowest concentration in the mineral water. [1]

(c) Complete the equation below to show how a calcium ion is formed from a calcium atom.

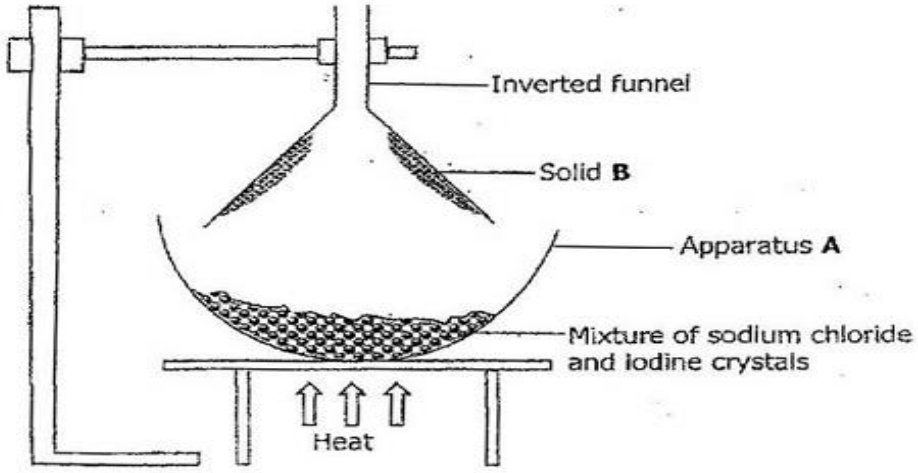
$Ca \longrightarrow Ca^{2+} + _____$ [1]

(d) When this sample of mineral water is evaporated to dryness. Various compounds remain as residues. Give the name of two of these compounds. [2]

(e) State the type of bonding found in the various compounds in the mineral water. [1]

2015 G.C.E QUESTION PAPER 3

74. The diagram below shows a method by which a mixture of sodium chloride and iodine crystals can be separated.



(a) Give the name of the separation technique shown in the the diagram above. [1]

(b) Name apparatus **A**. [1]

(c) Identify solid **B** [1]

(d) Explain why sodium chlrode solution is regaeded as a mixture. [1]

75. Study the nuclide of a sodium isotope below and answer the questions that follow; $^{24}_{11}Na$.

(a) What is meant by the term nucleon number? [1]

(b) The table below describes the composition of the nucleus of sodium. Complete the table by writing the total number of each particle in one atom of the element. [3]

Name of particle	**Total number in one atom**
Neutron	
Protons	
Electron	

(c) Write the formula of sodium ion. [1]

(d) Write the electronic configuration of the sodium ion. [1]

76. Explain the meaning of

(a) (i) a filtrate (ii) a residue [2]

(b) Describe **two** ways of separating a solid from a liquid in a suspension. [4]

(c) (i) State one difference between simple distillation and fractional distillation. [2]

(ii) Name two industries in Zambia where fractional distillation is used. [2]

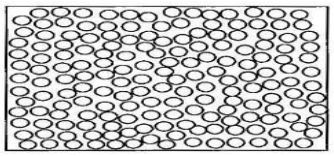
2016 G.C.E QUESTION PAPER 3

77. All matter is made up of small particles which are referred to as the basic units of matter. The small particles of matter are always in random motion.

(a) State any two basic units of matter. [2]

(b) Suggest two pieces of evidence which show that matter is made up of tiny, randomly moving particles. [2]

(c) The diagram below represents one of the physical states of matter.



Name this physical state of matter and explain why it has a fixed volume. [2]

78. Each of the following statements is incorrect. But one scientific term is replaced by a correct one, the statement then becomes true. Identify the wrong word (term), underline it and in the space provided write the correct word.

(a) Condensation is the change of state of matter from solid to liquid. [1]

(b) The basic units of matter that exist in sodium chloride are molecules. [1]

(c) Isotopes are compounds with the same molecular formula but different structural formulae. [1]

(d) Calcium hydroxide is classified as an amphoteric substance because it dissolves in sodium hydroxide solution. [1]

(e) Zinc and chlorine bond covalently. [1]

79. Two elements **A** and **D** are represented by the notations shown below.

$$^{23}_{11}A \qquad ^{16}_{8}D$$

(a) Draw electron-shell diagrams to show the arrangement of electrons in the atoms of **A** and **D** [2]

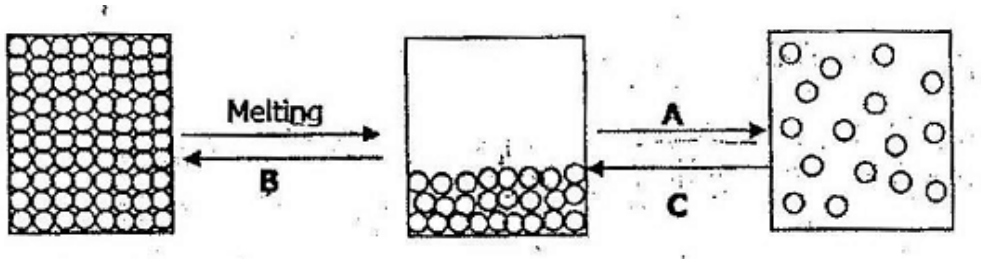
(b) (i) When atoms of element A and atoms of element D react together, what type of bonding occurs? [1]

(ii) Using the dot and cross diagram, show the structure of the compound formed between **A** and **D** showing all the electron shells. [2]

(c) State any **two** physical properties of the compound formed between **A** and **D**. [2]

2016 QUESTION PAPER 2

80. The diagram below shows how the molecules are arranged in three states of matter, solid, liquid and gas.



(a) State the name given to the change of state labeled **A** and **B** [2]

(b) Describe the movement of .molecules in a gas. [1]

(c) Which of the changes A, B or C is endothermic? Explain your answer. [2]

81. The table shows some apparatus used in the laboratory.

Separating funnel	Liebig condenser	Pipette
Desiccator	Volumetric flask	Laboratory thermometer
Evaporating dish	Bunsen burner	Burette

State the apparatus used for the following:

(a) Burning and heating ……...……[1]

(b) Preparation of standard molar solution……...[1]

(c) Obtaining a distillate from vapour of a liquid……..……..[1]

(d) Drying substances or keeping them free from moisture……...……..[1]

(e) Measuring exactly, 25.0 cm^3 of solution……...…….[1]

82. Given below is a list of substances.

Aluminium	Nitrogen	Cement	Potassium chloride
Bronze	Water	Methanol	Sea water

(a) Which substance is an element? ……..……...[1]

(b) Which substance is a mixture?.....[1]

(c) Which substance is a single compound?.....[1]

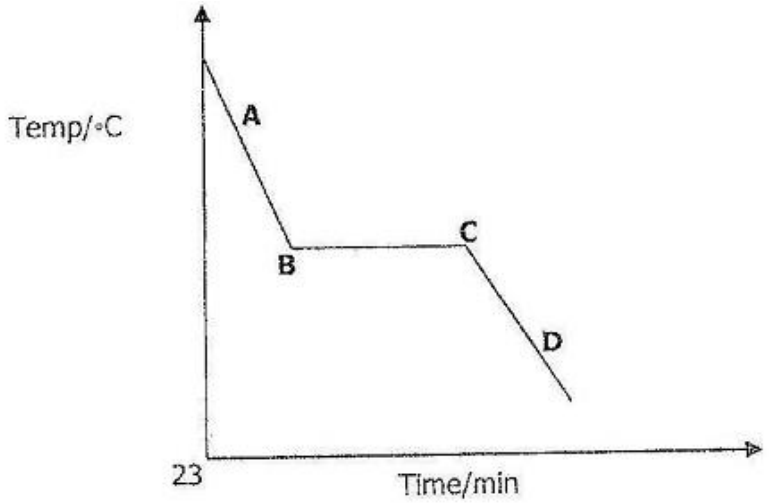
(d) Which substance is an element that conducts electricity?.....[1]

(e) From the above list, select one:

(i) Ionic compound……. (ii) Alloy …… [2]

2017 QUESTION PAPER 2

83. The diagram below shows the cooling curve for a liquid.



(a) In what state of matter is the substance in area A? [1]

(b) What name is given to the point labeled B? [1]

(c) In what states of matter is the substance between point B and C? [1]

(d) Explain the reason why thermometer reading remained constant between points B and C. [2]

(e) Explain what happens during cooling in relation to the heat content of the substance. [1]

84. Beryllium burns in fluorine to form a white solid, beryllium fluoride.

(a) Name the type of bonding in beryllium fluoride…….[1]

(b) In the space below, draw a 'dot' and 'cross' diagram to show the bonding in beryllium fluorine. Show all electrons. [2]

(c) Suggest any **two** physical properties of compounds that have similar bonding as beryllium fluoride. [2]

85. A learner wanted to obtain clear water from muddy water.

(a) Name the process that the learner would use to obtain the clear water…… [1]

(b) Draw a large labeled diagram to show the arrangement of the apparatus the learner would use. [3]

(c) Give an industrial application of the process named in (a) above. [1]

2017 G.C.E QUESTION PAPER 2

86. Study the following list of processes: melting, chemical change, sublimation, condensation, evaporation, dissolving. Which of the processes listed above best describe what is taking place in each of the following?

(a) The formation of water droplets on the window pane on a cold day. [1]

(b) The formation of liquid sodium chloride from solid sodium chloride due to strong heating [1]

(c) The formations of iodine vapour from solid iodine on heating. [1]

(d) Adding sugar to hot tea and stirring it. [1]

(e) The formation of calcium oxide when calcium is heated in the air. [1]

87. Define the following terms:

(a) (i) Endothermic reaction…… [1]

(ii) Exothermic reaction…….. [1]

(b) Give an example of each type of reaction in (a) in nature.

Endothermic reaction…… [1]

Exothermic reaction…….. [1]

(c) Describe the effect of exothermic reactions in industries on the environment. [1]

88. (a) Give **two** reactions why chemistry is important in industry. [2]

(b) State any **two** laboratory safety rules. [2]

2019 G.C.E QUESTION PAPER 2

89. The table below shows some common gases and their particulars.

Gas	Relative molecular mass	Vapour density
Ammonia	17	8.5
Carbon dioxide	44	22
Carbon monoxide	28	14
Helium	4	2
Nitrogen	28	14

Choose from the list given, the gas (es) that best suits the following descriptions. A gas may be chosen once, more than once or not at all.

(a) (i) The gas that diffuses fastest. [1]

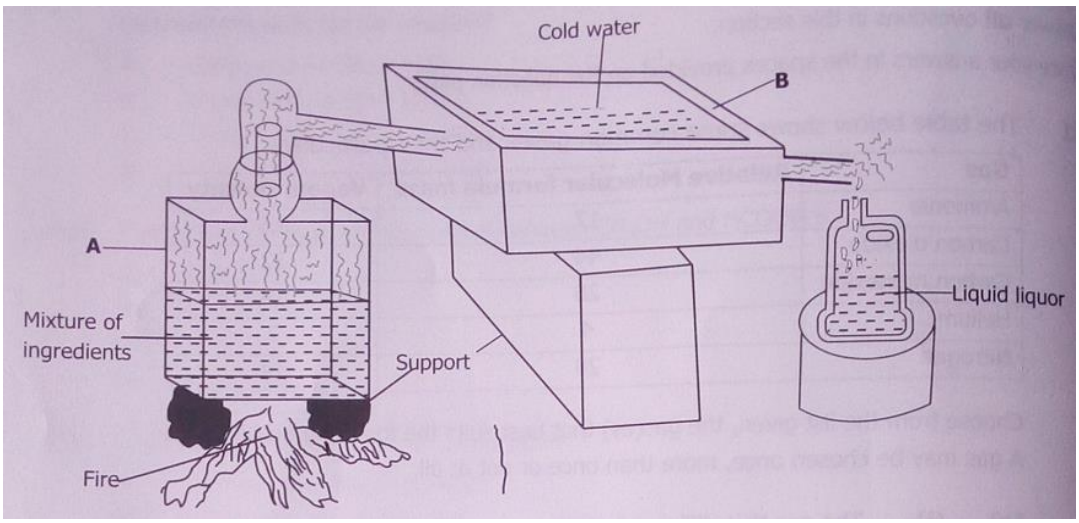
(ii) The names of the gases that diffuse at the same rate. [1]

(b) State the name of the gas that would diffuse faster than any other gas shown in the table. [1]

(c) What is the relationship between relative molecular mass of a gas and its vapour density? [1]

(d) Under what condition would carbon dioxide diffuse faster than ammonia both of equal volume? [1]

90. The diagram below shows the preparation of liquor in a village.



(a) Name the process demonstrated above. [1]

(b) (i) Name apparatus **B**. [1]

(ii) If this experiment was done in the school laboratory, what name would be apparatus **A** and **B**? [2]

(c) What is the scientific term for the liquor/liquid in the experiment? [1]

91. Aluminium metal was reacted with aqueous copper (ii) sulphate.

(a) Construct a balanced chemical equation for the above reaction. Include state symbols. [2]

(b) Deduce an ionic equation from (a). [2]

GRADE 11 E.C.Z QUESTIONS

2009 QUESTION PAPER1

1. Limestone, CaCO3 decomposes into lime, CaO according to the equation,

   $CaCO_3\,(s) \longrightarrow CaO\,(s) + CO_2\,(g)$

   What mass of limestone would produce 11.2g of lime?

   A. $\frac{100\ X\ 11.2}{56}\,g$ B. $\frac{100\ X\ 56}{11.2}\,g$ C. $\frac{100\ X\ 56}{100}\,g$ D. 11.2 x 56 x 100g

2. Which of the following salts can be crystallized from an aqueous solution?

   A. Barium sulphate B. Lead (II) sulphate

   C. Silver chloride D. Ammonium sulphate

3. Solution **R** forms a white precipitate with little amount of aqueous ammonia. The precipitate dissolves in excess aqueous ammonia to form a colourless solution. Which cation is present in **R**?

   A. Ca^{2+} B. Al^{3+} C. NH^{4+} D. Zn^{2+}

4. Which of the following is true about chlorine, bromine and iodine?

   A. They are good conductors of electricity. B. When in the gas phase, they have no smell.

   C. They are all coloured. D. They are non-poisonous.

2010 QUESTION PAPER 1

5. What is the formula mass of magnesium nitrate, $Mg\,(NO_3)_2$?

   A. 54 B. 74 C. 148 D. 296

6. How many moles of oxygen molecules are there in 64 grams of oxygen, O_2?

   A. 2 moles **B**. 4 moles **C**. 8 moles **D**.16 moles

7. Hydrogen burns in oxygen to form water. The equation for the reaction is:

   $2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(g)}$

   How much oxygen is needed to burn 1 gram of hydrogen?

   A. 2g B. 4g C. 5g D. 8g

8. Water at 25°C was used to dissolve two compounds. Immediately after the compounds had dissolved, the temperature of each solution was measured.

Compound	Temperature (^OC)	
	Water	Solution
NH_4Cl	25	15
$CaCl_2$	25	45

   Which of the following is true about the compounds?

A. The temperature change for NH_4Cl *is* +10° **B**. The temperature change for $CaCl_2$ is +20°C

C. $CaCl_2$ dissolves endothermically **D**. NH_4Cl dissolves exothermically

9. Reaction that is reversible, is shown by using the symbol ….

A. ≈ B. = C. ↔ D. ⇌

10. The lower the pH number of a substance the ...

A. more OH^- ions it contains. B. fewer H^+ ions there are

C. more H^+ ions there are. D. less acidic it is.

11. Which of the following does not produce a salt when it reacts with an acid?

A. Metal oxide B. Ammonia solution **C**. Metal hydrogen carbonate D. Pure water

12. Which of the following is **not** reactive?

A. Phosporus B. Sulphur **C**. Chlorine D. Argon

13. Which of the following is a neutral oxide?

A. Nitrogen dioxide B. Magnesium oxide C. Carbon dioxide D. Carbon monoxide

2011 QUESTION PAPER1

14. How many oxygen atoms are in 1.6g of Sulphur trioxide, SO_3?

A. 3 B. 4.8 C. 3.6×10^{22} **D**. 9.6×10^{23}

15. Acids act alike because they all contain…….

A. sulphate ions. B. hydrogen ions. C. hydroxide ions. D. cations.

16. Solution **X** formed a white precipitate with silver nitrate which was insoluble in nitric acid. What could solution **X** contain?

A. Silver chloride B. Sodium carbonate **C**. Barium nitrate **D**. Ammonium chloride

17. How many elements are in period 6 of the Periodic Table?

A. 8 B. 10 C. 18 D. 32

18. In which of the following are the halogens correctly arranged as solid, liquid or gas?

	Chlorine	**Iodine**	**Bromine**
A.	gas	solid	liquid
B.	gas	liquid	solid
C.	liquid	gas	solid
D.	solid	gas	liquid

2012 QUESTION PAPER1

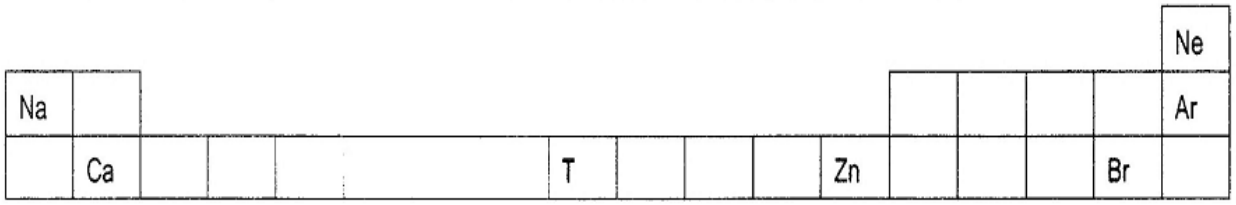
19. What mass of calcium metal reacts completely with 9.0g of water according to the equation? $Ca\ (s) + 2H_2O\ (l) \longrightarrow Ca(OH)_2\ (aq) + H_2O\ (g)$

    A. 40g B. 20g C. l0g D. 5.0g

20. Which of the following is likely to be the molecular formula of a hydrocarbon containing 85.7% carbon and 14.3% hydrogen by mass?

    A. C_2H_6 B. C_3H_8 C. C_5H_{12} D. C_6H_{12}

21. The diagram below shows part of the Periodic Table of elements. T represents an element in the periodic Table but it is not the actual symbol of the element.



Which of the following statements is true about **T?**

A. **T** has a high density and high melting point.

B. **T** forms covalent compounds with bromine.

C. **T** is in Period 2 of the Periodic Table.

D. **T** is likely to be a gas at r t p

22. What is the ionic equation for the neutralisation reaction between sodium hydroxide and sulphuric acid?

A. $2Na^+(aq) + SO_4^{2-}(aq) \longrightarrow Na_2SO4_{(aq)}$

B. $Na^+(aq) + OH^-(aq) \longrightarrow NaOH(aq)$

C. $H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$

D. $2H^+(aq) + SO_4^{2-}(aq) \longrightarrow H_2SO_4(aq)$

23. Which of the following oxides dissolves in both acid and alkali?

    A. ZnO B. CuO C. SiO D. Fe_2O_3

2013 QUESTION PAPER1

24. Methane burns completely in oxygen according to the equation below:

    $CH_4.(g) + 2O_2\ (g) \longrightarrow CO_2\ (g) + 2H_2O\ (g)$

    If 0.2 mol of methane is burned completely, which volume of carbon dioxide measured at r.t.p is formed?

    A. $0.2dm^3$ B. $0.6dm^3$ C. $2.4dm^3$ D. $4.8dm^3$

25. Potassium is in the same Group of the Periodic Table as lithium. Which of the following is a property of lithium?

A. It is a poor conductor of electricity. B. It forms an acidic oxide.

C. It forms an ionic chloride, $LiCl_2$. D. It reacts with water, liberating hydrogen.

26. How many Groups in the Periodic Table contain both metals and non-metals?

A. 2 B. 4 C. 6 D. 8

27. Which of the salts below can be prepared by a titration method?

A. Copper (II) sulphate. **B**. Aluminium nitrate.

C. Ammonium chloride. **D**. Barium sulphate.

2015 QUESTION PAPER1

28. How many atoms are there in 6.0g of carbon atoms?

A. 3×10^{23} B. 6×10^{23} C. 1.2×10^{24} D. 6×10^{24}

29. Calcium carbonate, CaCO3 decomposes according to the following equation;

$CaCO3\ (s) \xrightarrow{heat} CaO\ (s) + CO_2\ (s)$

What volume of carbon dioxide, measured at room temperature and pressure is produced when 50.0g calcium carbonate is decomposed?

A. $12.0dm^3$ B. $24.0dm^3$ C. $48.0dm^3$ D. $120dm^3$

30. In the Periodic Table hydrogen (H) is not placed in any of the Groups. Which of the following is the best explanation for its position? It…

A. has no neutrons.

B. is the lightest of all the elements.

C. has properties of both Group I and Group VII elements.

D. has only one electron in its only single energy level.

31. Which ions form the net ionic equation when aqueous solutions of ethanoic acid and sodium hydroxide react together?

A. Ethanoate ions and sodium ions. C. Ethanoate ions and hydroxide ions.

B. Hydrogen ions and hydroxide ions. D. Hydrogen ions and sodium ions.

32. A solution of pH less than 7 is…..

A. acidic. B. amphoteric. C. basic. D. neutral.

33. Which one of the following salts can be suitably prepared by precipitation method?

A. $BaSO_4$ B. $BaCl_2$ C. $Ba(NO_3)_2$ D. $Pb(NO_3)_2$

2016 QUESTION PAPER1

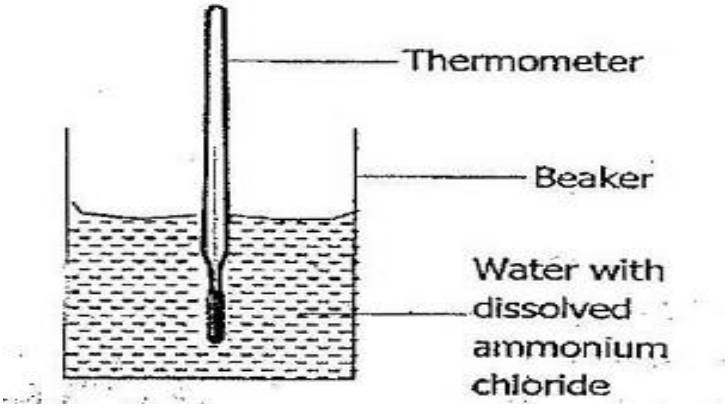
34. Given that the relative formula mass of a compound $Na_2X_2O_3$ is 158, what is element X?

A. Chlorine B. Copper C. Phosphorus D. Sulphur

35. What is the mass of 0.2 moles of chlorine (Cl_2) molecules?

A. 142g B. 71g C. 14.2g D. 7.1g

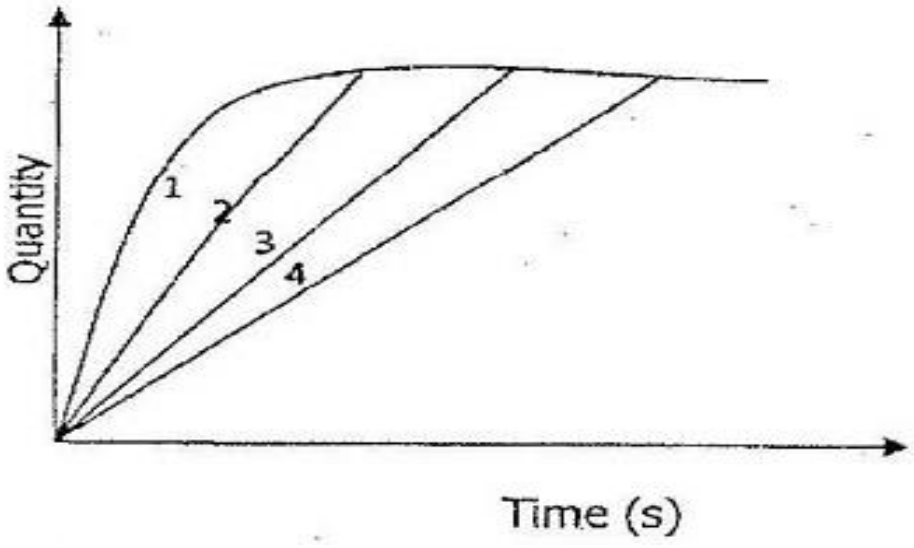
36. 2g of ammonium chloride (NH4C*l*) is dissolved in 20cm^3 of tap water whose initial temperature is 23°C.



Which reading below suggest the temperature of the solution after all the ammonium chloride has dissolved?

A. 25°C B. 23°C C. 20°C D. 10°C

37. The graph below shows the rate of reaction between a metal and a dilute acid.



Which reaction on the graph was the fastest?

A. 1 only B. 4 only C. 1 and 4 D. 2 and 3

38. Dilute hydrochloric add reacts with both magnesium oxide and calcium carbonate. What is common about these reactions?

A. Carbon dioxide is produced **B**. Water is produced

C. Water vapour is condensed **D**. A white precipitate is formed

39. Barium sulphate is one of the insoluble salts which can be prepared by precipitation. Suggest the **two** possible reagents that can be used to prepare this salt.

A. Barium oxide and sodium sulphate. B. Barium nitrate and lead (II) sulphate

C. Barium iodide and sodium sulphate D. Barium chloride and sodium sulphate

40. Which one of the following statements about the Periodic Table is **not** true?

A. The reactivity of the halogens decrease down the group.

B. The elements within the group have the same number of shells.

C. The elements within a group have the same number of electrons in the outermost shell.

D. The elements in a period have the same number of shells.

41. One mole of a sample of hydrated sodium sulphide contains 162g of water of crystallization. What is the correct chemical formula of this compound?

A. $Na_2S.7H_2O$ B. $Na_2S.9H_2O$ C. $Na_2S.3H_2O$ D. $Na_2S.5H_2O$

42. On strong heating copper (II) nitrate decomposed to produce copper (II) oxide, nitrogen dioxide and oxygen according to the balanced chemical equation below;

$2Cu(NO_3)_2 \longrightarrow 2CuO + 4NO_2 + O_2$

Calculate the mass of copper (II) oxide obtained when 56.4g of copper (II) nitrate decomposes.

A. 24.0g B. 40.0g C. 80.0g D. 160.0g

2017 QUESTION PAPER2

43. Which statement best describes the rate of a chemical reaction?

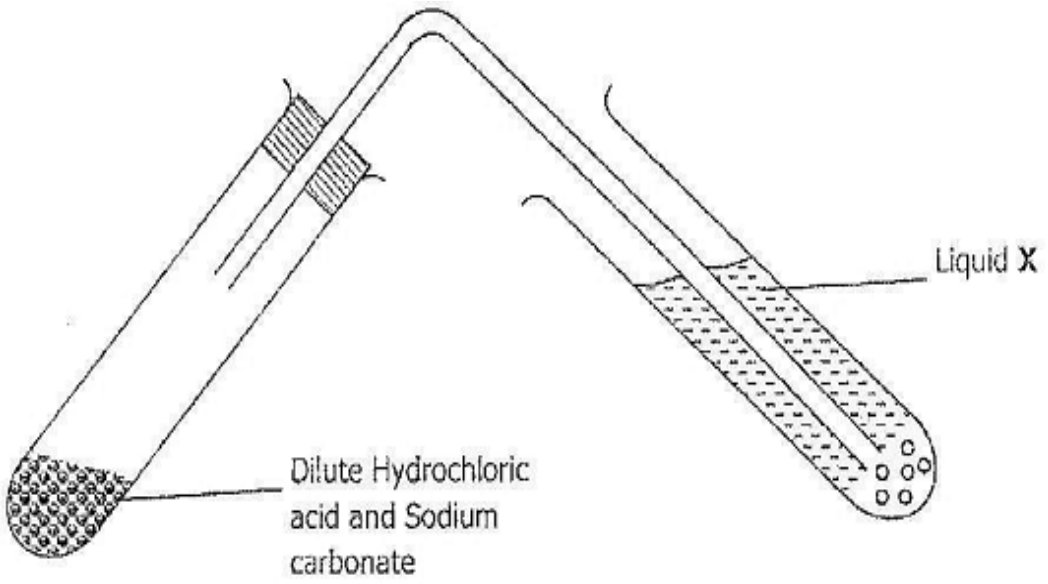
A. The time taken for reactants to be used up.

B. The time taken for one of the reactants to finish.

C. The time taken for products to be formed.

D. The increase in the concentration of a product per unit time.

44. Study the diagram below;



During the experiment a gas and a white precipitate were formed. What is the identity of liquid X and the white precipitate?

	Liquid X	**White precipitate**
A	Water	Calcium carbonate
B	Dilute nitric acid	Calcium oxide
C	Lime water	Calcium hydrogen carbonate
D	Lime water	Calcium carbonate

45. Which of the following salts can be crystallized?

A. Sodium sulphate B. Barium sulphate

C. Lead (II) sulphate D. Silver chloride

46. Halogens play an important role in industry. The halogen which is used in photography is ...

A. Bromine. B. Chlorine. C. Fluorine. D. Iodine.

47. An element is in period 3 and group VII of the Periodic table. Which statement about this element is correct?

A. It forms a cation with a 2+ charge.

B. It is a gas at room temperature and pressure.

C. It is a liquid at room temperature and pressure.

D. It forms an anion with a 2— charge.

48. Solution **P** forms a white precipitate with a little amount of aqueous ammonia solution. The precipitate dissolves in excess ammonia solution to form a colourless solution. The cation present in solution **P** is …..

A. Al^{3+} B. Ca^{2+} C. $NH4^{+}$ D. Zn^{2+}

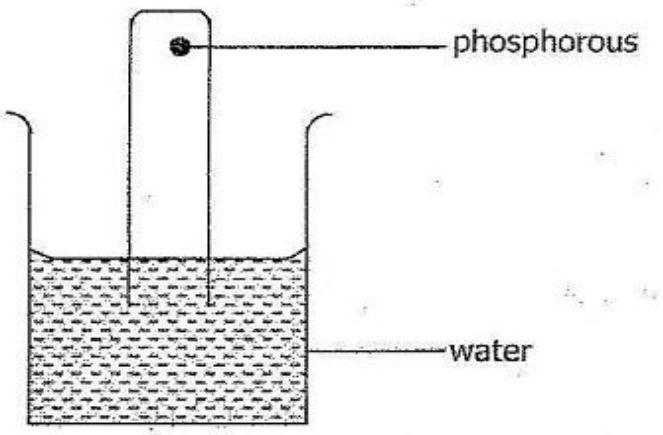
49. A compound X leaves behind a black solid when heated. What is the identity of compound X?

A. Copper (II) hydrogen carbonate B. Magnesium carbonate

C. Sodium hydrogen carbonate D. Calcium carbonate

2017 G.C.E QUESTION PAPER

50. The diagram shows an apparatus used to measure the percentage composition of gases in the atmosphere. Phosphorous reacts with oxygen in the air to form phosphorous (V) oxide which dissolves in water. The initial volume of gas in the tube is 80 cm^3.



What volume of gas remained after all the phosphorous had burned?

A.16 cm^3 **B**. 40 cm^3 **C**. 60 cm^3 **D**.64 cm^3

51. Determine the relative molecular mass of lead (IV) chloride, $PbCI_4$.

A. 249 B. 278 C. 349 D. 378

52. Which change will **not** increase the rate of a chemical reaction? An increase in ,..

A. concentration- of aqueous reactants. B. pressure of gaseous reactants.

C. temperature of a reaction system. **D**. the particle size of solid reactants.

53. Choose a substance which when added in excess to acidic soil will increase its pH without making it alkaline.

A. $CaCl_2$ B. $CaCO_3$ C. CaO D. $Ca(OH)_2$

54. An acid differs from a base in that an acid ...

A. Turns a red litmus paper blue. B. has a pH value above 7.

C. has a sour taste. **D**. turns a blue litmus paper red.

55. Which set of elements exist as diatomic molecules at room temperature?

A. Hydrogen, oxygen, helium. **B**. Nitrogen, chlorine, neon.

C. Nitrogen, oxygen, fluorine. **D**. Oxygen, chlorine, helium.

56. Two elements are in the same group of the periodic table.--Which property will be the same for both elements?

A. Their boiling points **B**. The number of shells

C. Their electronic structure **D**. The charge on their ions

57. When the temperature of a chemical reaction is increased, the kinetic energy of particles increases and the….

A. number of effective collisions increases.

B. number of effective collisions decreases.

C. particles become far apart from each other.

D. particles become closer to each other.

2019 G.C.E QUESTION PAPER2

58. Which one of the following is **not** an example of mineral acid?

A. Sulphuric acid B. Nitric acid

C. Hydrochloric acid **D**. Citric acid

59. Lead (ii) iodide is yellow insoluble salt. Which of the following should be reacted with lead (ii) nitrate solution to produce lead (ii) iodide?

A. Iodine crystals B. Iodine solution

C. Copper (II) chloride solution D. Potassium iodide solution

60. Zinc hydroxide can react with sodium hydroxide because it is ...

A. acidic. B. amphoteric. C. basic. D. neutral.

61. Calcium burns in oxygen according to the following equation;

$2Ca\ (s) + O_2\ (g) \longrightarrow 2CaO\ (s)$

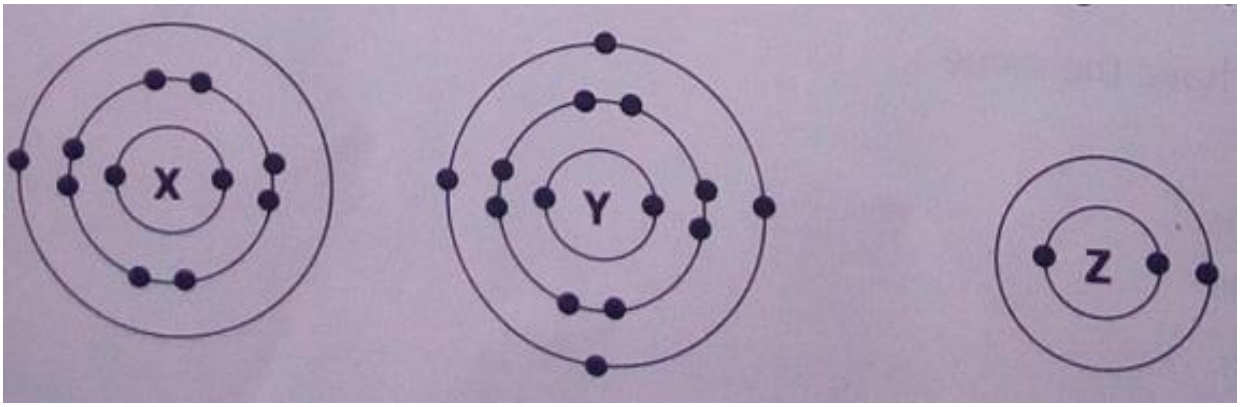
If 5.2g of calcium burns completely, what will be the mass of calcium oxide produced?

A. 6.8g B. 7.3g C. 7.8g D. 8.0g

62. Barium sulphate is best prepared by …..

A . Crystallisation. B . Neutralisation. C . Oxidation. D . Precipitation

63. The order of the reactivity of the elements below starting from the least reactive is .



A. X, Y, Z B. Z, X, Y C. Y, Z, X D. Y, X, Z

64. Which of the following explains the importance of catalysts in chemical reactions? They…

A. Enable the activation energy to be lowered.

B. Enable reactants to be consumed completely.

C. Help to conserve reactants and products.

D. offer more energy for the reaction to take place

65. An element was described as shown in the table below. Which description is correct about the element at r.t.p?

	Number of shells	**valency**	**period**	**Group**	**State**
A	3	4	5	3	Solid
B	3	3	3	6	Gas
C	3	6	6	3	Liquid
D	3	5	3	5	solid

66. Which of the following observations most strongly suggests that a solid element **X** is a non-metal?

A. **X** reacts vigorously with chlorine B. **X** is a conductor of electricity

C. **X** forms an acidic oxide. D. **X** has more than one valency.

2009 QUESTION PAPER 3

67. Part of the process for the extraction of Uranium uses the reaction of Uranium Tetrafluoride (UF4) with magnesium.

$$UF_4 + 2Mg \longrightarrow 2MgF_2 + U$$

(a) State whether Uranium is more or less reactive than Magnesium, explain your answer. [1]

(b) (i) Calculate the relative molecular mass of Uranium Tetracfluoride (UF_4) [1]

(ii) How many tonnes of Uranium can be produced in the above reaction using 24 tonnes of magnesium? [2]

(c) Natural Uranium has several Isotopes. Define the term Isotopes. [1]

68. (a) Work out the relative formula mass, Mr of the following:

(i) $Ca(HCO_3)_2$ (ii) $Al_2(SO4)_3$ [2]

(b) When water containing dissolved calcium hydrogencarbonate is boiled, the calcium hydrogencarbonate decomposes according to the equation below:

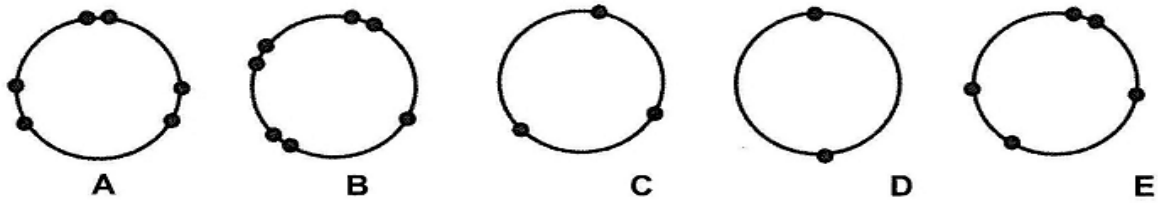
$$Ca(HCO3)_2(aq) \longrightarrow CaCO_3(s) + H_2O\ (l) + CO_2(g)$$

(i) Name the white solid formed in this reaction. [1]

(ii) If the water boiled contained 16.2g of Calcium hydrogencarbonate, calculate the mass of $CaCO_3$ formed. [2]

69. The diagrams below show the electron arrangement in the outer shell of five elements **A** to **E.**

All elements are from Period 3 of the Periodic Table.



(a) Put the letters A to E in the table to shows which elements are metals and which are non-metals.

Metals	**Non-metals**

(b) Which element is most likely to be in Group VI of the Periodic Table? [1]

(c) Which element will form an ion of the type X^{2+}? [1]

(d) Which element has an atomic number of 15? [1]

70. Barium sulphate is an insoluble. Describe briefly how a pure, dry sample of barium sulphate can be prepared using sodium sulphate as one of the reagents.

2010 QUESTION PAPER 3

71. Part of the Periodic Table is shown below. Use it to answer the questions that follow.

H							He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Aℓ	Si	P	S	Cl	Ar

(a) State the chemical symbol for:

(i) An element which is a noble gas. [1]

(ii) The most reactive metal. [1]

(iii) The most reactive halogen. [1]

(iv) The element which supports burning. [1]

(b) **(i)** Write the formula of the compound which would be formed if the element whose atomic number is 13 reacted with an element whose atomic number is 8. [1]

(ii) State the type of bonding which would be present in the compound in **(b)(i)**above. [2]

72. A spillage of 9.8 tonnes of sulphuric acid results from an accident by a road tanker. Slaked lime is used to neutralize the acid.

(a) State the effect of the acid on the vegetation. [1]

(b)The chemical equation for the neutralization reaction is given below:

$$H_2SO_4\,(aq) + Ca(OH)_2\,(s) \longrightarrow CaSO_4\,(s) + H_2O\,(l)$$

(i) Balance the equation. [1]

(ii) Calculate the mass of slaked lime needed to neutralize 9.8 tonnes of spilt sulphuric acid. [2]

(**iii)** State one use of lime in agriculture. [1]

(C) Explain why sulphuric acid is said to be a strong acid. [1]

73. **(a)** Define a salt and give one example. [2]

(b) Iron (ii) sulphate (FeSO4) can be prepared by reacting iron metal and dilute sulphuric acid.

(i) Write a balanced equation for the reaction. [2]

(ii) Which reactant should be in excess? Give a reason for your answer. [2]

(c) How would you obtain fairly pure dry crystals of iron (ii) sulphate from its solution? [3]

(d) Name a salt that can be prepared by precipitation. [1]

2011 QUESTION PAPER 3

74. Use the Periodic Table to answer this question.

(a) Give the symbol of:

(i) a non-metal used to sterilise water, [1]

(ii) an element which forms diatomic molecules, [1]

(iii) An element which reacts with water to give an alkaline solution. [1]

(iv) an element which forms an ion of the type X^{2-} [1]

(b) (i) Oxygen, sulphur and selenium are in Group VI of the Periodic Table. At room temperature oxygen is a gas and sulphur is a solid. Predict whether selenium is a liquid, a solid or a gas, at room temperature. [1]

(ii) The trend in reactivity of Group VI is similar to that in Group VII. Suggest the most reactive element in Group VI. [1]

75. Urea, $(NH_2)_2CO$ and water are formed when ammonia reacts with carbon dioxide. Urea is obtained as a solid from the reaction mixture.

(a) (i) Write a balanced chemical equation including state symbols for this reaction. [3]

(ii) How many atoms of hydrogen are present in one molecule of urea? [1]

(b) What mass of urea in tonnes can be formed from 34 tonnes of ammonia? [2]

76. Caesium, lithium, potassium and sodium are all in Group 1 of the Periodic Table.

(a) Place these metals in order of reactivity, **starting** with the **most** reactive. [1]

(b) All Group **1** elements react in a similar manner with water. [1]

(i) Name the chemical products of the reaction between caesium and water. [2]

(ii) Write a chemical equation for the reaction of caesium with water. Include state symbols. [3]

(iii) What **three** things would you expect to **see** if small pieces of caesium were dropped in water in a glass trough? [3]

(c) What is the other name for Group 1 elements? [1]

77. Barium sulphate (BaSO4) is an insoluble salt which is prepared by precipitation.

(a) Using sodium sulphate as one of the reactants:

(i) Name the other reactant you would use to prepare barium sulphate. [1]

(ii) Write a balanced chemical equation for the reaction. Include state symbols [2]

(iii) Write an ionic equation for the reaction. [1]

(b) Briefly explain how you would obtain a fairly pure dry sample of the salt. [3]

(c) Name **one** salt that can be prepared by the reaction of a metal with a dilute acid. [1]

(d) Calcium chloride (CaCl2) can be prepared by reacting calcium carbonate and dilute hydrochloric acid as shown in the equation below:

$$CaCO_3\,(s) + 2HCl\,(aq) \longrightarrow CaCl_2\,(aq) + CO_2(g) + H_2O\,(l)$$

Calculate the mass of calcium chloride produced by 150g of calcium carbonate. [2]

2013 QUESTION PAPER 3

78. The table below shows the properties of an alkali, an acid and a salt solution.

(a) Complete the table. [5]

Solution	Approximate pH	Ions present
(i)	(ii)	Na^+, OH^-
Hydrochloric acid	(iii)	(iv)
Sodium chloride	7	(v)

(b) Name the two solutions from the table which, when mixed together, form a solution of sodium chloride. [2]

79. (a) Define relative molecular mass (Mr). [1]

(b) Potassium carbonate reacts with dilute sulphuric acid to form potassium sulphate, carbon dioxide gas and water.

(i) Construct a balanced chemical equation, including state symbols for the reaction above. [2]

(ii) Calculate the mass of potassium sulphate produced when 2.0g of potassium carbonate reacts with an excess of dilute sulphuric acid. [2]

(c) What is the identity test of the gas produced in the reaction above? [1]

80. Copper (II) sulphate ($CuSO_4$) is a soluble salt prepared by using sulphuric acid as one of the starting materials.

(a) Define a **salt.** [1]

(b) (i) Name one other reagent which must be reacted with dilute sulphuric acid to form copper **(ii)** sulphate. [1]

(ii) Construct a balanced chemical equation for the reaction in **b (i)** above. [2]

(c) Describe briefly how a dry sample of copper (ii) sulphate can be prepared from the named reagents in **(b).** [4]

(d) Will the method in part **(c)** above be suitable to prepare a sample of potassium chloride? Give a reason for your answer. [2]

2014 QUESTION PAPER 3

81. Salts are prepared in a laboratory using several general methods. ***Table 4.1*** below shows some different methods of preparing salts.

(a) Complete the table.

Reaction		Name of salt formed	Name of reactants
(i)	Precipitation	Barium sulphate	_____ and _____
(ii)	Neutralisation		Potassium hydroxide and sulphuric acid
(iii)	Acid and carbonate	Copper (II) nitrate	_____ and _____

(b) Write down the ionic equation for the precipitation reaction in the *Table 4.1* above. Include state symbols.

82. Magnesium carbonate decomposes on heating to form magnesium oxide and carbon dioxide.

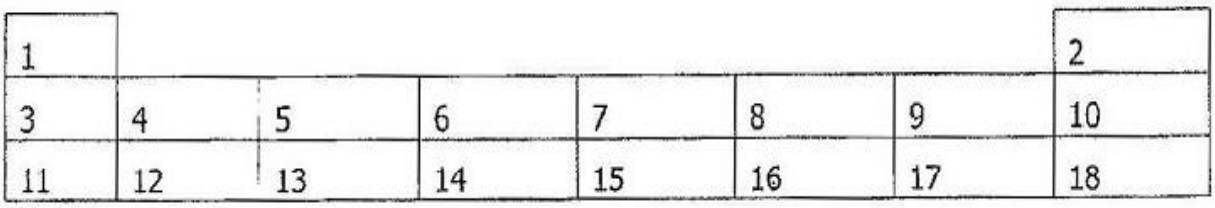
(a) Construct a chemical equation for the decomposition of magnesium carbonate. Include state symbols.

(b) Calculate the relative formula mass of magnesium carbonate.

(c) Given that 21kg of magnesium carbonate is heated, calculate the mass of magnesium oxide formed.

(d) Name a metal carbonate that would not decompose on heating like magnesium carbonate.

83. The grid below represents part of the Periodic Table with atomic numbers of some elements



(a) Using the grid above, state the atomic number of an element which is;

(i) An alkali metal [1]

(ii) A more reactive halogen [1]

(iii) A noble gas used in electric bulbs [1]

(iv) A solid element at room temperature which forms an ion with a charge of negative two. [1]

(b) (i) Use the actual symbols to construct a chemical formula formed by elements with atomic number 13 and 8.

(ii) Give the physical state in which the compound in **b (i)** above will exist at room temperature and pressure.

84. Sodium is an element in Group I of the Periodic Table.

(a) Draw an atomic structure of sodium. [1]

(b) Sodium chloride and hydrogen chloride gas are both compounds of chlorine.

(i) Draw a '**dot**' and '**cross**' diagram to show the bonding in sodium chloride. Show outer shell electrons only. [2]

(ii) Explain why sodium chloride is a solid at room temperature and pressure while hydrogen chloride is a gas. [2]

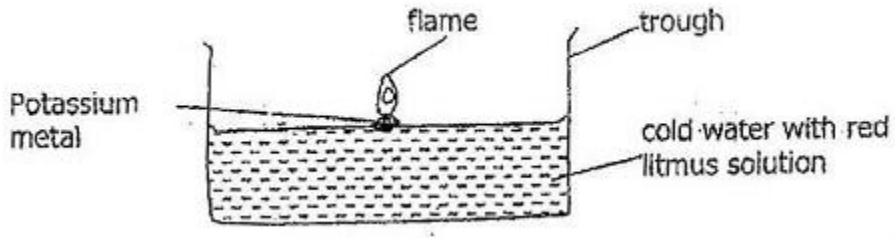
(c) Compare the thermal decomposition of lithium nitrate with that of caesium nitrate. [2]

(d) Explain why the density of Group I metals increases down the Group. [1]

(e) Describe how Group I elements are stored. Give a reason for your answer. [2]

2015 G.C.E QUESTION PAPER 3

85. The diagram below illustrates the reaction that occurs when a small piece of potassium metal is dropped into cold water mixed with a little red litmus solution.



(a) The potassium metal reacts explosively with cold water'. Predict the expected colour change of the litmus solution.

(b) Give a reason for the colour change observed in (a) above. [1]

(c) Bubbles of a colourless gas are produced during this reaction.

(i) Name the gas [1]

(ii) Describe the identity test of this gas [1]

(d) Construct a balanced chemical equation for the reaction between cold water and potassium metal. [2]

86. Pure dry crystals of magnesium sulphate can be made by reacting excess magnesium powder with dilute sulphuric acid.

(a) (i) Explain why excess magnesium powder is used.

(ii) How is excess magnesium powder removed from the reaction mixture?

(b) The chemical equation for the reaction between magnesium and sulphuric acid is given below.

$$Mg(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + H_2(l)$$

Calculate the mass of hydrogen gas produced when 96g of magnesium reacts with dilute sulphuric acid.

87. Bromine is in Group VII of the Periodic Table. It reacts with aqueous potassium iodide to form potassium bromide and iodine.
    (a) Describe what you would see when bromine gas is bubbled through aqueous potassium iodide.
    (b) Construct a balanced chemical equation for the reaction between bromine and aqueous potassium iodide.
    (c) Name the type of reaction in **(b)** above:
    (d) Explain why iodine does not react with a solution of potassium bromide.
    (e) State the similarity in the electron arrangement of halogens.
88. **(a)** Name the black solid reagent that can be reacted with dilute sulphuric add to obtain Copper (II) sulphate solution.
    (b) Describe how you can prepare Copper (II) sulphate crystals. [4]
    (c) Construct a balanced chemical equation with state symbols for the reaction in (a) above. [2]
    (d) In another reaction Zinc oxide was reacted with dilute sulphuric acid according to the equation below.
    $ZnO\ (s) + H_2SO_4\ (aq) + 6H_2O\ (l) \longrightarrow ZnSO_4.7H_2O$
    Calculate the mass of zinc sulphate heptahydrate, $ZnSO_4.7H_2O$, crystals that can be formed from 19.6g of sulphuric acid.
    (e) Define water of Crystallisation.

2016 G.C.E QUESTION PAPER 3

89. When carbon dioxide gas is bubbled through aqueous calcium hydroxide (lime water), the following reaction occurs:
    $Ca(OH)_2\ (aq) + CO_2\ (g) \longrightarrow CaCO_3\ (s) + H_2O\ (l)$
    (a) Write a word equation for this reaction.
    (b) If $1.2dm^3$ of carbon dioxide gas measured at r.t.p was bubbled through excess calcium hydroxide solution. Calculate the:
        (i) Mass of calcium hydroxide solution which reacted. [2]
        (ii) Mass of calcium carbonate formed. [2]
        (iii) Describe what would be seen as the carbon dioxide gas is bubbled through aqueous calcium hydroxide. [2]

90. The diagram below shows part of the Periodic Table.

																H	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	C	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn						Kr

State the name of an element from this part of the Periodic Table which

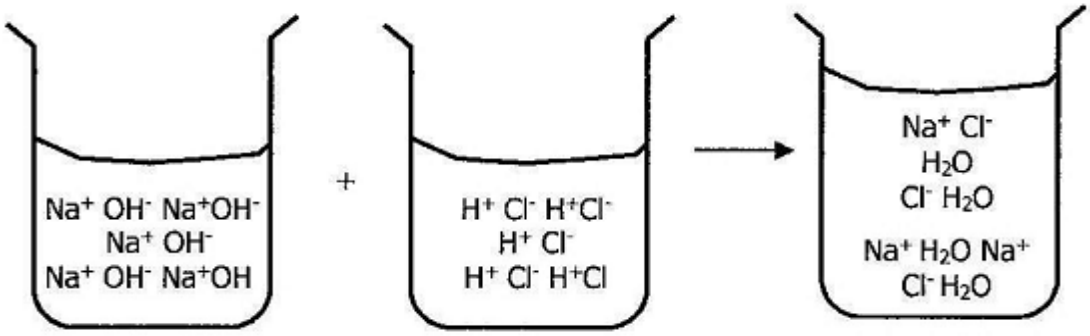
(a) Forms univalent cations. [1]

(b) Reacts most vigorously with cold water. [1]

(c) Forms oxides of formulae XO and X_2O, where **X** represents the chemical symbol of the element.

(d) Does **not** react with oxygen. [1]

91. Solutions of hydrochloric acid and sodium hydroxide are mixed together and they react as shown below.



(a) Write a balanced chemical equation for this reaction. [1]

(b) State the name for this type of reaction and write an ionic equation for it. [2]

(c) Describe how crystals of sodium chloride can be obtained from the solution formed during the reaction. [2]

2016 QUESTION PAPER 2

92. (a) Explain what is meant by limiting reactant. [1]

(b) 2.4g of magnesium reacts with 0.30 mol of hydrochloric acid.

(i) Write the balanced chemical equation for the reaction. [2]

(ii) Determine the limiting reactant. [2]

(iii) Calculate the mass in excess for the substance which is in excess. [1]

93. 20g of marble chips were reacted with 50crn3 of 3.0M hydrochloric acid. The total volume of a gas formed was recorded at various time, intervals and results were tabulated as shown in the table below.

Time (s)	0	15	45	60	75	90
Volume of carbon dioxide (cm^3)	0	40	56	60	60	60

(a) Plot a graph of volume of carbon dioxide against time: [3]

(b) (i) Work out the rate of production of the gas at 30 seconds and 50 seconds. [2]

(ii) What do you conclude about the rate of reaction as time progressed? [1]

94. What do you understand by each of the following?

(a) (i) An anhydrous salt. [1]

(ii) An efflorescent substance. [1]

(b) Give an example of the following:-

(i) An efflorescent substance. [1]

(ii) Deliquescent substance. [1]

95. Bodium, Bo, is a newly discovered element. It is a solid at room temperature and pressure. It can be cut with a knife, reacts vigorously with water liberating a flammable gas and forming a solution with a high pH.

(a) (i) To Which group of the Periodic Table does bodium belong? [1]

(ii) Write a balanced chemical equation for the reaction between bodium with-water

(b) State whether you would expect bodium carbonate to decompose on heating. Give a reason. [1]

2017 G.C.E PAPER 2

96. A spillage of 15.5 tonnes of sulphuric add results from an accident of a road tanker. Slaked lime is used to neutralise the acid according to the equation below:

$$H_2SO_4\,(aq) + Ca(OH)_2\,(s) \longrightarrow CaSO_4(s) + H_2O\,(l)$$

(a) Balance the equation above. [2]

(b) Determine the relative formula mass of $Ca(OH)_2$. [1]

(c) Use the balanced equation to determine the mass of calcium sulphate formed during the neutralization of the spilt acid. [1]

(d) Calcium hydroxide is a base, which ion present in the compound is responsible for its basic properties? [1]

97. Use the following list of elements to answer the questions below. Iron, Lithium, Mercury, Oxygen, Potassium, Sulphur. Each element can be used once, more than once or not at all. Which element
    (a) Is used as a catalyst in the manufacture of ammonia in the Haber process? [1]
    (b) Is lower than sodium in the reactivity series? [1]
    (c) Is a non-metallic solid, whose atoms contain only six valency electrons? [1]
    (d) Is in Period 6 of the Periodic Table? [1]
    (e) Forms an oxide which is amphoteric? [1]
98. Calcium chloride, $CaCl_2$ is a soluble salt that can be prepared in the laboratory.
    (a) Suggest suitable reactants for its preparation in the laboratory. [2]
    (b) Describe how you would prepare a pure dry sample of calcium chloride in the laboratory. [4]
    (c) Lead (II) iodide is an insoluble salt.
        (i) What method can be used to prepare it? [1]
        (ii) Write an ionic equation for the reaction used in the preparation of the salt, include state symbols.

2017 QUESTION PAPER 2

99. When caesium metal is reacted with water, there is a rise in temperature.
    (a) (i) How would you detect the rise in temperature?
        (ii) What type of a reaction takes place?
        (iii) Give a reason for your answer in *(a) (ii)* above.
    (b) Potassium is found in the same group of the Periodic table as caesium.
        (i) Compare the reaction of the two metals with water.
        (ii) Give a reason for your answer in (b) (i) above.
100. Chlorine, Bromine and Iodine are elements in Group VII of the Periodic Table.
    (a) Describe the change in the states of the elements at room temperature and pressure as the atomic numbers increase.
    (b) Why is chlorine used in water treatment?
    (c) Write an ionic equation for the reaction between chlorine and aqueous potassium bromide solution. Include state symbols.

101. A solution of aqueous sodium hydroxide was added from a burette to 25.0cm^3 of dilute sulphuric acid solution in a conical flask. The pH of the mixture was measured during the addition of sodium hydroxide.

(a) Describe how the pH value changed.

(b)(i) Name the type of reaction that took place between sodium hydroxide and sulphuric acid.

(ii) Write a balanced chemical equation for the reaction above (include state symbols)

(c) Sulphuric acid is a strong acid. What does this mean?

102. (a) Define the term concentration.

(b) Calculate the concentration of a solution made by dissolving 60g of sodium hydroxide, (NaOH) pellets in 300cm^3 of water.

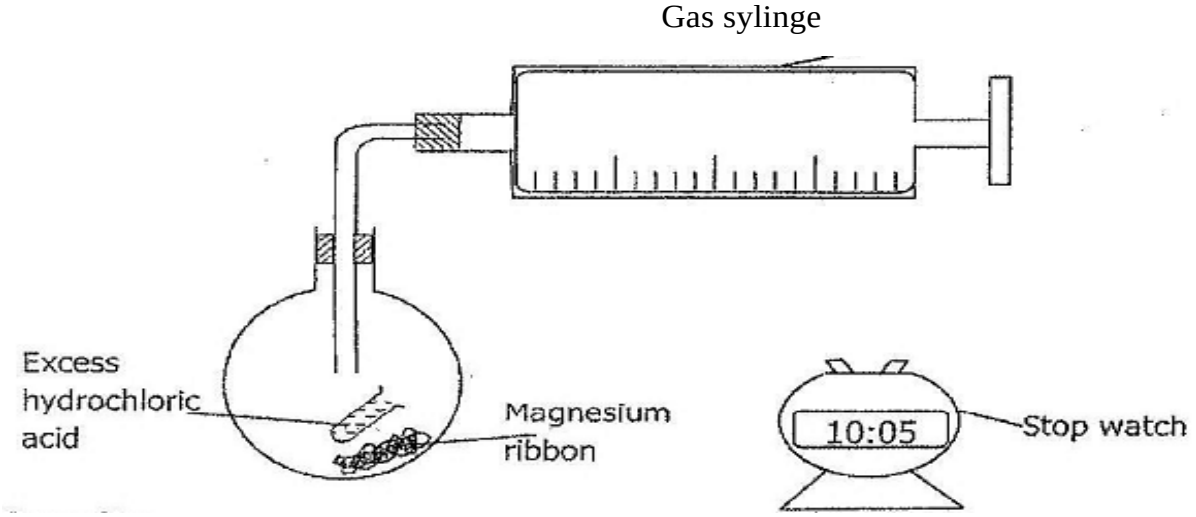
(c) What is the effect of increasing the concentration of the reactants on the rate of a chemical reaction?

103. Draw diagrams to show the arrangement of particles in:

(a) (i) Aluminium metal (ii) Hydrogen chloride

(b) Which of the two substances has a lower melting point? Give a reason.

104. A piece of Magnesium ribbon was made to react with dilute hydrochloric acid. The volume of the hydrogen gas collected in a syringe was measured at intervals. The results are shown below.



RESULTS

Time/minutes	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
Volume of hydrogen gas (cm^3)	0	8	14	20	25	35	33	36	38	39	40	40	40

(a) Write a balanced chemical equation for the reaction between magnesium and dilute hydrochloric acid (include state symbols). [3]

(b) Plot a graph of the results (volume against time) on the graph paper provided. [3]

(c) Which result should be rejected as being an error? [1]

(d) What was the maximum volume of hydrogen produced in this reaction? [1]

(e) From the graph, how can you tell when the reaction came to an end? [1]

(f) What is the average rate of this reaction? [1]

105. Iron (II) sulphate crystals can be prepared from the reaction between iron metal and warm dilute sulphuric acid.

(a) (i) Construct a balanced chemical equation for the above chemical reaction. [2]

(ii) What is the importance of warming the acid? [1]

(iii) How do you ensure that the iron (II) sulphate obtained is free of sulphuric acid? [1]

(iv) Describe how you can obtain pure crystals of iron (II) sulphate from the above reaction. [3]

(b) When an iron nail is placed in an aqueous solution of copper (II) sulphate, a reaction takes place.

(i) Construct an ionic equation for the reaction. [1]

(ii) State two observations you would make during the reaction. [2]

2019 G.C.E QUESTION PAPER 2

106. Aluminium metal was reacted with aqueous Copper (II) sulphate.

(a) Construct a balanced chemical equation for the above reaction. Include state symbols

(b) Deduce an ionic equation from (a). [2]

107. Below is a list of substances which are either deliquescent or hygroscopic and some which are not. Aluminium hydroxide, calcium oxide, sodium hydroxide, sulphuric acid, anhydrous calcium chloride, anhydrous copper (II) sulphate, iron (III) oxide and sodium carbonate.

(a) From the list given above, choose **one** substance which is

(i) Deliquescent.

(ii) Hygroscopic.

(iii) Neither deliquescent nor hygroscopic.

(b) What is the difference between a deliquescent and hygroscopic substance?

(c) Some substances are said to be efflorescent. What is an efflorescent substance?

108. A learner reacted a certain mass of magnesium with 150cm^3 of dilute hydrochloric acid. 240cm^3 of hydrogen gas evolved at room temperature and pressure.

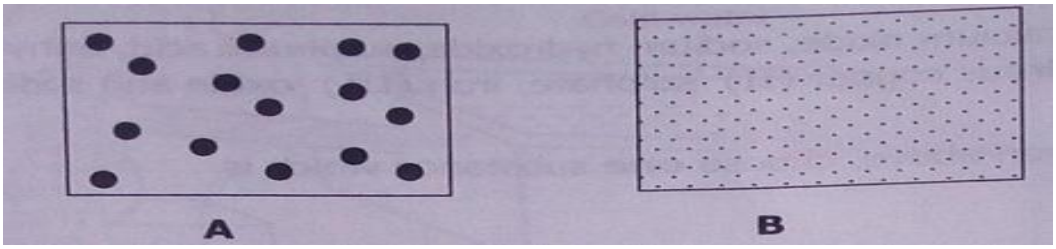
(a) Construct a balanced chemical equation for the reaction.

(b) Calculate

(i) The number of moles of hydrogen in 240cm^3 of the gas.

(ii) The mass of magnesium used in the experiment to produce 240cm^3 of hydrogen at room temperature and pressure.

109. Study the two diagrams below showing different sizes of sodium metals.



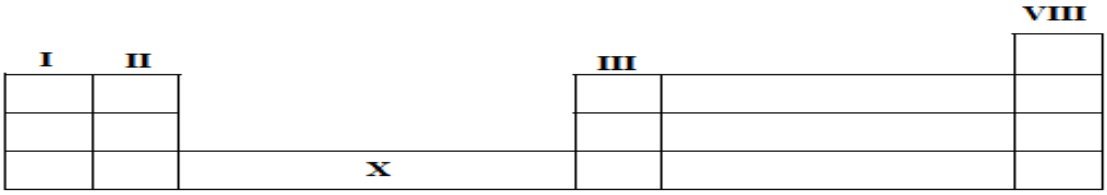
(a) (i) Which diagram would the sodium react faster if put in a beaker containing water?

(ii) Give a reason for your answer in **(a) i.**

(b) State **one** other factor that can affect the rate of this chemical reaction.

(c) State the effect of a catalyst on the activation energy.

110. Below is a skeleton diagram of the Periodic Table.



(a) Give the name of the region marked **X** on the Periodic Table above. [1]

(b) Give any **two** general properties of elements found in the region marked **X.** **[2]**

(c) State any **two** uses of the elements found in the region marked **X.** **[2]**

111. 8.0g of impure sodium hydroxide solution reacted with exactly 40.0cm^3 of 2.0mol/dm^3 sulphuric acid.

(a) (i) Write the balanced chemical equation for the reaction. Include state symbols. [3]

(ii) Construct a simplified net ionic equation for the reaction. [1]

(b) Calculate the number of moles for sulphuric acid that reacted with 8.0g of impure sodium hydroxide. [2]

(c) Find the mass of pure sodium hydroxide from the impure solution that reacted with 40.0cm3 of sulphuric acid.

(d) Determine the percentage purity of sodium hydroxide. [2]

GRADE 12 E.C.Z QUESTIONS

2009 QUESTION PAPER 1

1. Thermal stability of a metal nitrate depends on the reactivity of the metal. Which of the following represents the change when potassium nitrate is heated?

   A. $4KNO_3 \rightarrow 2K_2O + 4NO_2 + O_2$ B. $2KNO_3 \rightarrow 2KNO_2 + O_2$

   C. $KNO_3 \rightarrow$ No change D. $2KNO_3 \rightarrow K_2O_2 + 2NO + O_2$

2. When hydrogen is fitted into the reactivity series of metals, it comes immediately after ……

A. Copper **B.** silver **C**. lead **D**. iron

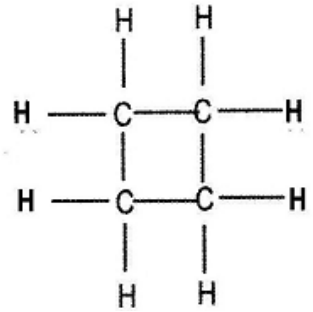
3. Which of the following is used in the manufacturing of margarine?

   A. Oxygen **B**. Nitrogen **C**. Propane **D**. Hydrogen

4. A sample of air of volume $200cm^3$ is enclosed in a tube containing moist iron filings. After the iron has stopped rusting, what volume of air would be remaining?

   A. $40cm^3$ B. $200cm^3$ C. 1 $60cm^3$ D. $200cm^3$

5. Cyclobutane has the structure ...



   Which of the following is true about Cyclobutane?

   A. It is alkene **B**. Its empirical formula is the same as that of all alkanes

   C. It is a saturated hydrocarbon **D**. It decolourizes bromine solution rapidly.

2010 QUESTION PAPER 1

6. No two metals can have exactly the same...

   A. volume. **B**. mass. **C**. properties. D. temperature.

7. During the Haber process, the unreacted nitrogen and hydrogen are ...

   A. scrubbed again. B. compressed further.

   C. pumped back to the catalyst. D. run into tanks to be stored as a liquid.

8. Which of the following is a neutral oxide?

   A. Nitrogen dioxide B. Magnesium oxide C. *Carbon dioxide* D. Carbon monoxide

9. Which of the following alkanes has the highest boiling point?

   A. CH_4 B. C_2H_6 **C**. C_3H_8 D. C_4H_{10}

10. Plastics which get soft only once-the first time they are heated are called ...

    A. thermoplastics. B. thermosets. **C**. isomers. D. monomers.

2011 QUESTION PAPER 1

11. Which of the following is not a physical property of metals?

    A. They react with oxygen to form oxides.

    B. They have high melting and boiling points.

    C. They are good conductors of electricity and heat.

    D. They have high densities.

12. The gases coming from a car's exhaust contain oxides of nitrogen. How are these oxides formed? Nitrogen reacts with

    A. carbon dioxide.

    B. carbon monoxide.

    C. oxygen.

    D. petrol.

13. *When* hydrogen is passed over black powder (Copper (II) oxide) the black powder turns pink. The reaction is shown in the equation below.

    Copper (II) oxide + hydrogen → copper + water

    In this reaction, hydrogen is...

    A. the oxidising agent.

    B. the reducing agent.

    C. one of the products.

    D. being reduced.

14. At the water works, the screen • • •

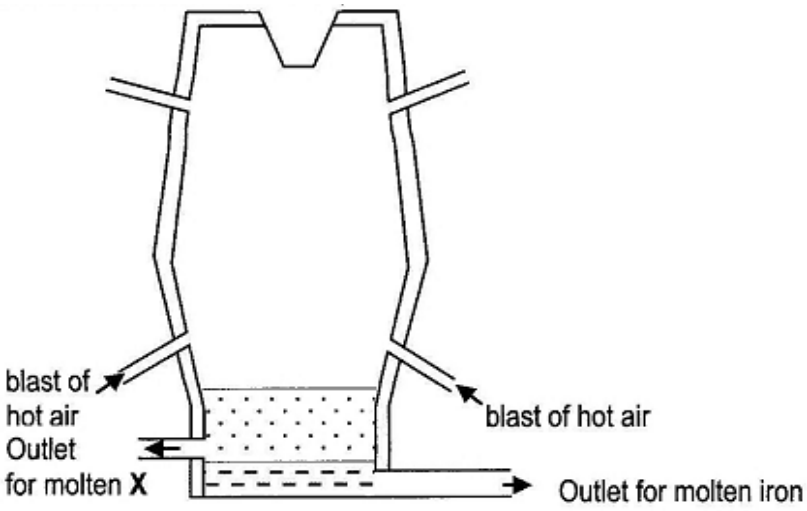
    A. gets rid of the large bits of rubbish.

    B. traps the larger particles of sand.

    C. makes smaller particles stick together.

    D. dissolves and kills any remaining bacteria

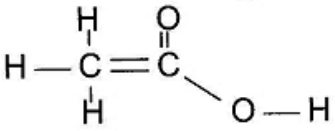
15. The diagram shows the blast furnace used to extract iron from haematite.

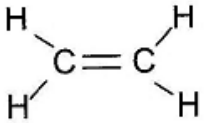


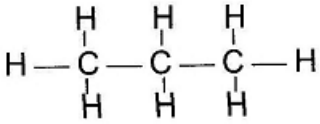
When substance X is drained and solidified, it is used mostly for

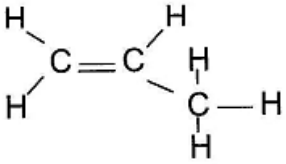
A. road building.
B. making electric wires.
C. making car bodies.
D. making water pipes.

16. Some structures of organic compounds are shown below.

1. 

2. 

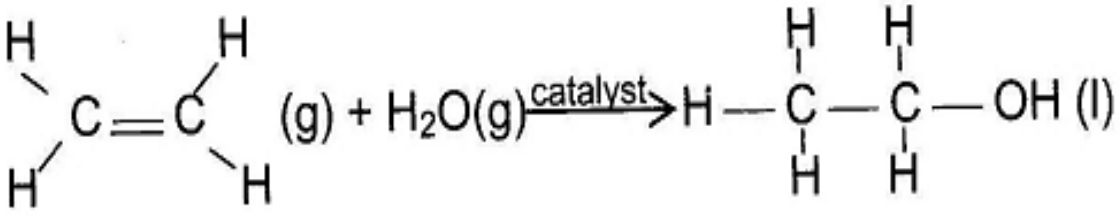
3. 

4. 

Which compound(s) decolourise bromine water?

A. 1 and 2
B. 2 and 4
C. 3 only
D. 1, 2 and 3

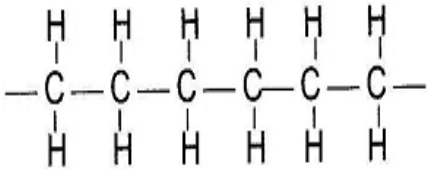
17. Ethanol for use as a solvent is made from ethene and steam as shown below. This reaction is called

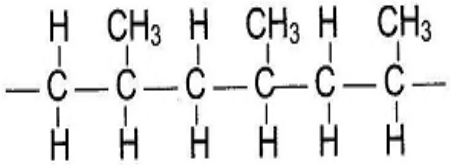


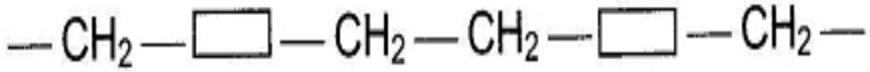
…

A. polymerisation.
B. fermentation.
C. addition.
D. hydrolysis.

18. Which of the following is the structure of polypropene?

A 

B 

C 

D 

19. Which of the following is **not** one of the reasons why a lot of aluminium metal is recycled?

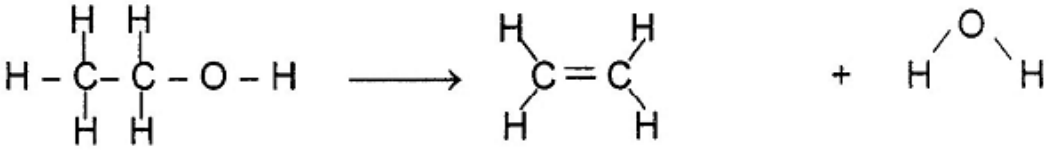
A Recycled aluminium is more resistant to corrosion.

B Recycling is a cheaper way of obtaining aluminium.

C Recycling reduces land degradation.

D Recycling conserves aluminium ores for future generation.

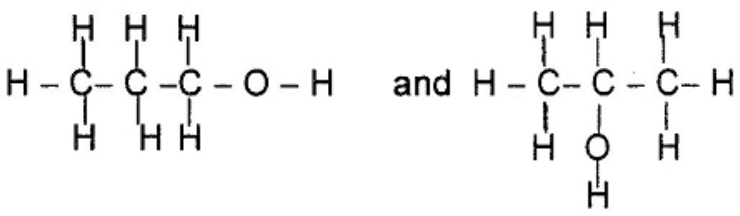
20. When ethanol vapour is passed over heated alumina, ethane is formed according to the equation below:

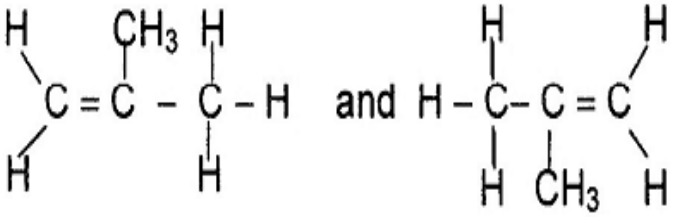


What term is used to describe the above reaction?

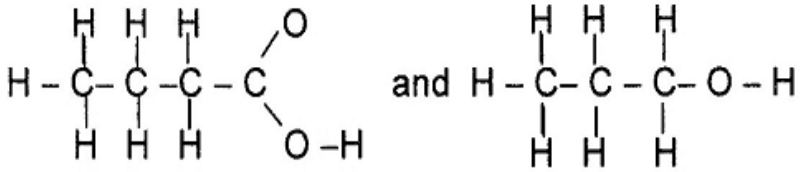
A Reduction **B** Hydrolysis **C** Substitution **D** Dehydration

21. Which of the pairs of compounds represents isomers?

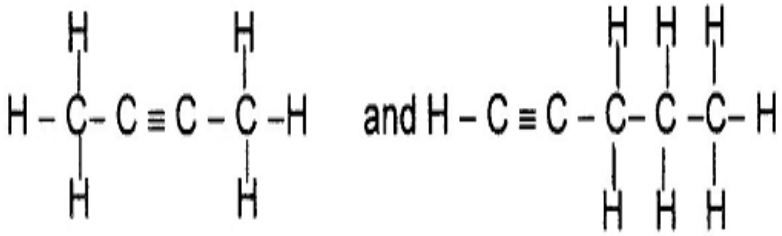
A 

B 

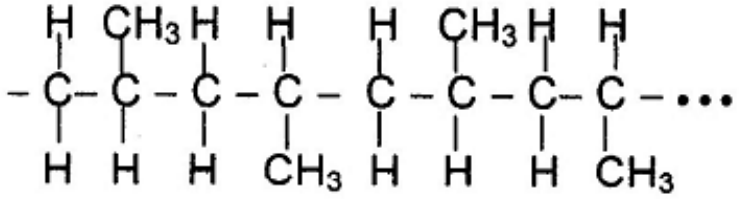
C



D

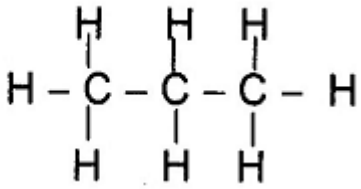


22. The structure below shows an addition polymer.

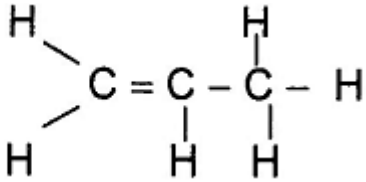


Monomer for the above polymer is…..

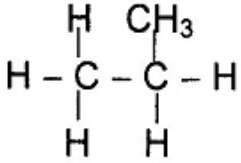
A



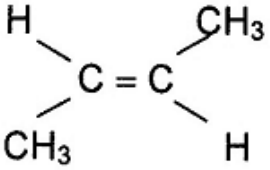
B



C



D



23. Which of the compounds below is likely to be present in the petrol fraction?

A. C_3H_8 B. C_6H_{14} C. $C_{12}H_{26}$ D. $C_{24}H_{56}$

24. In which of the following reactions is the **underlined** substance being oxidised?

A $\underline{CO_2}$ (a) + C (s) --> 2CO (g)

B $\underline{MgO}$ (s) + H_2SO_4 (aq) ---> $MgSO_4$ (aq) + H_2O(i)

C $\underline{HCI}$ (g) + NaOH (aq) → NaCI (aq)

D $\underline{H_2O_2}$ (aq) + Ag_2O (s) → H_2O (i) + Ag (s) + O_2 (g)

25. Which of the following pollutant gases is the major cause of acid rain?

A Nitrogen monoxide **B** Chlorofluorocarbons

C Carbon monoxide **D** Sulphur dioxide

26. Ammonia is manufactured by the Haber process using the reaction represented by the equation below. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

What are the conditions of temperature and pressure which are used to obtain a satisfactory yield of ammonia?

	Temperature	Pressure
A	300°C	450atm.
B	450°C	250atm.
C	1000°C	20atm.
D	450°C	700atm

27. Aluminium cooking utensils are used in many kitchens. What property of aluminium is **not** important for this use?

A. It has a high melting point. **B**. It is a good conductor of electricity.

C. It is a good conductor of heat. **D**. It is resistant to corrosion.

28. Which row in the table below gives a correct use for the metal stated?

A. aluminium - making water pipes B. Copper - galvanising dustbins

C. mild steel - making car bodies D. Zinc - manufacture of aircrafts

29. What is the identity test of the gas formed when an ammonium salt reacts with an alkali?

A. It relights a glowing splint. **B**. It turns damp red litmus paper blue.

C. It burns with a 'pop' sound. **D**. It turns limewater milky.

30. Methane, CH_4, the first member of the alkane homologous series, has the boiling point -161°C. Which molecular formula and boiling point could be correct for another alkane?

A	C_2H_4	-88
B	C_2H_6	-185
C	C_3H_5	-69
D	C_3H_5	-42

31. Which of the following is not a use of ammonia?

A. manufacture of ammonium nitrate **B**. Manufacture of nitrogen

C. manufacture of nitric acid **D**. Manufacture of ammonium chloride.

32. One physical property of all metals is that they are all ...

A. Hard with high melting points. C. reactants forming coloured compounds.

B. Never found native. D. good electrical conductors.

33. Which pair suits the metal and its ore from which it is extracted?

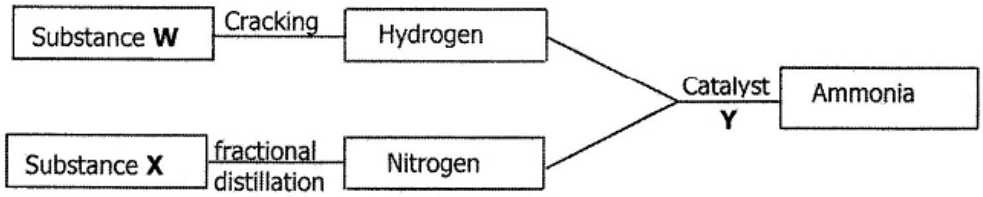
	Metal	**Ore**
A	copper	haematite
B	aluminium	haematite
C	iron	bauxite
D	iron	haematite

34. Which one of the following compounds contains two elements essential to plant growth?

A. Ammonium nitrate **C**. Potassium sulphate

B. Potassium nitrate **D**. Sodium phosphate

35. The diagram below shows processes that lead to manufacturing of ammonia.



Which of the following identifies **W, X** and **Y?**

	W	**X**	**Y**
A	Oil	Air	Vanadium (V) oxide
B	Oil	Air	Iron
C	Air	Oil	Iron
D	Air	Oil	Vanadium (V) oxide

36. Which compound is formed by reacting ethene with steam in the presence of hot phosphoric acid catalyst?

A. Ethane B. Ethanol C. Propane D. Propanol

37. Which of the following processes involves formation of small molecules from large molecules?

A. Formation of starch from glucose. C. Polymerisation of ethene.

B. Hydrogenation of ethene. D. Fermentation of sugar.

2016 QUESTION PAPER 2

38. The metal used to protect galvanized iron from rusting is ...

A. Chromium. B. copper. C. magnesium. D. zinc.

39. One major stage in the extraction of copper metal from its ore, copper pyrite, (CuS) is by roasting. Which of the chemical reactions occur during roasting?

A. $2CuS + 3O_2 \rightarrow 2CuO + 2SO_2$ **B**. $CuS + 2O_2 \rightarrow CuO + SO_3$

C. $CuS + O_2 \rightarrow Cu + SO_2$ **D**. $2CuS + O_2 \rightarrow 2Cu + 2SO$

40. Which one of the following gases is not found in the atmosphere?

A. Argon B. Carbon dioxide C. Hydrogen D. Water vapour

41. Which one of the following would be used in the chemical test for water?

A. Anhydrous copper II sulphate B. Lead II sulphate

C. Litmus paper D. Universal indicator.

42. The two chemical reactions below takes place in the commercial production of a useful gas.

$CH_{4\,(g)} + H_2O_{(g)} \rightarrow CO_{(g)} + 3H_{2\,(g)}$

$CO_{(g)} + H_2O_{(g)} \rightarrow CO_{2\,(g)} + H_{2\,(g)}$

Which of the following processes use the product of the reactions above?

A. Cracking B. Harber process C. Osward process D. Steam reforming

43. Ethane and ethene are hydrocarbons belonging to two different homologous series. These can be distinguished by ….

A. a lighted splint.

B. aqueous barium chloride.

C. aqueous bromine.

D. lime water

44. Polymers are made up of monomers. Identify the correct set of monomers for the respective polymer from the pairs below.

A. Fatty acid and glycerol. **B**. Fatty add and glucose.

C. Glycerol and glucose. ***D***. Glucose and amino acid.

2017 G.C.E QUESTION PAPER 2

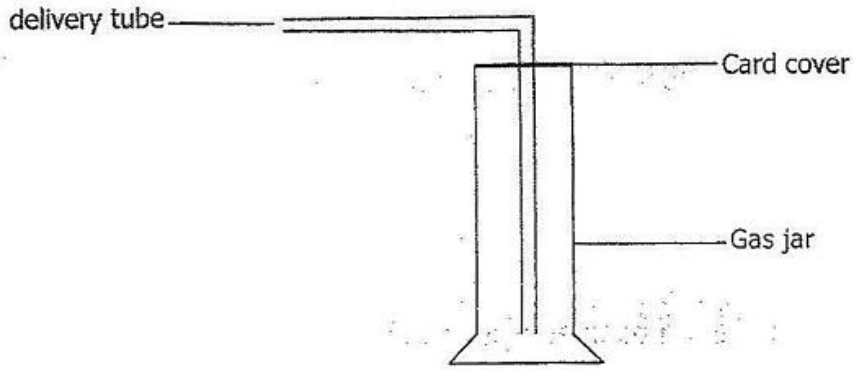
45. Which metal is extracted from its ore by reduction of its oxide by carbon?

A. Aluminum **B**. Copper **C**. Sodium **D**. Zinc

46. Identify the substance which undergoes decomposition because of high temperature in the blast furnace?

A. Calcium silicate **B**. Calcium carbonate C. Coke **D**. Slag

47. A colourless. gas can only be collected using the method shown below



What does this tell you about the gas? It is...

A. denser than air and insoluble in water. **B**. denser than air and soluble in water.

C. less dense than air and insoluble in water. **D**. less dense than air and soluble in water.

48. When ethene is bubbled through aqueous bromine, the solution turns

A. brown. **B**. colourless. C. purple. **D**. red.

49. Methane is a greenhouse gas, which process releases methane into the air?

A. Combustion of petrol B. Decay of vegetable matter C. Volcanic activity D. Photosynthesis

50. When the temperature of a chemical reaction is increased, the kinetic energy of particles increases and the …

A. number of effective collisions increases. B. number of effective collisions decreases.

C. particles become far apart from each other. D. particles become closer to each other.

2017 QUESTION PAPER 2

51. A compound X leaves behind a black solid when heated. What is the identity of compound X?

A. Copper (II) hydrogen carbonate

B. Magnesium carbonate

C. Sodium hydrogen carbonate

D. Calcium carbonate

52. The identity test for the element which is immediately above copper in the reactivity series is that it ...

A. Puts off a burning splint with a pop sound

B. Re-lights a glowing splint.

C. Puts off a glowing splint with a pop sound.

D. Re-lights a burning splint.

53. Graphite powder is used as a lubricant for machinery. What property makes graphite suitable for this use?

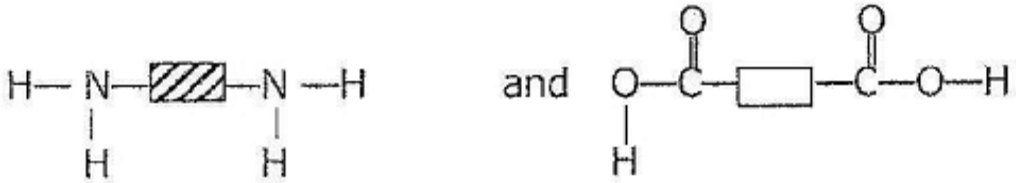
A. It contains many ions.

C. It consists of layers of atoms which slide over each other.

D. It has a structure of small molecules.

B. Its atoms are spherical.

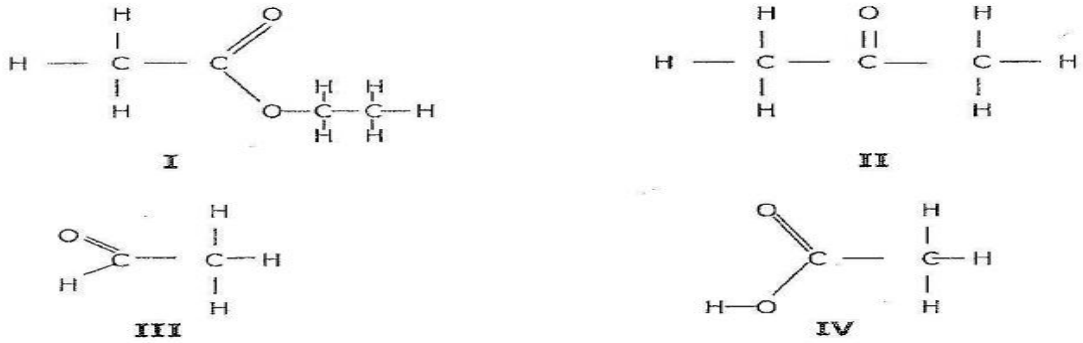
54. A pupil reacted the monomers shown below.



What name is given to the product of the reaction between the two monomers above?

A. Nylon B. Protein C. Starch D. Terylene

55. Below are some structures of organic compounds. Which organic compound will react with rubidium?



A. **IV** B. **III** C. **II** D. **I**

2019 G.C.E QUESTION PAPER 2

56. Which of the following sets corresponds to a metal and the main ore from which it is extracted?

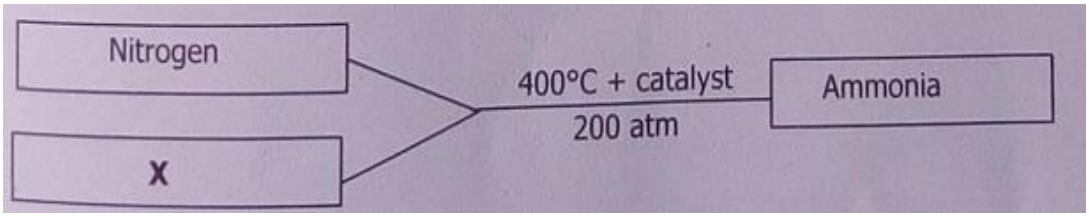
	Metal	**Ore**
A	Zinc	Calamine
B	Iron	Bronze
C	Copper	Magnetite
D	Aluminium	Haematite

57. Which of the following observations most strongly suggests that a solid element **X** is a non-metal?

A. **X** reacts vigorously with chlorine B. **X** is a conductor of electricity

C. **X** forms an acidic oxide. D. **X** has more than one valency.

58. Nitrogen is used to produce ammonia as shown in the diagram below.



What is **X**?

A. Air B. Hydrogen C. Oxygen D. Water

59. Ethene, C_2H_4 burns completely in air to form balanced equation for the reaction?

A. $C_2H_4 + O \longrightarrow CO_2 + H_2O$ B. $C_2H_4 + O_2 \dashrightarrow CO + H_2O$

C. $C_2H_4 + 2O_2 \dashrightarrow 2CO_2 + 2H_2O$ D. $C_2H_4 + 3O_2 \longrightarrow 2CO_2 + 2H_2O$

60. The organic product of the reaction between CH_3OH and HCOOH is …

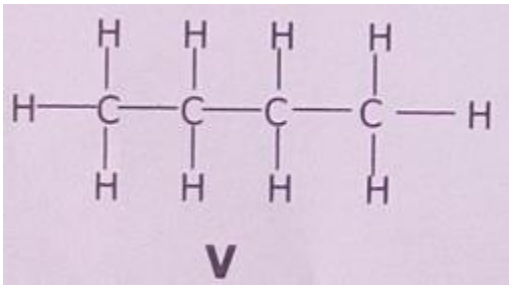
A. CH_3OCH_3 B. $HCOOCH_3$ C. $CHOCH_3$ D. CH_3OOH.

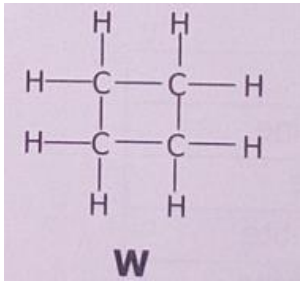
61. Choose the correct monomers for terylene.

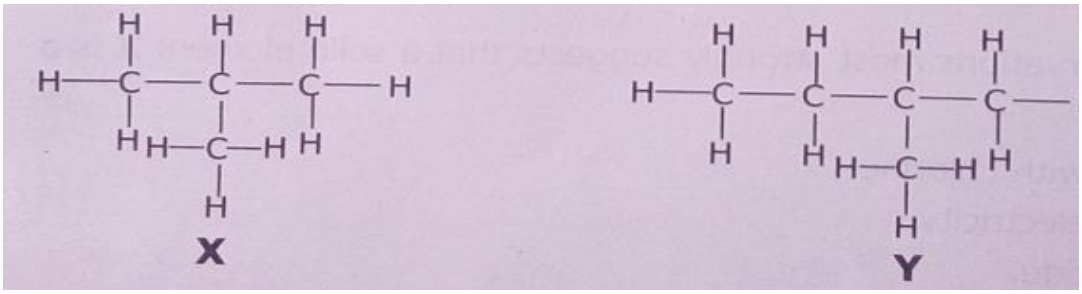
A. Alcohol and carboxylic acid. C. Amine and carboxylic acid.

B. Ethene and diol. D. Ethene and carboxylic acid.

62. The displayed structural formulae below are of different compounds. Which ones are isomers?



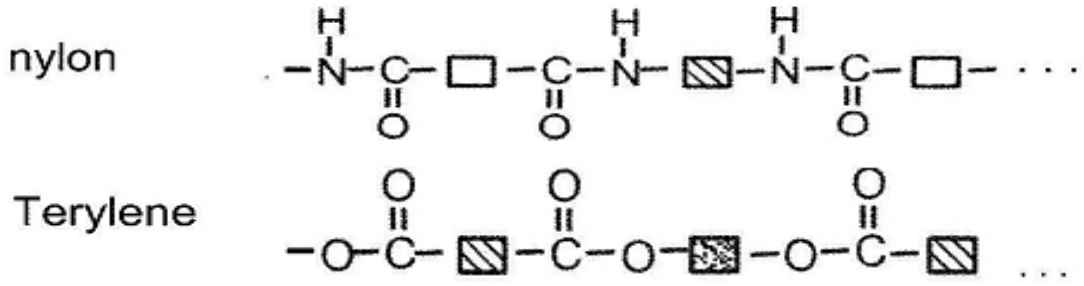




A. **X** and **Y** B. **V** and **X** C. **Y** and **W** D. **V** and **Y**

2009 QUESTION PAPER 3

63. The diagrams below show sections of the polymer chain of two condensation polymers.



(a)(i) Draw a circle around an amide linkage in the diagram. Label this amide linkage. [1]

(ii) Draw a circle around an ester linkage in the diagram. Label this ester linkage. [1]

(b) Name a type of naturally occurring polymer that has a similar linkage to nylon. [1]

(c) Why are nylon and terylene known as condensation polymers? [1]

d) Fishing nets used to be made of natural fibres but many nets are now made from nylon. Suggest one advantage other than strength and one disadvantage of using nylon in place of natural fibres to make fishing nets. [2]

64. Nitrogen and oxygen are the two main gases present in the air. Both gases are obtained from air.

(a) By what process are the two gases obtained from the air? [1]

(b) Nitrogen is used in the manufacture of ammonia by the Haber process. State any **2** important conditions for the reaction in which ammonia is formed by the Haber process. [2]

(c) State one important use of ammonia. [1]

65. The reaction below takes place during the production of calcium oxide when calcium carbonate is thermally decomposed.

$CaCO_3$ (s) → CaO (s) + CO_2 (g)

(a) Give a common name for:

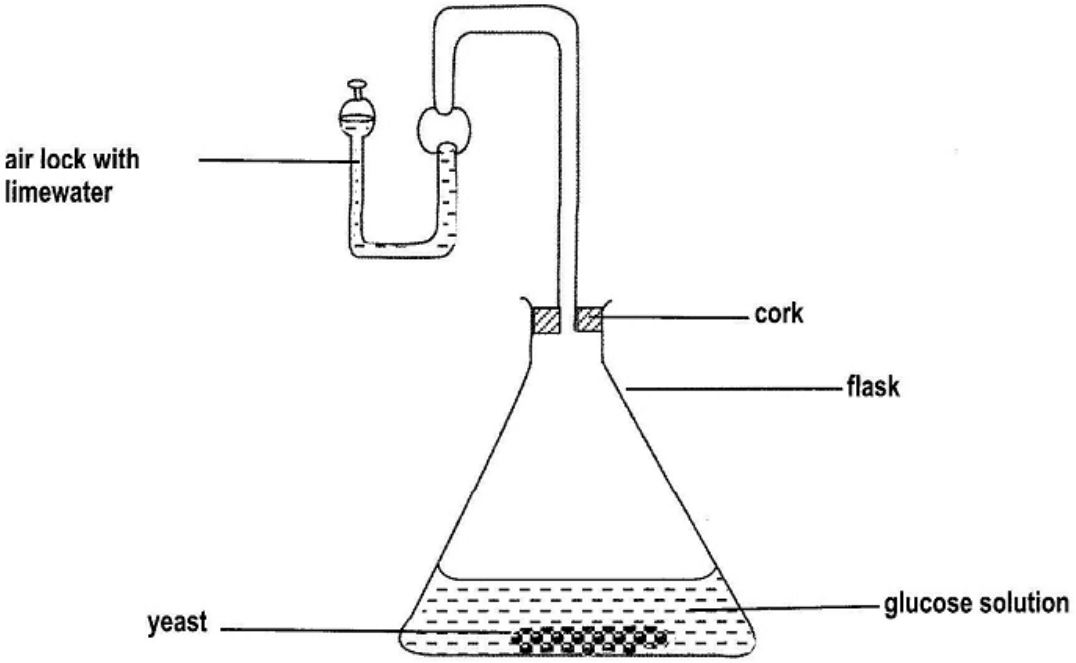
(i) Calcium carbonate.

(ii) Calcium oxide.

(iii) Calcium hydroxide. [3]

(b) What is the Chemical Formula of Calcium hydroxide? [1]

66. A student set up an experiment to produce ethanol from glucose, she dissolved glucose in warm water, added yeast and left the mixture in a warm place for about 7 days in the apparatus shown below.



(a) Write a word equation for the formation of ethanol from glucose. [1]

(b) What is the scientific name for this reaction? [1]

(c) Suggest a suitable temperature at which the process occurs. [1]

(d) The airlock prevents air from going into the flask but allows carbon dioxide gas to go out.

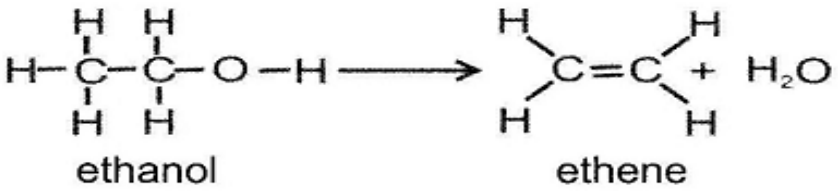
(i) Why must air not be allowed into the flask?

(ii). Describe the change which will be seen in the airlock. [3]

(e) Ethanol can be oxidised to ethanoic acid by boiling ethanol with acidified potassium dichromate (VI). Give two reasons why the conversion of ethanol to ethanoic acid is an oxidation process.

[2]

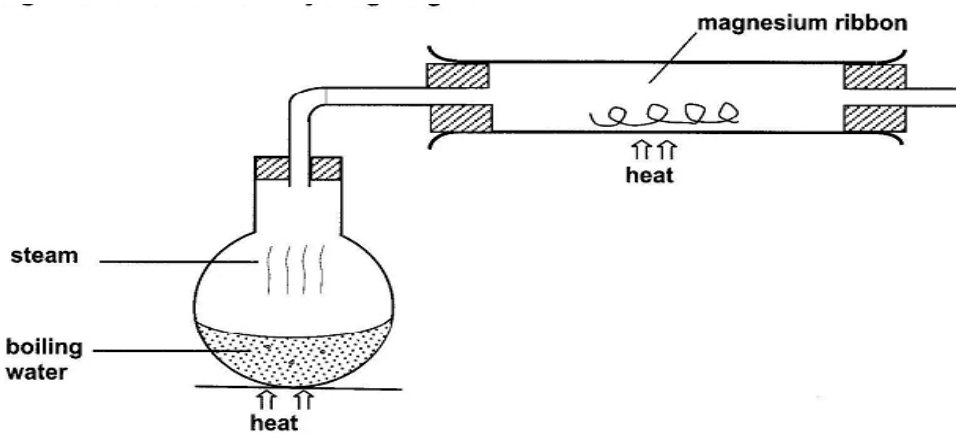
(f) When concentrated sulphuric acid is added to ethanol, the following reaction



(i) What type of reaction is this?

(ii) What test can be done to show that ethene is formed? [2]

67. When magnesium is heated in a stream of steam, it reacts vigorously forming a white solid and hydrogen gas.



(a) Complete the diagram to show how hydrogen gas can be collected. [2]

(b) Write a balanced chemical equation for the reaction of magnesium with steam. [3]

(c) Choose two other metals from the reactivity series of metals which will also react with steam and one metal which will not react with steam. [3]

(d) Suggest a method which can be used to extract magnesium from its ores. [1]

(e) When a piece of magnesium is added to copper (ii) sulphate solution, copper metal is formed according to the word equation below:

Magnesium + Copper (II) sulphate → magnesium sulphate + copper.

(i) Write a balanced chemical equation for the reaction. [1]

(ii) What type of a reaction is this? [1]

68. (a) The increase in industrial activity in Zambian cities has results in high levels of atmosphere (air) pollution. Carbon monoxide is one of the pollutants of the air.

(i) How is carbon monoxide produced? [1]

(ii) Explain why carbon monoxide is a dangerous gas. [1]

(iii) Name the pollutant gas which is mainly responsible for producing acid rain. [2]

(b)(i) State the two main stages in the purification of domestic water supply.

(ii) Give two industrial uses of water. [2]

(c) When steam is passed over very hot carbon, a mixture of carbon monoxide and hydrogen gas is formed.

H_2O (g) + C (s) → H_2 (g) + CO_2 (g)

State which substance is

(i) Reduced. (ii) Oxidised. [2]

2010 QUESTION PAPER 3

69. Iron, calcium and copper are metals. The table below describes the reactions of these metals with cold water and steam.

(a) (i) Put a tick (✓) if a reaction will take place and a cross (x) if a reaction will not take place. [3]

Metal	Reaction of metal with cold water	Reaction of metal with steam
Copper		
Iron		
Calcium		

(ii) Place these three metals in order of chemical activity, starting with the most reactive. [1]

(b) Before experimenting with aluminium to place it in the above series, the surface of the aluminium must first be scraped. Why is this necessary? [2]

(c) Give two reasons why it is important to recycle metals. [2]

70. Oxidation can be described as either the addition of oxygen to a substance or the removal of hydrogen from a substance. Study the reactions given below and state whether the substance underlined has been reduced or oxidized.

(a) (i) Copper (ii) oxide + <u>ammonia</u> → copper + nitrogen + water

(ii) <u>Carbon dioxide</u> + carbon → Carbon momoxide.

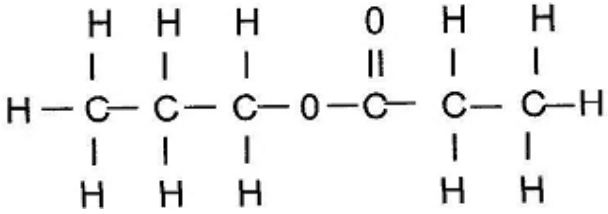
(iii) Iron (ii) oxide + <u>Aluminium</u> → Aluminium oxide + Iron [3]

(b) Steam reacts with carbon as shown in the chemical equation below.

H_2O (g) + C (s) → H_2 (g) + CO(g)

Identify the oxidizing agent. Give a reason for your answer. [2]

71. Below is a structure of an ester made in a reversible reaction between a carboxylic acid and an alcohol.



(a) (i) Draw the structure of the carboxylic acid used in the reaction. [1]

(ii) State one condition necessary for the formation of the ester above. [1]

(b) A student carried out an experiment to compare the relative strengths of dilute methanoic acid and dilute sulphuric acid.

(i) Describe a test that can be used to distinguish between the two acids. [2]

(ii) Name a metal that will react with both acids. Describe what you would see during the reaction. [2]

72. A carbohydrate is formed from the reaction of many small molecules, one of which is represented by the diagram below:



(a) (i) What is the general name of the small molecules which combine to form very large molecules?

(ii) Show how two of the small molecules like the one drawn above would join together to form a bond.

(b) *Starch is* hydrolysed to *glucose* by the *enzymes in yeast and* the *glucose* is then converted to an alcohol by a second process. Name:

(i) the second process.

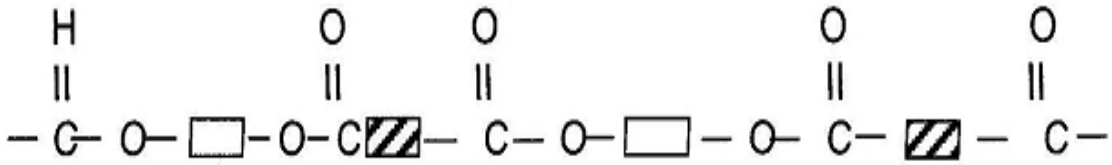
(ii) the alcohol produced.

(c) State two uses of alcohol named in **(b) (ii)** above.

(d) Terylene is an ester.

(i) state one use of terylene.

The structure of terylene is represented in the diagram below.



(ii) Draw a box around a repeating unit in this structure.

(iii) Why does terylene cause pollution?

<u>2011 QUESTION PAPER 3</u>

73. (a) Name three components of clean, dry air. [3]

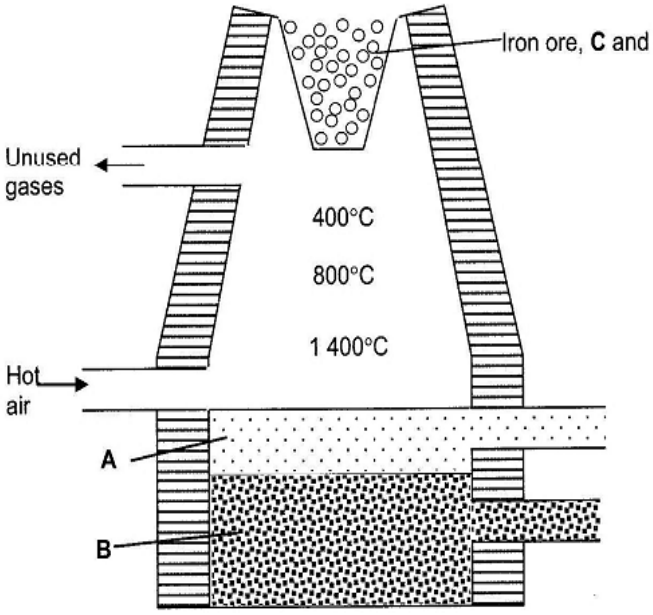
(b) Air can be polluted various chemicals:

(i) Give the chemical name for one of these air pollutants. [1]

(ii) Name the source of the pollutants mentioned in b (i) above. [1]

(iii) State the process by which it is produced. [1]

74. Study the diagram below on extraction of iron.



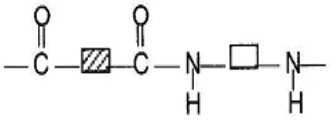
(a) (i) Apart from iron ore, name **two** other materials **(C** and **D)** that are fed into the reaction chamber. [2]

(ii) Write the balanced chemical equation for the reduction reaction of iron ore to iron metal [2]

(b) Name substances **A** and **B.** [2]

(c) State **one** physical characteristic of the brick lining in the furnace. [1]

75. Plastics are organic polymers. The table below describes two types of plastics. Complete the table. Part of the table has been completed for you as an example.

Name	Repeating unit	Use	Type of polymerisation used in manufacture
Poly(ethene)	(a)	Making plastic bag	(b)
(c)		(d)	Condensation polymerisation

76. (a) Organic compounds form homologous series.

Give two characteristics of members of any homologous series. [2]

(b)(i) Draw the structure of an alkane with **two** carbon atoms in the molecule. [2]

(ii) Calculate the percentage by mass of hydrogen in this alkane. [2]

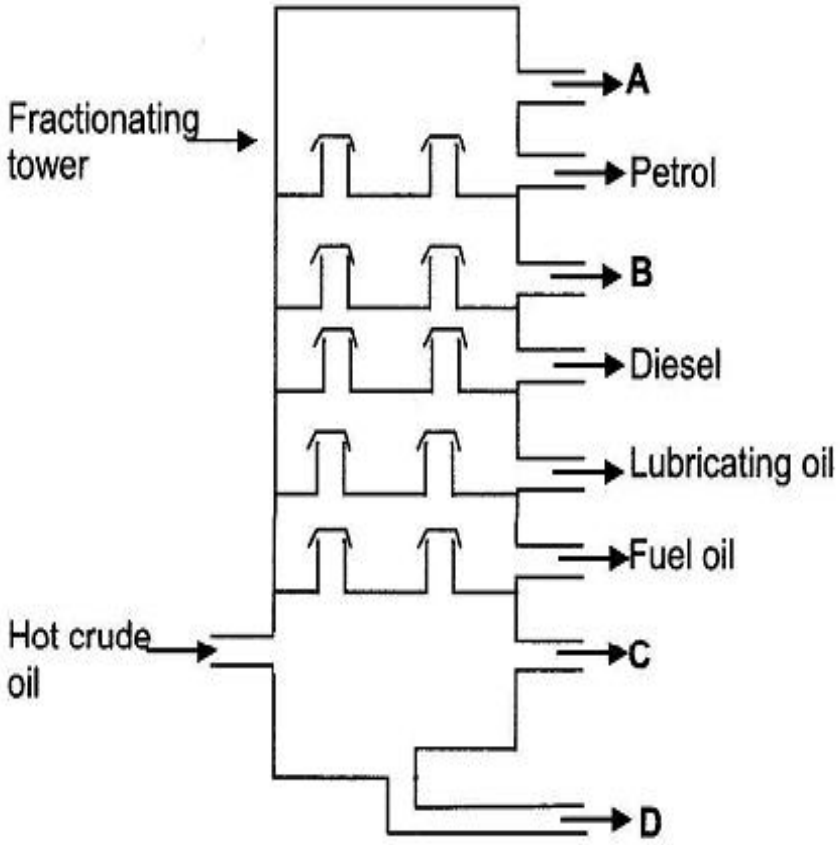
77. Oxygen and nitrogen are the major gases present in the clean air. Other gases which are present in the clean air in small quantities include carbon dioxide and argon.

(a) State the percentage of oxygen and nitrogen in the air. [2]

(b) Briefly describe how you would show that carbon dioxide is present in the air. [2]

(c) Argon is a noble gas. What chemically makes it unreactive? [1]

78. Crude oil is an important raw material that we need in modern life. Engineers process and refine crude oil in a tower to produce a number of fractions as shown in the diagram below.



(a) Name the process used to separate the fractions of crude oil. [1]

(b) Name the **two** major elements that are found in crude oil. [2]

(c) State **two** differences in physical properties between fractions extracted at **C and B.** [2]

(d) A large proportion of fraction **A** has the molecular formula CH_4. Write a balanced chemical equation for the complete combustion of the compound with chemical formula CH_4. [2]

(e) Crude oil is a non-renewable energy source. Explain what is meant by a non-renewable source. [1]

(f) Give **two** reasons why the sun is a better source of energy than crude oil. [2]

2012 QUESTION PAPER 3

79. The list below shows metals arranged in ascending order of reactivity: Silver, Zinc and Aluminium Sodium. Using metals from this list **only,** name:

(a) a metal which can be displaced by copper. [1]

(b) a metal which reacts with cold water to produce an alkaline solution. [1]

(c) a metal which forms an amphoteric oxide when burnt. [1]

(d) a metal whose carbonate does **not** decompose when heated. [1]

(e) a metal which forms a stable oxide layer. [1]

(f) Write a balanced chemical equation for the reaction between sodium and water. [2]

80. Organic acids are a homologous series of compounds having the carboxylic group — COOH joined to an alkyl radical.

(a) What is the general formula for organic acids? [1]

(b) Draw the structure of butanoic acid. [1]

(c) A reaction between an alcohol and an organic acid is described as esterification and this is similar to neutralisation. [3]

(i) Ethylethanoate is an ester. Name two reagents used to prepare it. [2]

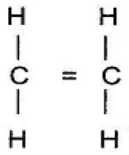
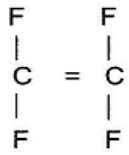
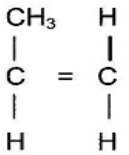
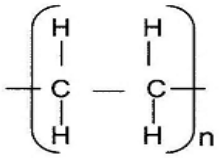
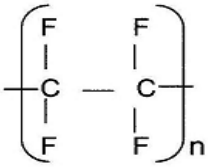
(ii) State two ways in which esterification is different from neutralisation. [2]

(iii) Write a balanced chemical equation for the esterification of ethylethanoate. [2]

(d) Calculate the mass of ethylethanoate formed from 15g of the organic acid. [2]

2013 QUESTION PAPER 3

81. The table below shows the structural formulae of various monomers and the polymers that can be made from them.

Monomer	 ethene	 tetrafluoroethene	 **W**
Polymer	 poly(ethene)	 poly (tetrafluoroethene)	**Y**

(a) Identify the monomers which are hydrocarbons. [2]

(b) Name the monomer **W.** [1]

(c) Draw the displayed structure for polymer **Y**. [1]

(d) (i) Identify the common feature among the monomers in the table above. [1]

(ii) Give the name of the compounds with this feature. [1]

(e) Decane $C_{10}H_{22}$, can be cracked to produce hexane, C_6H_{14} and butane, C_4H_8

Explain the meaning of the word cracking. [1]

82. (a) Give **two** ways by which rusting of iron can be prevented. [2]

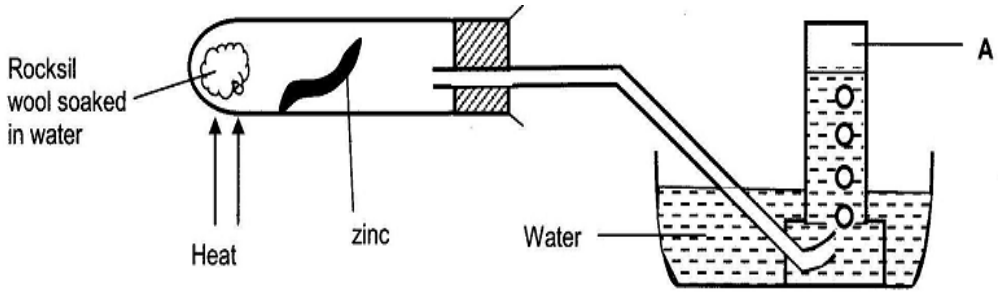
(b) Burning and rusting are two chemical processes. Give

(i) One similarity (ii) One difference [2]

(c) (i) Name **two** non-metallic oxides which cause atmospheric pollution. [2]

(ii) State **two** sources of these pollutants. [2]

83. Zinc can be reacted with steam using the apparatus shown below in the reaction of zinc with steam, gas **A** and solid **B** are formed.



(a) Name gas **A**. [1]

(b) State the product formed when gas **A** burns in air. [1]

(c) Give **two** uses of gas **A.** [2]

(d) (i) What is solid **B?** [1]

(ii) Construct a balanced chemical equation with state symbols for the reaction between substances **B** and dilute hydrochloric acid. [3]

84. Below is a list of some organic compounds represented by the letters **A** to **E.**

A	**B**	**C**	**D**	**E**
CH_4	C_4H_{10}	C_4H_9OH	C_4H_8	CH_3COOH

(a) Give the name of compound **A.** [1]

(b) Identify the **two** compounds, from the list, that belong to the same homologous series? [2]

(c) (i) State the compound which has the same functional group as ethanol. [1]

(ii) Draw the displayed molecular structure of ethanol. [1]

(iii) Construct a chemical equation for the hydration of ethene to form ethanol. [1]

(d) (i) Give **one** compound that is an unsaturated hydrocarbon, from the list above. [1]

(ii) Describe a chemical test for an unsaturated hydrocarbon. [1]

(e) Compound **E** is acidic.

(i) State the name of the compound **E.** [1]

(ii) Describe a test to show that compound **E** is acidic. [1]

2014 QUESTION PAPER 3

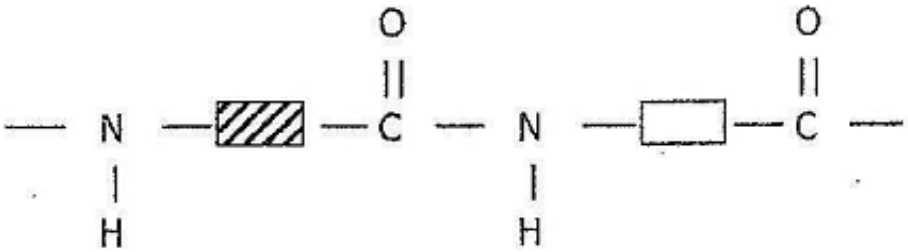
85. Proteins are naturally occurring macromolecules.

(a) (i) Define the term macromolecule,

(ii) Name another naturally occurring macromolecule.

(b) Proteins can be hydrolysed to amino acids. State a suitable reagent and condition for this hydrolysis.

(c) The structure of a section of a protein can be represented as



(i) Describe one similarity in the structure of a protein and the structure of nylon.

(ii) Describe one way in which the structure of a protein differs from the structure of nylon.

86. Complete the *Table 8.1* below on the properties and the use of the metal that depend on the property.

(a) The first part has been done for you**.**

PROPERTY OF METALS	***USE***
High melting/boiling point	Filament in an electric bulb
(i)	Making bells/drums
Ductility	**(ii)**
(iii)	Making ornaments
Free electrons	**(iv)**

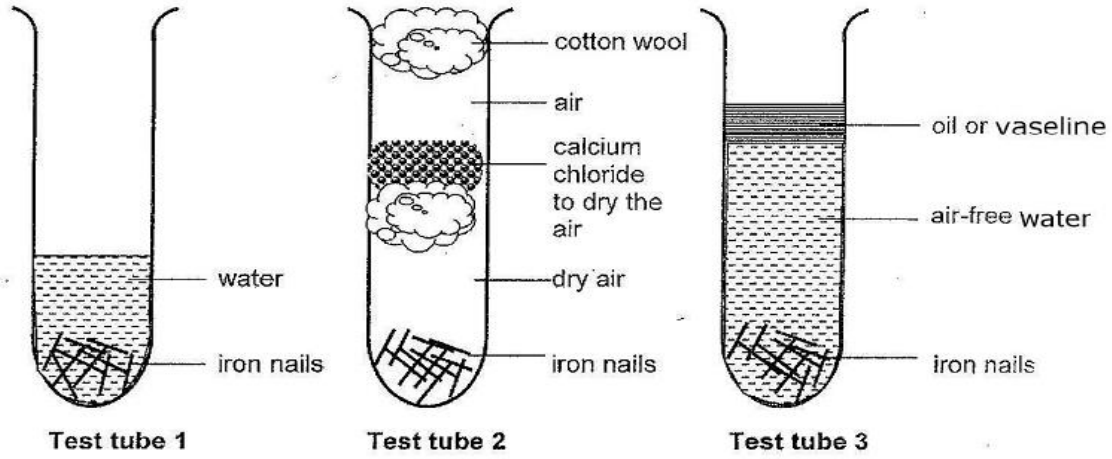
(b) Magnesium reacts with dilute nitric acid to form magnesium nitrate and a colourless gas. Construct a balanced chemical equation for this reaction.

87. Oxygen can be made on a large scale by fractional distillation of liquid air.

(a) Briefly outline the three major stages in this process.

(b) Describe the test for oxygen.

(c) The diagrams below show the apparatus set up to study the process of rusting.



The test tubes were left to stand for three days.

(i) Describe what is observed in test tubes 1, 2 and 3.

(ii) Give a reason for each observation in test-tubes 2 and 3.

88. Ethene, C_2H_4, is a hormone found in most plants. Tomatoes release ethene as they ripen. Fruit growers use ethene as a ripening agent.

(a) Draw a displayed structural formula of ethene. [1]

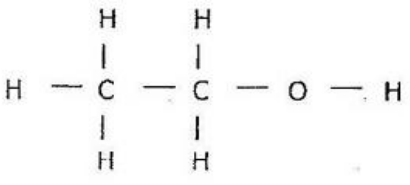
(b) (i) State the homologous series to which ethene belongs. [1]

(ii) Give the general formula of this homologous series. [1]

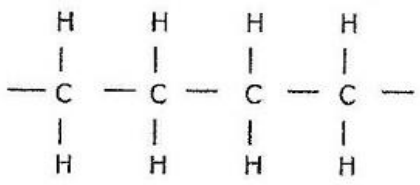
(iii) State the molecular formulae of the fourth and fifth members of this homologous series respectively. [2]

(c) State **one** major source of ethene and the method by which it is obtained. [2]

(d) Two of the substances made from ethene are shown below.



Substance **R**



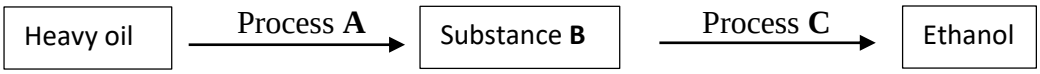
Substance **S**

(i) Name the substances **R** and **S.** **[**2]

(ii) State **one** use of substance **S.** [1]

2015 G.C.E QUESTION PAPER 3

89. Study the flow diagram below carefully.



(a) Name Process **A** and Process **C**. [2]

(b) Name Substance B. [1]

(c) Give **one** significance of Process **A.** **[1]**

90. Aluminium, copper, sodium and zinc are a few of the common metals.

Use **only** these metals to answer the questions below.

(a) Arrange the metals above in order of decreasing reactivity. [1]

(b) (i) Select the metal which is obtained from its ore by reduction with carbon. [1]

(ii) Name the ore from which the metal in **b (i)** is obtained. [1]

(c) Explain why aluminium is unreactive even though it is high in the reactivity series. [2]

91. Propanoic acid, CH_3CH_2COOH, is a weak acid.

(a) Explain what is meant by the term weak acid.

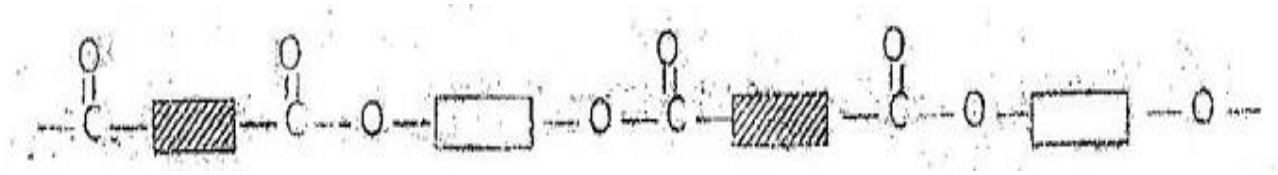
(b) Propanoic acid reacts with potassium carbonate. Write a balanced chemical equation for this reaction.

(c) Calcium reacts with propanoic acid to form calcium propanoate and hydrogen gas as shown below.

$Ca\ (s) + 2CHCH_2CO_2H\ (aq) \rightarrow (CH_3CH_2CO_2)_2Ca\ (aq) + H_2\ (g)$

Calculate the mass of hydrogen produced in the reaction shown above when calcium is added to excess propanoic acid.

(d) Terylene has the simplified structure shown below.



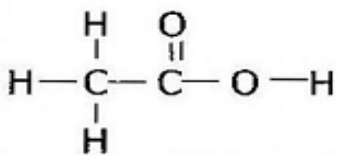
(i) State the functional groups on the monomers used to make terylene.

(ii) State the type of polymerisation that occurs when terylene is made.

(e) Many problems are caused by the disposal of plastics. Describe one method of disposal of plastic and a problem caused by this method.

2016 G.C.E QUESTION PAPER 3

92. The following table shows some information about the **second** member in the respective homologous series. Complete the table by filling **in** the correct information.

homologous series formula	**Name of the member**	**Displayed full structural formula**
Alkane	(i)	(ii)
(iii)	Ethanol	(iv)
(v)	(vi)	

93. Copper is extracted from one of the ores, copper pyrites, $CuFeS_2$. The copper pyrites is first converted to copper (I) sulphide before copper metal is produced.

(a) Write down the equations leading to the formation of copper metal from copper pyrites. [2]

(b) Explain why this extraction leads to air pollution. [1]

(c) State the process used to purify copper. [1]

(d) State any **two** large scale uses of copper and give reasons why copper is used in that way. [4]

(e) (i) Copper occurs 'native' in some countries such as Zambia.

Explain what is meant by the word 'native'. [1]

(ii) Name **one** alloy made from copper metal. [1]

94. When a mixture of ammonium sulphate and sodium hydroxide is heated, the reaction represented by the word equation below occurs.

Ammonium sulphate + sodium hydroxide → sodium sulphate + water + ammonia.

(a) Write a balanced chemical equation for the above reaction. [2]

(b) Describe a chemical test you can carry out to show that ammonia is formed. [2]

(c) On a large scale, ammonia is produced by the Haber process:

(i) What are the reactants used in the Haber process? [2]

(ii) Write a balanced chemical equation for the Haber process. [2]

(iii) State the essential conditions used in order to obtain an economical yield of the ammonia gas. [2]

95. Propene ($CH_3 - CH = CH_2$) reacts with steam to give a major product propan -2 -ol.

(a) State the essential conditions in this reaction. [2]

(b) State the kind of reaction that occurs between propene and steam. [1]

(c) However, there is also a minor by-product formed in such a reaction. [1]

(i) Give the full (displayed) structural formula and the systematic name of this by-products. [2]

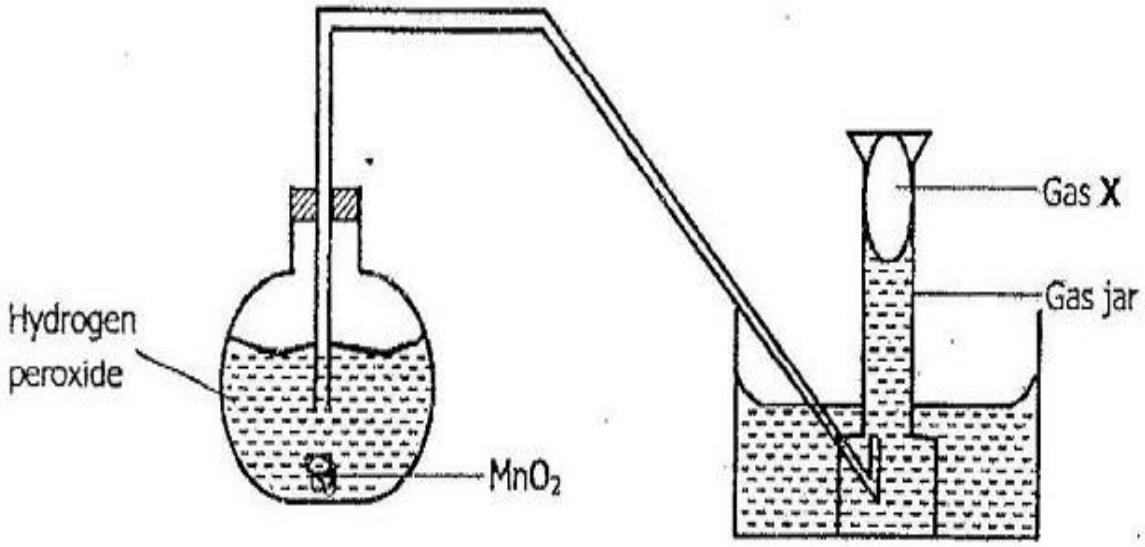
(ii) How is the major product separated from the minor product? [1]

(iii) If 2.1kg of propene is fully reacted with steam. Calculate the mass of propan -2-ol that would be formed

(d) State the general formula for alkenes. [1]

2016 QUESTION PAPER 2

96. The diagram below shows the setup of the apparatus that would be used to prepare gas **X.**



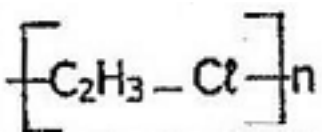
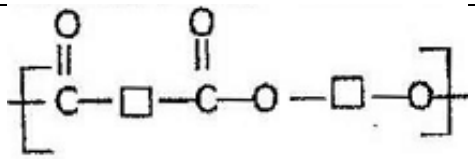
(a) (i) Identify gas X. [1]

(ii) Describe the chemical identity test for gas **X**. [1]

(b) (i) What is the purpose of adding manganese (iv) oxide to hydrogen peroxide? [1]

(ii) Explain why this gas is obtained using the method shown in the diagram. [1]

97. (a) The table below shows some information on two polymers. Complete the table below.

Name of polymer	Formula	Type of polymer
Polyvinylchloride		**A**
B		**C**

[3]

(b) What is the best method of disposing of polyvinyichioride and polymer **B?** [1]

(c) Name the products of protein hydrolysis. [1]

98. Study the following sequence of reactions.

Glucose → ethanol → ethene → polyethene

(a) (i) What name is given-to the process by which glucose changes into ethanol? [1]

(ii) Name the enzyme responsible for the change in (1) above.

(iii) What gas is produced during this process? [2]

(b) Name the process and the catalyst used when ethene changes into polyethene. [2]

(c) Draw the structure of polyethene with three repeating units. [1]

(d) When polyethene is burnt in excess air, two products are formed, Name the products. [2]

(e) Polyethene is a non-biodegradable substance. Explain what is meant by the term non-biodegradable and state the effect of such polymers on the environment. [2]

99. One major ore of capper is upper pyrite, 'CuFeS_2, to extract copper-from-the ore the ore is crushed, then undergoes froth-floatation and finally roasted in the air. The metal is then purified by electrolysis

(a) Explain why:-

(i) the copper ore has to undergo froth-floatation. [1].

(ii) the copper ore is roasted in air. [1]

(b) Write a balanced chemical equation for the reaction that occurs when the copper ore is roasted.

(c) The physical properties of copper can be explained in terms of metallic bonding.

(i) Describe with the aid of a labelled diagram the metallic bonding in copper. [3]

(ii) Explain how metallic bonding-makes copper to be Malleable. [2]

(d) State one use of copper. [1]

100. Nitric acid is an important acid in everyday life.

(a) Name the commercial process used to prepare nitric acids. [1]

(b) State the catalyst used during the commercial preparation of nitric acid. [1]

(c) The equation below shows the initial stage in the industrial preparation of nitric acid.

NH_3 (g) + O_2 (g) → NO (g) + H_2O (g)

Copy and balance the equation.

(d) The nitrogen monoxide produced as shown by the equation in (c) above reacts with oxygen to give nitrogen dioxide. Construct the balanced chemical equation. [2]

(e) How is nitrogen dioxide converted to nitric acid in this process? [1]

(f) Explain why the concentration of nitric acid cannot be made more than 68% even if it is distilled. [1]

(g) Explain why higher concentrations of nitric acid can only be obtained if it is mixed with concentrated sulphuric acid and redistilled. [1]

(h) Give any two uses of nitric acid. [2]

2017 G.C.E QUESTION PAPER 2

101. Most metals are not found as pure elements in the earth's crust, and iron is one such metal. Iron is extracted from its ore in a blast furnace.

(a) Name two other raw materials added to the blast furnace other than haematite. [2]

(b) Write a balanced chemical equation for the reduction of the iron ore to the metal. [2]

(d) State two conditions necessary for rusting to occur. [2]

102. The complete combustion of petrol produces carbon dioxide, water vapour and sulphur dioxide. The exhaust gases from cars contain oxides of nitrogen.

(a) State the source of these oxides. [1]

(b) The sulphur dioxide and oxides of nitrogen from cars cause an environmental problem.

(i) State what this problem is. [1]

(ii) What is the effect of this problem on buildings painted with lime? [1]

(c) Carbon monoxide is produced when there is incomplete combustion of carbon containing fuels like petrol. Name **one** solid product of incomplete combustion of petrol in car engines. [1]

103. Choose a word from the box to match the chemical reactions expressed below.

Combustion, Synthesis, Decomposition, Displacement, Neutralisation, Precipitation

(a) $AgNO_3(aq) + HCl(aq) \rightarrow AgCl(s) + HNO_3(aq)$ [1]

(b) $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$ [1]

(c) $H^+(ag) + OH^- \rightarrow H_2O(i)$ [1]

(d) $C(s) + O_2(g) \rightarrow 3CO_2(g)$ [1]

(e) $NH_3(g) + HCl(aq) \rightarrow NH_4Cl(S)$ [1]

104. The table below shows the properties of elements **W, X, Y** and **Z.**

Elements	**Reaction with**			**Effect of heat on their carbonates**
	Cold water	Oxygen	Dilute hydrochloric acid	
W	X	X	X	√
X	X	√	√	√
Y	√	√	√	X
Z	√	√	√	√

Key:

(√) Chemical change occurs (X) **NO** Chemical change occurs

Use the letters to answer the following questions.

(a) Arrange the elements in the increasing order of their reactivity. [1]

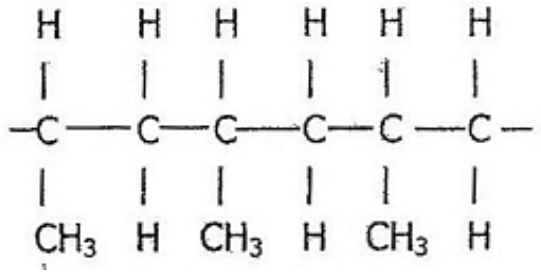
(b) Which element is used in the making of car bodies? [1]

(c) Suggest a method by which element **W** can be extracted from its ore. [1]

(d) Using the letter **Z,** construct an equation to illustrate the effect of heat on its carbonate. [1]

(e) State the element which is suitable for making ornaments. Explain your answer. [2]

105. The structure below is for a polymer.



(a) (i) Name the polymer.

(ii) Name and draw the structural formula of the monomer for the polymer. [3]

(b) (i) State the type of polymer that is shown above? [1]

(ii) Give **one** use of the polymer. [2]

(c) Ethanol is an alcoholic beverage which can be brewed from cassava. Outline the process by which ethanol can be prepared. [3]

(d) Ethanol is used as a fuel. Construct a balanced chemical equation for its complete combustion. [2]

106. The exhaust fumes from an internal combustion engine contain the pollutant gases carbon monoxide and nitrogen dioxide.

(a) Many vehicles have a catalytic converter fitted on their exhaust systems.

(i) Describe the chemical reactions which occur in the catalytic converter to reduce the emissions of carbon monoxide and nitrogen dioxide. [3]

(ii) Write a balanced chemical equation for one of the reactions that occurs in the catalytic converter. [2]

(b) Briefly explain the effect of carbon monoxide on human beings. [3]

(c) Suggest **two** other ways of minimizing pollution of the air by these two gases other than using a catalytic converter. [2]

2017 QUESTION PAPER 2

107. Below are chemical formulae of organic compounds.

C_2H_4 CH_3OH $CH_3\,CH_2CO_2H$ HCO_2H C_2H_6 CH_3COCH_2OH

(a) Name the compound which reacts with steam. [1]

(b) Draw the structural formula of a compound which turns blue litmus paper red. [1]

(c) Choose two compounds which are isomers. [2]

(d) Which two compounds can undergo esterification? [1]

108. On the packet of a particular opaque beer is written, "Contains 5% alcohol per unit volume".

(a) (i) Name the alcohol found in the opaque beer. [1]

(ii) Describe, in outline, how this alcohol you have named in (a) above is commercially produced from starch. [5]

(iii) Write down a balanced chemical equation for the formation of the alcohol from glucose with state symbols.

(b) Give **one** use and **one** bad effect of the alcohol found in beer. [2]

2019 G.C.E QUESTION PAPER 2

109. Generally some metals easily corrode when exposed to certain conditions.

(a) Explain what is meant by corrosion. [1]

(b) Which of the metals zinc, potassium or copper corrodes fastest when exposed to certain conditions? [1]

(c) Give **two** conditions that easily promote corrosion. [2]

(d) Give **two** methods of preventing corrosion of metals. [2]

110. Carbon is asymmetrical and exists in many forms of allotropes.

(a) What are allotropes? [1]

(b) Give **two** crystalline allotropes of carbon. [2]

(c) Give **one** property and the use, based on the property, of one of the allotropes you have named in (b) above. [2]

111. The equations below show the thermal decomposition of the nitrates of three metals represented by **W**, **X** and **Y** (not their real chemical symbols).

$$2\mathbf{W}NO_3 \rightarrow 2\mathbf{W} + 2NO_2 + O_2$$

$$2\mathbf{X}NO_3 \rightarrow 2\mathbf{X}NO_2 + O_2$$

$$2\mathbf{Y}(NO_3)_2 \rightarrow 2\mathbf{Y}O + 4NO_2 + O_2$$

(a) Which one of the three metals is the;

(i) Most reactive? Give a reason for your answer. [2]

(ii) Least reactive? Give a reason for your answer. [2]

(b) Which one of the three metals is likely to be sodium? [1]

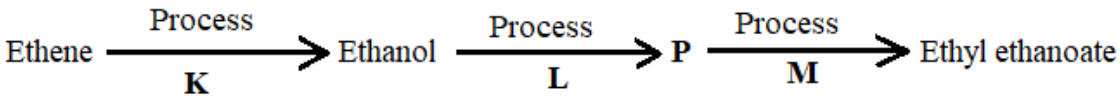
(c) Which one of the three metals is

(i) more reactive than zinc? Give a reason for your answer. [2]

(ii) less reactive than zinc? Give a reason for your answer. [2]

(d) State the chemical test for oxygen. [1]

112. The series of processes below show the links to the formation of an ester known as ethyl ethanoate.



(a) From the reaction scheme above, name processes **K, L** and **M.** [3]

(b) Construct the chemical equation describing process **K.** Include state symbols. [2]

(c) State both the chemical name and common name for substance **P.** [2]

(d) State the general formula of the series where substance **P** belongs. [1]

(e) Give any **two** uses of ethyl ethanoate. [2]

ANSWERS FOR GRADE 10 E.C.Z QUESTIONS

2009	***2010***	***2011***	***2012***	***2013***	***2015***	***2016***	***2017***	***2017***	***2019***
1. B	*8. C*	*14. B*	*19. C*	*25. B*	*31. C*	*36. C*	*41. C*	***G.C.E***	***G.C.E***
2. C	*9. A*	*15. C*	*20. A*	*26. C*	*32. A*	*37. B*	*42. A*	*48. B*	*54. B*
3. A	*10. D*	*16. C*	*21. D*	*27. C*	*33. D*	*38. D*	*43. C*	*49. C*	*55. C*
4. C	*11. B*	*17. D*	*22. C*	*28. A*	*34. C*	*39. A*	*44. B*	*50. C*	*56. C*
5. C	*12. D*	*18. A*	*23. C*	*29. A*	*35. C*	*40. B*	*45. B*	*51. A*	*57. C*
6. A	*13. B*		*24. D*	*30. D*			*46. C*	*52. D*	*58. A*
7. C							*47. B*	*53. D*	

ANSWERS FOR 2009 QUESTION PAPER 3

59. (a) **d** = freezing / solidification, **e** = sublimation, **f** = condensation

(b) By putting substance A into a freezer compartment of a refrigerator.

(c) Ions, atoms or molecules

ANSWERS FOR 2010 QUESTION PAPER 3

60. (a) Water molecules from wet clothes gains kinetic energy and evaporate faster.

(b) Intermolecular forces become weak and particles start to slide over each other.

(c) High collisions of particles in hot water due to their high kinetic energy than in cold water.

(d) Sugar molecules move from high concentration to lower concentration hence fill the whole region.

61. (a) (i) Crystallisation (ii) Distillation (iii) Filtration

(b) (i) Esterification (ii) Combustion

62. (a) (i)

Name of particles	Number of particles
Protons	3
Neutrons	4

(ii) **Mass**: both have same mass of 1 a.m.u

Electrical charge: Proton is positively (+) charged while neutron has **no** charge.

(b) Difference in number of neutrons which is 1.

ANSWERS FOR 2011 QUESTION PAPER 3

63. (a) (i) Tungsten / aluminium (ii) Because they conduct electricity

(b) It has variable melting point (35 ^{o}C – 50 ^{o}C)

(c) (i) Sodium chloride

(ii) Substance made of two or more atoms that are chemically combined.

64. **(a)** Filtration **(b)** Use of separating funnel

(c) Chromatography **(d)** Magnetism

(e) Filtration

65. (a) Nitrogen (b) Ammonia

(c) NH3 (d) Covalent bond

(e) The gas is very volatile.

(f) Carbon dioxide (CO_2)

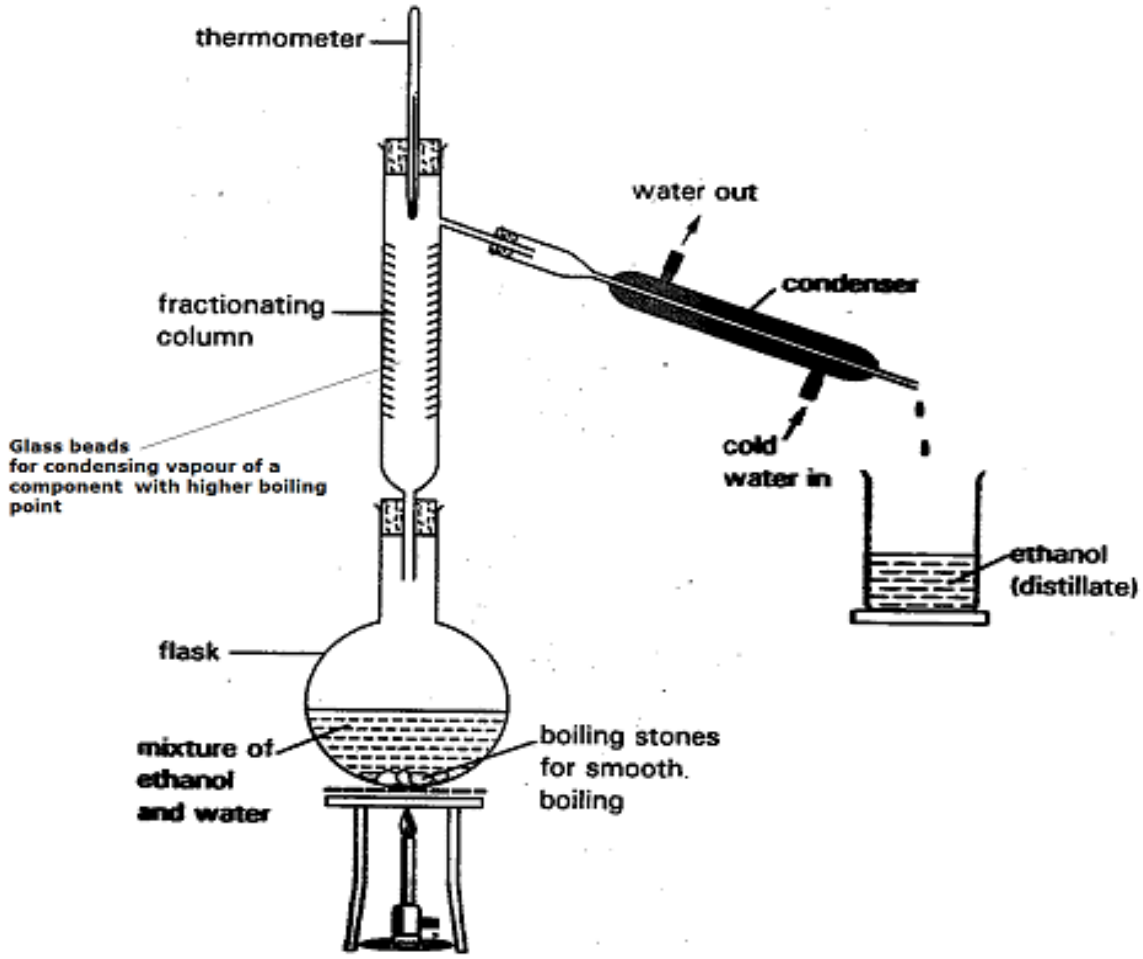
66. (a) Solid – it has a fixed shape. - It has a fixed volume.

(b) Liquid – has a fixed volume. - It takes the shape of a container

(c) Solid – no fixed volume. - No fixed shape

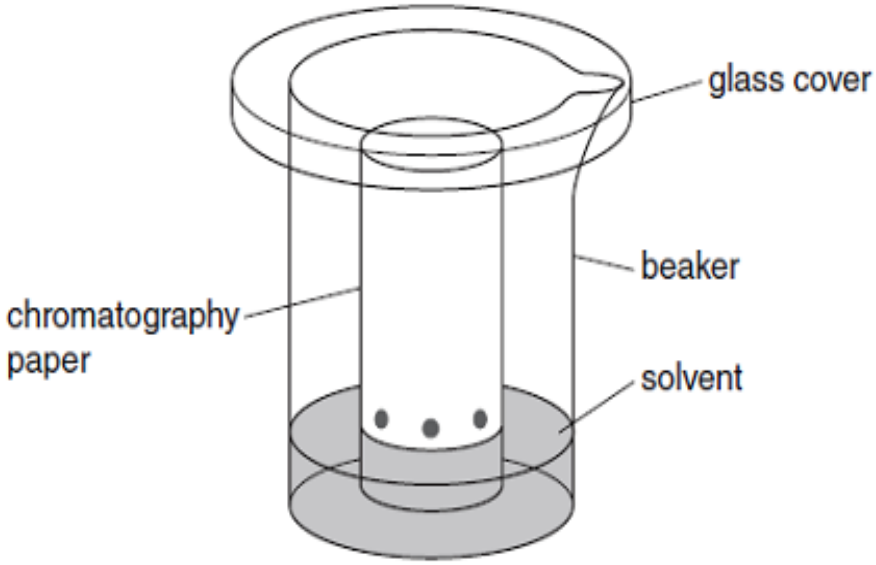
67. (a) Fractional distillation

(b)



(c) Heat sodium sulphate solution in the evaporating dish until it has saturated. Then cool the saturated solution of sodium sulphate to form crystals and leave it to dry on a filter paper.

68. (a)



(b) ink D (c) Ink A

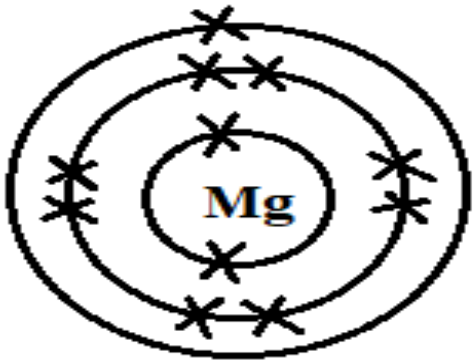
69. (a) (i) Z = 11 Protons (ii) Neutron = 12

(b) (i) Group one

(ii) Because it has one electron in its outermost shell.

70. (a) (i) Mr of magnesium is 24g/mol. (ii) It has 20 protons in its nucleus and 20 electrons in the shells.

(b)



(c) both have 2 electrons in their outermost shell

(d) (i) No charge (Neutral) (ii) Positively charge (+) (e) $2A + Cl_2 \longrightarrow 2AC$

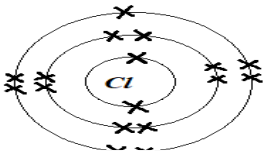
(f) Ionic bond. It has high melting and boiling point, it conduct electricity when it is molten state.

ANSWERS 2013 QUESTION PAPER 3

71. (a) **Isotopes** are atoms of the same element having same numbers of proton but different numbers of neutron.

(b) (i) **Chlorine – 35** = 18 neutrons, and **Chlorine – 37** = 20 neutrons.

(c) (i)



(c) (ii) Cl^-

(d) Because the average atomic mass of the two isotopes of chlorine – 35 and chlorine – 37 with their percentage abundance 75% and 25% respectively is 35.5.

ANSWERS FOR 2014 QUESTION PAPER 3

72. (a) (i) C (ii) B (iii) D (iv) A

(b) Separating amino acid from proteins. Identify flavor components in food stuffs.

73. (a) An **ion** is a charged particle. (b) **Sodium ion** = Na^+

(c) $Ca \longrightarrow Ca^{2+} + \underline{2e^-}$ (d) -Sulphate, - Bicarbonate (e) covalent bond

ANSWERS FOR 2015 QUESTION PAPER 3

74. (a) Evaporation (b) Evaporation dish (c) Iodine

(d) *NaCl* can be separated from water by physical reaction

75. (a) Nucleon number is the total number of protons and neutrons in an atom.

(b)

Number of particle	**Total number in one atom**
Neutrons	12
Protons	11
Electrons	11

(c) Na^+ (d) Na^+_{10}: 2, 8

76. (a) (i) **Filtrate** is a solution or liquid that has passed through a filter paper during filtration.

(ii) **Residue** is substance that remained on a filter paper during the process of filtration.

(b) – by decantation; solid and liquid are let to settle and pour out liquid and solid remain.

- By filtration; liquid pass through a filter paper and solid remain as residue.

(c) (i) **Fractional distillation** has a fractionating column while **Simple distillation** doesn't have.

(ii) - Separation of crude oil. – Separation of fermented liquor into ethanol and water.

77. (a) ions, atoms or molecules

(b) – Spray perfume at one corner of the room but its smell fill the whole room.

- smelling the aroma (delicious food) from the distance.

(c) Solid; has fixed volume because particles are very closely packed and are held together by very strong attraction force between particles.

78. (a) Condensation is the change of state of matter from solid to liquid. **Melting**

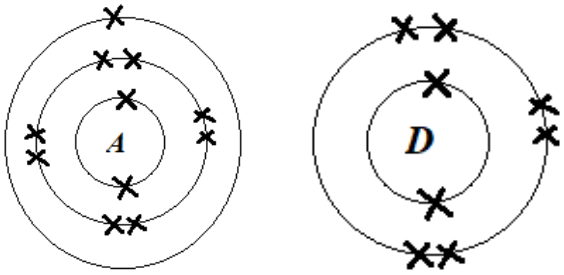
(b) The basic units of matter that exist in sodium chloride are molecules. **Ions**

(c) Isotopes are compounds with the same molecular formula but different structural formulae. **Isomers**

(d) Calcium hydroxide is classified as an amphoteric substance because it dissolves in sodium hydroxide solution. **Aluminium oxide**

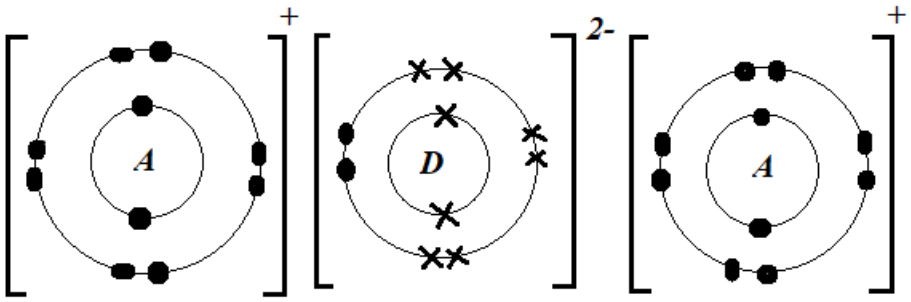
(e) Zinc and chlorine bond covalently. **Ionic bond**

79. (a)



(b) (i) ionic bond

(ii)



(c) – it conduct electricity in molten or aqueous state.

- it has high melting and boiling point

ANSWERS FOR 2016 QUESTION PAPER 2

80. (a) A – Boiling / evaporation B – Freezing / Solidification.

(b) Particles move randomly at high speed.

(c) **A**; because during this process (boiling) particles absorb heat energy which overcome the force at attraction between particles. Then the particles start moving randomly and very far apart.

81. (a) Bunsen burner (b) Volumetric flask (c) Liebig condenser (d) Evaporation dish (e) Pipette

82. (a) Aluminium / Nitrogen (b) Bronze / sea water (c) Nitrogen (d) Aluminium

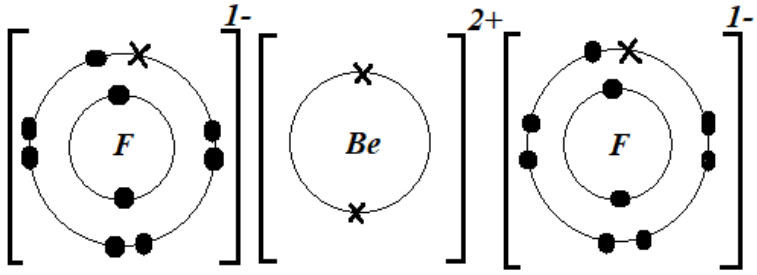
(e) (i) **Ionic compound** - potassium chloride (ii) **Alloy** - Bronze

83. (a) A – Liquid (b) Freezing point (c) Liquid and solid

(d) Particles lose energy and move close together as the lost energy is used to change state of a substance

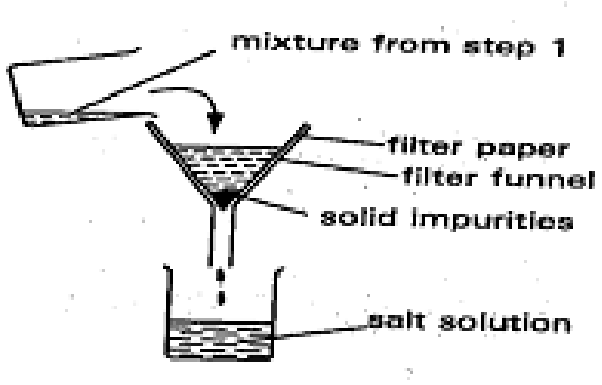
(e) The substance loses its heat content and its temperature falls, then it changes from liquid to solid.

84. (a) Ionic bond (b)



(c) – they are solid at room temperature. – they are non volatile. – they are insoluble in organic liquid

85. (a) Filtration (b)



(c) Removal of solid impurities from water in treatment works. – Separation of penicillin from yeast.

ANSWERS FOR 2017 G.C.E QUESTION PAPER 2

86. (a) Condensation (b) Melting (c) Sublimation (d) Dissolving (e) Chemical change

87. (a) (i) **Endothermic reaction** – this is a reaction in which energy is absorbed from the surrounding.

(ii**) Exothermic reaction** – this is reaction in which energy is released to the surrounding.

(b) **Endothermic reaction** –e.g. Photosynthesis, melting of ice, absorption of light in photography.

Exothermic reaction – Respiration, combustion (burning of papers), freezing of liquid.

(c) Combustion produces carbon monoxide which causes air pollution.

88. (a) **Agriculture** – farmers use fertilizers made from chemical elements such as nitrogen, phosphorous to improve soil fertility.

- **Mining industry** – chemistry is applied in the extraction of metals such as copper from copper ores.

(b) - You should always wear shoes in the laboratory. – do not use broken glassware

ANSWERS FOR 2019 G.C.E QUESTION PAPER 2

89. (a) (i) Helium (ii) Carbon monoxide and nitrogen (b) Hydrogen gas

(C) The relative molecular mass is twice larger than the vapour density.

(b) Under diffusion medium – ammonia must be in liquid state and carbon dioxide in gaseous state

90. (a) Simple distillation

(b) (i) A dish (bucket)

(ii) A – Round bottomed flask, B – Liebig condenser (c) Ethanol

91. (a) $2Al_{(s)} + 3\,CuSO_{4\,(aq)} \longrightarrow Al_2(SO_4)_{3\,(aq)} + 3Cu_{(s)}$

(b) $2Al_{(s)} + 3\,CuSO_{4\,(aq)} \longrightarrow Al_2(SO_4)_{3\,(aq)} + 3Cu_{(s)}$

$Al_{(s)} + 3\,Cu^{2+} + \cancel{3SO^{2-}}_{4\,(aq)} \longrightarrow 2Al^{3+} + \cancel{3SO^{2-}}_{4(aq)} + Cu_{(s)}$

$Al_{(s)} + 3\,Cu^{2+}_{(aq)} \longrightarrow 2Al^{3+}_{(aq)} + Cu_{(s)}$

ANSWERS FOR GRADE 11 E.C.Z QUESTIONS

2009 P1	9. D	***2012 P1***	***2015 P1***	37. A	47. B	57. A
1. A	10. C	19. C	28. A	38. B	48. D	***2019 G.C.E P2***
2. D	11. D	20. *	29. A	39. D	49. A	58. A
3. D	12. D	21. A	30. C	40. D	***2017 G.C.E P1***	59. D
4. C	13. D	22. C	31. B	41. B	50. D	60. B
2010 P1	***2011 P1***	23.A	32. A	42. A	51. B	61. B
5. C	14. D	***2013 P1***	33. A	***2017 P2***	52. D	62. D
6. A	15. B	24. D	***2016 P1***	43. D	53. D	63. B
7. D	16. A	25. D	34. D	44. D	54. D	64. A
8. B/C/D	17. C	26. B	35. C	45. C	55. C	65. *
	18. A	27. C	36. A	46. A	56. D	66. C

2009 PAPER 3

67. (a) Uranium is less reactive, because it has be displaced by magnesium.

(b) (i) Mr of UF4 = (1x238) + (4 x 19) = ***314g/mol***

(ii) Mass of U = $\frac{24\ x\ 238}{48}$ = ***119 tonnes***

(c) **Isotopes** are atoms of the same element,but different numbers of neutrons.

68. (a) (i) Mr = (1x40)+(2x1)+(2x12)+(3x16) = ***114g/mol***

(ii) Mr = (2x27)+(2x32)+(12x16) = ***310g/mol***

(b)(i) Calcium carbonate ($CaCO_3$) (ii) m = $\frac{16.2\ x\ 100}{162}$ = ***10g***

69. (a) Metals; **C** and **D** /Non-metals; **A**, **B**, and **E** (b) **B** (c) **D** (d) **E**

70. *Take 100cm^3 of water in the beaker and dissolve a small amount of sodium sulphate. Take 100cm^3 of water in another beaker and dissolve barium chloride. Mix the two solutions. Sodium sulphate reacts with barium chloride forming a white precipitate barium sulphate.*

2010 PAPER 3

71.(a) (i) Either **He**, **Ne** or **Ar** (ii) **Na** (iii) **F** (iv) **O** (b) (i) Na_2O (ii) Ionic bonding

72.(a) it wear out the cuticle on the leaves hence a plant loss excess water by evaporation.

(b) (i) $H_2SO_4(aq) + Ca(OH)_2(s) \longrightarrow CaSO_4(s) + \mathbf{2}\ H_2O(l)$

(ii) Mass = $\frac{74\ x\ 9.8}{98}$ = ***7.4 tonnes*** (iii) To neutralize the acidic soil.

(c) It ionize completely.

73.(a) Salt is a compound having a metal and a non-metal from acid.

(b) (i) $Fe(s) + H_2SO_4(aq) \longrightarrow FeSO_4(aq) + H_2(g)$

(ii) Iron metal because it has to make a saturated solution of iron (ii) sulphate.

(c) Filter the saturated solution of iron (ii) sulphate to remove undissolved iron metals. Heat the filtered solution to evaporate excess water. Then put it on evaporating dish to evaporate water of Crystallisation. (d) Barium sulphate

2011 QUESTION PAPER 3

74. (a) (i) *Cl* (ii) N (iii) Na (iv) O (b) (i) Solid (ii) Oxygen

75. (a)(i) $2NH_3(g) + CO_2(g) \longrightarrow (NH_2)_2CO(aq) + H_2O(l)$

(ii) 4 atoms of hydrogen (b) Mass = $\frac{34\ x\ 60}{34}$ = ***60 tonnes***

76. (a) Caesium, Potassium, Sodium, Lithium (b)(i) Caesium hydroxide and hydrogen gas

(ii) $Cs(s) + H_2O(l) \longrightarrow CsOH(aq) + H_2(g)$

(iii) – colourless solution

- Blue flame
- splashing of water

(c) Alkaline metals

77. (a)(i) Barium chloride

(ii) $Na_2SO_4(aq) + BaCl_2(aq) \longrightarrow BaSO_4(s) + 2NaCl(aq)$

(iii) $SO_4^{2-}(aq) + Ba^{2+}(aq) \longrightarrow BaSO_4(s)$

(b) Mix sodium sulphate with barium chloride and wait until solid precipitates formed. Then filter the solution to remove precipitates. Then put it on evaporating dish to evaporate water of Crystallisation.

(c) Copper (ii) sulphate (d) Mass = $\frac{150\ x\ 111}{100}$ = ***166 g***

78 (a)(i) NaOH (ii) 13 (iii) 1 (iv) H^+, Cl^- (v) Na^+, Cl^- (b) NaOH and, HCl

79.(a) This is the mass of a compound.

(b)(i) K_2CO_3 (s) + H_2SO_4 (aq) ⟶ K_2SO_4 (aq) + CO_2 (g) + H_2O (l)

(ii) m = $\frac{2\ X\ 174}{138}$ = ***2.52g*** (c) it turn lime water milky

80. (a) Salt is a compound consists of metal and a non-metal from an acid.

(b)(i) copper (ii) oxide

(ii) CuO (s) + H_2SO_4 (aq) ⟶ $CuSO_4$ (aq) + H_2O (i)

(c) Heat sulphuric acid in a beaker. Add copper (ii) oxide while stirring until the solution is saturated. Filter the saturated solution to remove undissolved copper (ii) oxide. Put the filtrate on evaporating dish to evaporate water of crystallization.

(d) No, because potassium is very reactive which can cause fire.

2014 QUESTION PAPER 3

81.(a)(i) Sodium sulphate and Barium chloride. (ii) Potassium sulphate

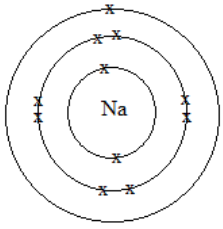
(iii) Copper carbonate and nitric acid (b) SO_4^{2-} (aq) + Ba^{2+} (aq) → $BaSO_4$(s)

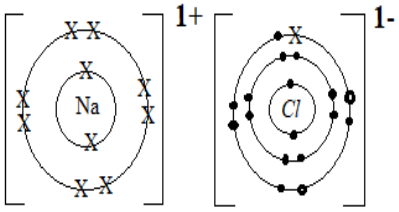
82.(a) $MgCO_3$ (s) → MgO (s) + CO_2(g) (b) Mr = 24 + 12 + 48 = ***84g/mol***

(c) m = $\frac{21\ x\ 40}{84}$ = **10 Kg** (d) Sodium carbonate

83. (a)(i) 3 (ii) 9 (iii) 18 (iv) 16 (b)(i) Al_2O_3 (ii) solid state

84.(a) (b)(i)





(ii) Because the attraction force holding the molecules between NaCl – NaCl is strong and it require huge amount of energy to break those forces.

(c) Lithium nitrate decomposes producing lithium oxide, nitrogen dioxide and oxygen gas while caesium nitrate decomposes forming caesium nitrate and oxygen gas.

(d) Because the atomic mass increases down the group.

(e) They are stored in oil, to prevent them react with oxygen in the air.

2015 G.C.E QUESTION PAPER3

85.(a) Blue (b)cause water turned into alkaline solution

(c) (i) Hydrogen (ii) it put off a burning splint with pop sound

(d) 2K (s) + $2H_2O$ (l) → 2KOH (aq) + H_2 (g)

86. (a)(i) because it will be added to sulphuric acid until the solution saturated.

(ii) Through filtration. (b) m = (96 x 2) ÷ 24 = ***8g***

87. (a) Potassium iodine is colourless when bromine gas bubbled in it, it will turned to shade brown because of potassium bromide which is formed.

(b) $2KI\ (aq) + Br_2\ (g) \rightarrow 2KBr\ (aq) + I_2\ (g)$ (c) Displacement reaction

(d) Iodine is less reactive than bromine. (e) 7 electrons in their outermost shell

88.(a) copper (ii) oxide

(b) Heat sulphuric acid in a beaker. Add copper (ii) oxide while stirring until the solution is saturated. Filter the saturated solution to remove undissolved copper (ii) oxide. Put the filtrate on evaporating dish to evaporate water of crystallization.

(c) $CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4\ (aq) + H_2O\ (i)$ (d) m = (19.6 x 287) ÷ 98 = ***57.4 g***

(e) *Water of Crystallisation* is water molecules present in one molecule of salt.

89. (a) Calcium hydroxide + Carbon dioxide → Calcium carbonate + water.

(b)(i) n = V ÷ Vm = 1.2 ÷ 24 = 0.05mol

m= n x mr = 0.05 x 44 = 2.2g

m = (74 x 12) ÷ 44 = ***3.7g***

(ii) m = (2.2 x 100) ÷ 44 = ***5g*** (iii) white precipitates will be seen

90. (a) Sodium (Na) (b) Potassium (K) (c) Copper (Cu) (d) Argon (Ar)

91.(a) $NaOH + HCl \rightarrow NaCl + H_2O$ (b) Neutralization reaction

(c) The solution is heated to evaporate the water to saturation point. The solution is cooled and crystals are formed. The pure crystals of sodium chloride are filtered, washed and dried between two filter papers.

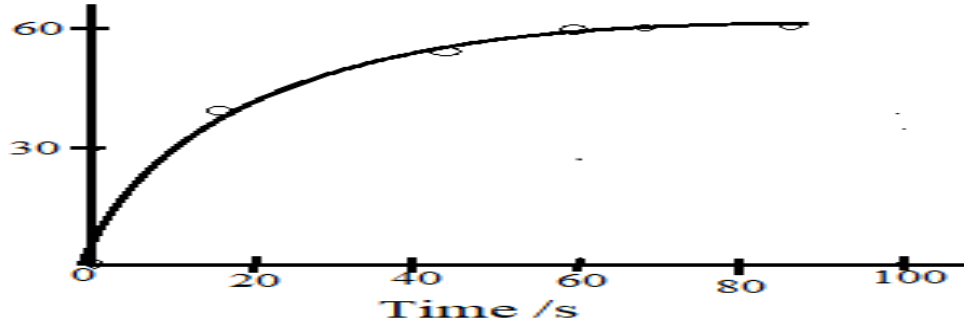
2016 QUESTION PAPER 2

92.(a) It is a reactant that determine how much products will be formed.

(b)(i) $Mg\ (s) + 2HCl\ (aq) \rightarrow MgCl_2\ (aq) + H_2O$ (ii) Magnesium metal

(iii) m = (2.4 x 74) ÷ 24 – 2.4 = ***4.9 g***

93. (a)



(b)(i) *At 30 seconds* - Rate of reaction = 45 ÷ 30 = ***1.5 cm^3/s***

At 50 seconds - Rate of reaction = 58÷50 = ***1.16 cm^3/s***

(ii) The rate of reaction was reaction is fastest for the first 20 seconds and started to reduce second 20 seconds and the reaction stopped for the last 20 second

94. (a)(i) *Anhydrous salt* is that which do not have water of crystallization

(ii) *Efflorescent substance* is that which releases its water of crystallization to the atmosphere.

(b)(i) Sodium Carbonate – 10-water ($Na_2CO_3.10H_2O$) (ii) Calcium chloride ($CaCl_2$)

95.(a)(i) Group one (1) (ii) $2Bo(s) + 2H_2O(i) \rightarrow 2BoOH(aq) + H_2(g)$

(b) No, because it forms stable carbonate

2017 G.C.E QUESTION PAPER 2

96.(a) $H_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2H_2O$

(b) Mr = 40 + 32 + 2 = 74g/mol (c) m = (15.5 x 74) ÷ 98 = 11.705 tonnes

(d) Hydroxide ion (OH^-)

97. (a) Iron (b) Lithium (c) Sulphur (d) Mercury (e) Iron

98 (a) - Calcium carbonate; - Hydrochloric acid

(b) The calcium carbonate is continuously added to dilute hydrochloric acid whilst stirring until the carbonate is in excess. The excess carbonate is then filtered off. The salt solution is then heated (evaporated) until a saturated solution is obtained. The saturated solution is cooled to grow salt crystals. The crystals are then filtered, washed and dried between two pieces of filter paper

(c)(i) Precipitation method (ii) $Pb^{2+}(aq) + 2I^-(aq) \rightarrow PbI_2(s)$

2017 QUESTION PAPER 2

99.(a)(i) By measuring the temperature of water using a thermometer.

(ii) Endothermic reaction (iii) because it has released heat energy to the surrounding

(b)(i) Caesium reacts very vigorous with water than potassium.

(ii) Because caesium is more reactive than potassium.

100. (a) It changes from gas, liquid to solid states.

(b) It form an acid when it reacts with water which kills the bacteria, viruses and germs.

(c) $2Br^-(aq) + Cl_2(g) \rightarrow 2Cl^-(aq) + Br_2$

101. (a) the pH value changes from 1 to 7 (b)(i) Neutralization reaction

(ii) $2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

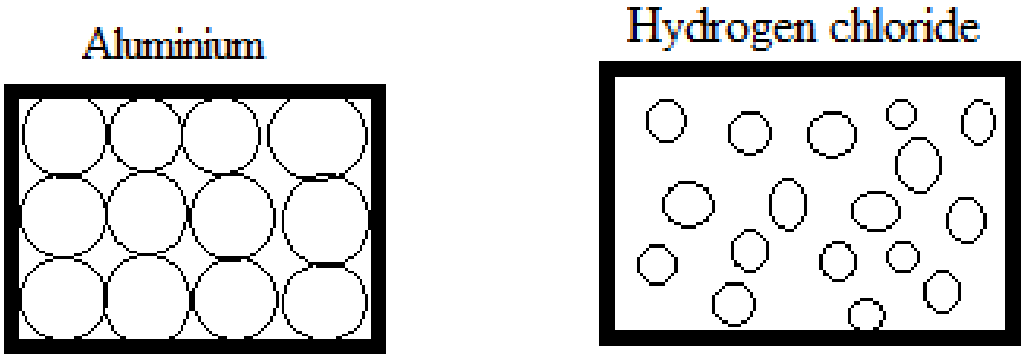
(c) It ionizes completely

102 (a) *Concentration* is the amount of solutes in $1dm^3$ of a solution.

(b) C = n ÷ V = 1.5 ÷ 0.3 = ***5 mol/dm³***

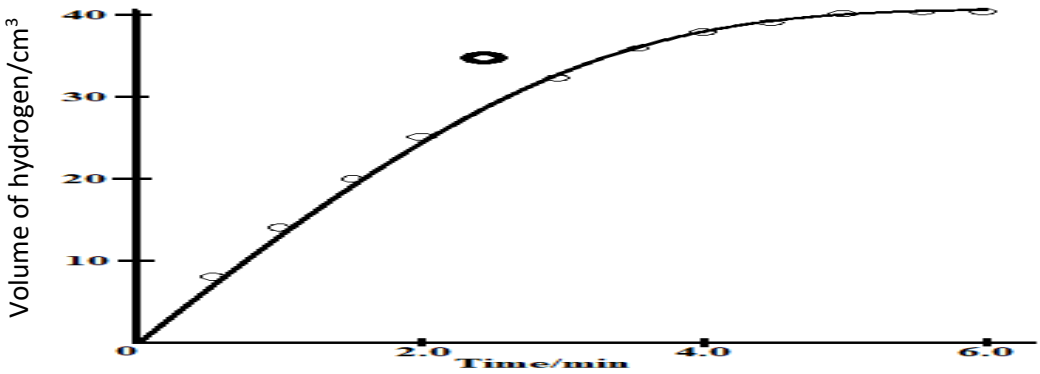
(c) Rate of reaction will increase / to speed up rate of chemical reaction.

103. (a)(i)



(b) Hydrogen chloride because the intermolecular forces holding HCl – HCl molecules is weak compared to the ones in holding Al – Al atoms.

104. (a) $Mg\ (s) + 2\ HCl\ (aq) \rightarrow MgCl_2\ (aq) + H_2\ (g)$



(c) (2.5, 35) error (d) 40 cm3 (e) When the graph flatten (f) Rate = 40 ÷ 6 = ***6.67 cm³/min***

105.(a)(i) $Fe\ (s) + H_2SO_4(aq) \rightarrow FeSO_4\ (aq) + H_2\ (g)$

(ii) To speed up the chemical reaction

(iii) By adding iron metal to sulphuric acid until the solution is saturated.

(iv)) Filter the saturated solution of iron (ii) sulphate to remove undissolved iron metals. Heat the filtered solution to evaporate excess water. Then put it on evaporating dish to evaporate water of Crystallisation.

(b)(i) $Fe\ (s) + Cu^{2+}\ (aq) \rightarrow Fe^{2+}\ (aq) + Cu\ (s)$

(ii) – brown coating copper on the surface of iron nails.

- solution changed from blue ($CuSO_4$) to light green ($FeSO_4$).

2019 QUESTION PAPER 2

106.(a) $2Al\ (s) + 3CuSO_4\ (aq) \rightarrow Al_2(SO_4)_3\ (aq) + 3Cu\ (s)$

(b) $2Al\ (s) + 3CuSO_4\ (aq) \rightarrow Al2(SO_4)_3\ (aq) + 3Cu\ (s)$

$2Al\ (s) + 3Cu^{2+}\ (aq) + \cancel{3SO_4^{2-}}\ (aq) \rightarrow 2Al^{3+}(aq) + \cancel{3SO_4^{2-}}\ (aq) + 3Cu\ (s)$

$2Al\ (s) + 3Cu^{2+}\ (aq) \rightarrow 2Al^{3+}(aq) + 3Cu\ (s)$

107 (a)(i)sodium hydroxide/calcium chloride

(ii) Calcium oxide (iii) Aluminium hydroxide

(b) *Deliquescent substance* are substance that absorbs enough water vapour from the atmosphere to form a solution

*Deliquescent substance*s are substances that absorb enough water vapour from the atmosphere to form a solution while hygroscopic substances are substances that absorb less water vapour from the atmosphere and fail to form a solution.

(c) Efflorescent substances are substances that lose water of crystallization when they are exposed to the atmosphere.

108. (a) $Mg\ (s) + 2\ HCl\ (aq) \rightarrow MgCl_2\ (aq) + H_2\ (g)$

(b)(i) $n = V \div Vm = 0.24 \div 24 =$ ***0.01 mol***

(ii) Mass of hydrogen gas

$m = n \times Mr$

$= 0.01 \times 2$

$m = 0.02g$

mass of magnesium used

$Mg \rightarrow H2$

$24 = 2$

$x = 0.02$

$x = (24 \times 0.02) \div 2 =$ ***0.24g***

109.(a)(i) **B.** (ii) It is in powdered form i.e the size particles of reactants are very small.

(b) Temperature/concentration/pressure

110. (a) Transition metals/block D elements

(b)- They have high densities; They have high melting and boiling points; They are good conductors of heat and electricity; They are solids at room temperature and pressure except mercury which is a liquid; They are ductile i.e. they can be drawn into wires; They are malleable i.e. they can be hammered into thin sheets.

(c) Tungsten is used to make filament in electric bulb

- Iron is used as a catalyst in the manufacturing of ammonia.

111.(a)(i) $2NaOH\ (aq) + H_2SO_4\ (aq) \rightarrow Na_2SO_4\ (aq) + 2H_2O\ (i)$

(ii) $2NaOH\ (aq) + H_2SO_4\ (aq) \rightarrow Na_2SO_4\ (aq) + 2H_2O\ (i)$

$\cancel{2Na}^+\ (aq) + 2OH^-\ (aq) + 2H^+\ (aq) + \cancel{SO}_4^{2-}\ (aq) \rightarrow \cancel{2Na}^+\ (aq) + \cancel{SO}_4^{2-}\ (aq) + 2H_2O\ (i)$

$2OH^-(aq) + 2H^+(aq) \rightarrow 2H_2O(l)$

(b) n = CV = 2.0 x 0.04 = ***0.08 mol***

(c) n = 2 x 0.08 = 0.16 mol

m = n x mr = 0.16 x 40 = ***6.4g***

(d) Percentage purity = (mass of pure ÷ mass of impure) x 100% = (6.4 ÷ 8.0) x 100

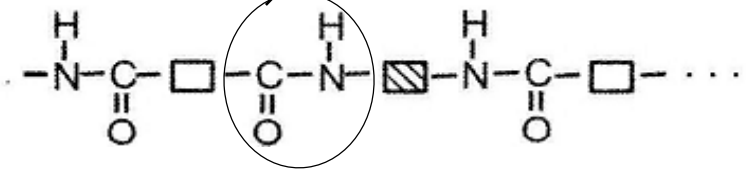
Percentage purity = ***80%***

ANSWERS FOR GRADE 12 E.C.Z QUESTIONS

2009 p1	9. D	18. B	28. C	***2016 p2***	46. B	55. A
1. B	10. B	19. A	29. B	38. D	47. B	***2019 G.C.E***
2. C	***2011 p1***	20. D	30. B	39. C	48. B	56. A
3. D	11. A	21. A	31. B	40. C	49. B	57. C
4. C	12. C	22. B	32. D	41. A	50. A	58. B
5. C	13. B	23. B	33. D	42. D	***2017 p2***	59. C
2010 p1	14. A	24. C	34. B	43. C	51. A	60. B
6. C	15. A	25. D	35. B	44. A	52. A	61. A
7. C	16. B	26. B	36. B	***2017 G.C.E***	53. C	62. B
8. D	17. C	27. B	37. D	45. D	54. D	

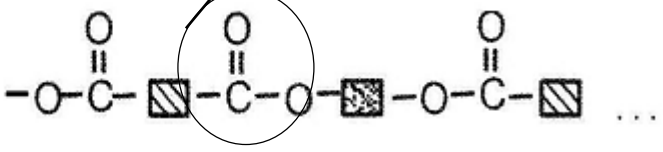
63.(a)(i) amide linkage

nylon



(ii) Ester linkage

Terylene



(b) fats, protein, etc

(c) because they are form after two different monomers join together using their functional groups

(d) **Advantage** - they don't rust , decay in water.

Disadvantage – *they are non-biodegradable making the disposal difficult resulting in pollution.*

64.(a) Fractional distillation

(b)catalyst (iron), pressure: 350 atm & temp. 350°C – 450°C

(c) it is used to make fertilizer like $(NH_4)_2SO_4$, NH_4NO_3,

65.(a)(i) Limestone (ii) Quicklime (iii) Slaked lime (b) $Ca(OH)_2$

66.(a)

$$\text{Glucose} \xrightarrow[\text{Zymase}]{\text{Yeast}} \text{Ethanol} + \text{Carbon dioxide}$$

(b) Fermentation (c) temperature: ***37°C***

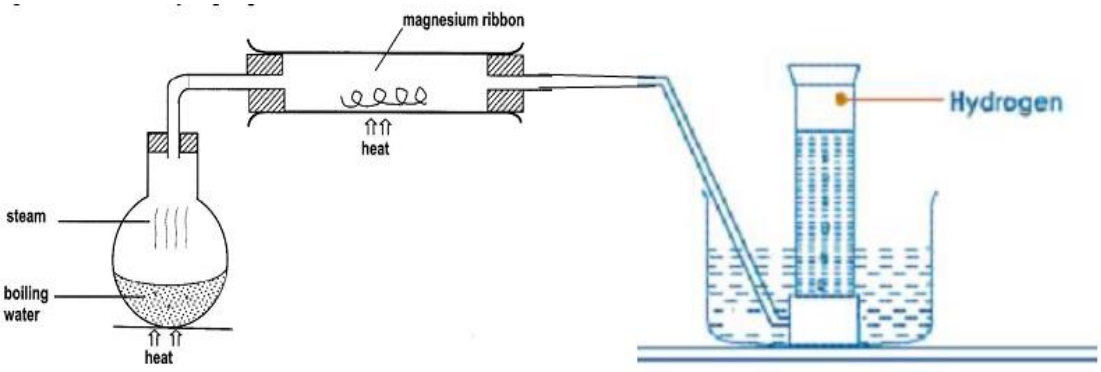
(d)(i) If air (oxygen) enters it will reacts with glucose forming carbon dioxide and water hence reducing the production of alcohol.

(ii) Caron dioxide will reacts with lime water forming white precipitate (calcium oxide) hence turning colourless lime water milky.

(e) - Oxygen is added to ethanol; - Hydrogen is removed from ethanol (f)(i) Dehydration

(ii) add it orange bromine and it will turn colourless.

67(a)



(b) Mg (s) + H_2O (g) → MgO (s) + H_2 (g)

(c) React with steam: *iron, zinc,* etc. Can't react with steam: *copper.*

(d) Electrolysis

(e)(i) Mg (s) + $CuSO_4$ (aq) → $MgSO_4$ (aq) + Cu (s) (ii) Displacement reaction

68.(a)(i) through incomplete combustion (burning)

(ii) It reacts with the haemoglobin in blood, and prevents it from carrying oxygen around the body – so you die from oxygen starvation.

(iii) *sulphur dioxide (SO_2)*

(b)(i) Sedimentation, Filtration or Chlorination

(ii) - polar solvent e.g dissolving ionic compound; - essential ingredient for products e.g beer

(c)(i) steam (H_2O) **(ii)** Carbon ($C_{(s)}$)

2010 QUESTION PAPER 3

69.(a)(i)

Metal	Reacts with cold water	Reacts with steam
Copper	X	X
Iron	X	√
Calcium	√	√

(ii) Calcium, iron, copper

(b) Because when aluminium exposes to air it reacts with oxygen in air forming a layer called aluminium oxide which protect aluminium further to react with oxygen in air.

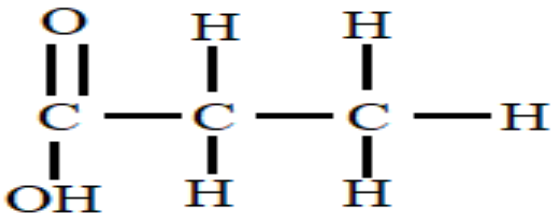
(c) – it a way of conserving minerals so that they last long.

- Less energy is used than full process of extraction.

70.(a)(i) Ammonia – oxidised (ii) carbon dioxide – reduced (iii) aluminium – oxidised

(b) Steam [$H_2O_{(g)}$], because it has caused carbon [$C_{(s)}$] to be oxidised and itself has being reduced.

71.(a)(i)



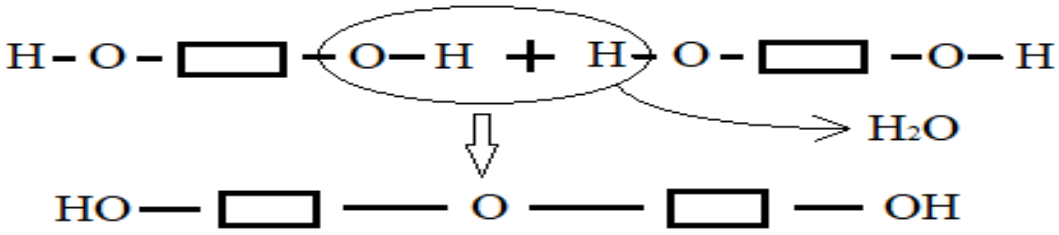
(ii) a catalyst called sulphuric acid

(b)(i) by using a pH scale, and if it gives a value less than 7 means it is an acid but greater than 7 means it is base.

(ii) Sodium – effervescence will be observed.

72.(a)(i) monomer

(ii)

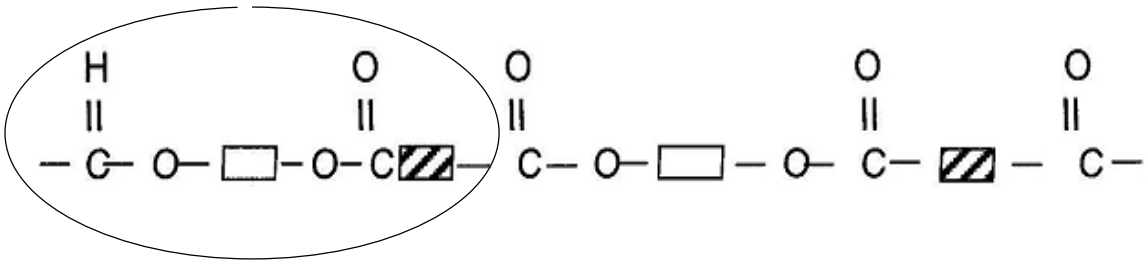


72.(b)(i) Fermentation (ii) Ethanol

(c) – it is used to make ester; used to make wine and liquor

(d)(i) used in perfumes (responsible for sweet smell)

(ii)



(iii)Terylene causes pollution because it is non-biodegradable

2011 QUESTION PAPER 3

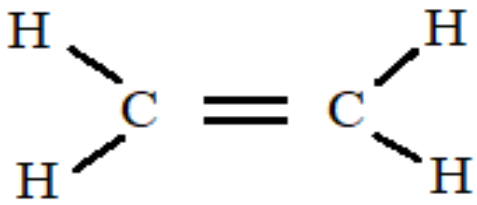
73.(a) Nitrogen, Oxygen and Carbon dioxide

(b)**(i)** Sulphur dioxide (SO_2) **(ii)** power stations/Car exhaust (iii) combustion of fossil fuels

74.(a)(i) C- Limestone D - Coke (ii) Fe_2O_3 (s) + 3CO (g) → $3CO_2$ (g) + 2Fe (l)

(b) A – slag D – molten iron (c) it has highest melting point over 2000 ^oC.

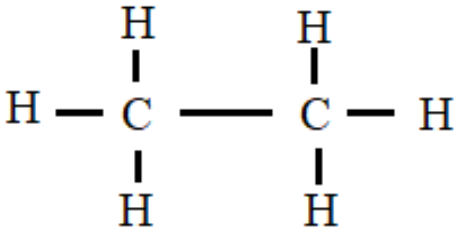
75.(a)



(b) Addition polymerization (c) Nylon (d) Used to in making fishing net.

76(a) – they have the same general formula; - They have similar chemical properties.

(b)(i) C_2H_6



(ii) % = H_6 ÷ C_2H_6 x 100 = 6 ÷ 30 x 100 = ***20%***

77.(a) % of oxygen – 21% ; % of nitrogen – 78%

(b) by blowing air in lime water, if it turns milky then carbon dioxide is present.

(c) because its outermost shell is filled up

78. (a) Fractional distillation

(b) Hydrogen element and carbon element

(c) Fraction at **C** has high boiling point while that at **B** has low boiling point. Fraction at **B** is high flammable while fraction at **C** has low flammable.

(d) CH_4 (s) + 2 O_2 (g) → CO_2 (g) + $2H_2O$ (l)

(e) It is a source of energy which if used up cannot be easily replaced.

(f) – cheap to run

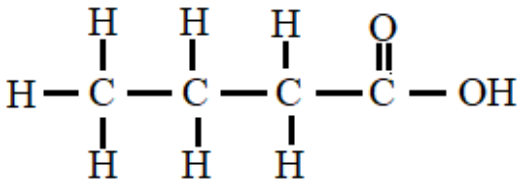
- non – pollutant

2012 QUESTION PAPER 3

79.(a) Silver (b) sodium (c) Zinc (d) sodium (e) Aluminium

(f) $2Na\ (s) + H_2O \rightarrow 2NaOH\ (aq) + H_2\ (g)$

80.(a) $C_nH_{2n+1}COOH$ (b)



(c)(i) Ethanol and ethanoic acid are reagents

(ii) - Esterification is the reaction between an organic acid and alcohol while neutralization is the reaction between any acid and a base. - In esterification an ester is formed while in neutralization salt is formed.

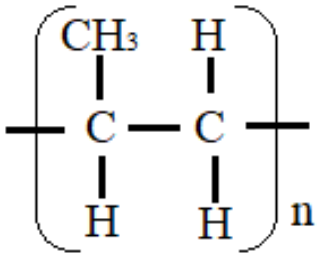
(iii) $CH_3CH_2OH + CH_3COOH \rightarrow CH_3COOCH_2CH_3 + H_2O$

(d) Mole of organic acid = 15 ÷ 60 = 0.25 mol

Mass of ester = n x Mr = 0.25 x 88 = ***22g***

2013 QUESTION PAPER 3

81.(a) Ethene and W (b) Propene (c)



(d)(i) they have double bond (ii) Alkenes

(e) Cracking is the process of breaking down large compound into smaller ones.

82.(a) Painting; Galvanising; Greasing and oiling

(b)(i) Both are exothermic reaction i.e they produce heat

(ii) Rusting can occur at any temperature while burning occurs at high temperature.

(c)(i) carbon monoxide; sulphur dioxide

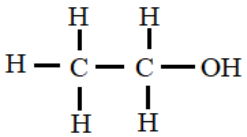
(ii) carbon monoxide – fire forest

Sulphur dioxide – burning of fossil fuels / car engine

83.(a) A – Hydrogen gas (b) Water (c) it is used to make ammonia; it is used as fuel in Jet

(d)(i) Zinc oxide (ii) $ZnO\ (s) + HCl\ (aq) \rightarrow ZnCl_2\ (aq) + H_2O\ (i)$

84.(a) A – Methane (b) A – CH_4 and B - C_2H_{10} (c) (i) C – C_4H_9OH (ii)



(iii) $C_2H_4 + H_2O \rightarrow C_2H_5OH$ (d)(i) D – C_4H_8

(ii) It turn orange bromine solution to colourless

(e)(i) Ethanoic acid (ii) It turn blue litmus paper red

2014 QUESTION PAPER 3

85(a)(i) Macromolecules are giant molecules formed by joining many small molecules known as monomers.

(ii) Fats/lipids/carbohydrate

(b) *Reagent* – hydrochloric acid; *Condition* – heating it using water bath.

(c)(i) both have amide linkage (ii) the monomers of nylon are amine and acid while that of $_{protein}$ are amino acids.

86.(a)(i) sonorous (ii) making wire (iii) strong and hard (iv) good conductor of electricity

(b) $Mg\ (s) + 2HNO_3\ (aq) \rightarrow Mg(NO_3)_2\ (aq) + H_2\ (g)$

87.(a) The gases are passed through the filler to remove dust. Then they are cooled to -80 oC to remove water vapour and carbon dioxide as solids. Liquid gases are passed now fractionating column and they are distilled.

(b) It relight the growing splint.

(c)(i) Test tube 1 – nails corrode and water turn dark brown

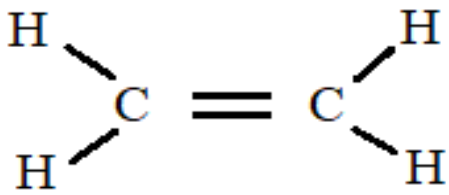
Test tube 2 – no corrosion hence nails remain the same

Test tube 3 - no corrosion hence nails remain the same

(ii) Test tube 2 – because of absence of moisture (water)

Test tube 3 – because of absence of oxygen

88.(a)



(b)(i) Alkene (ii) C_nH_{2n}

(iii) Fourth member – C_4H_8; Fifth member – C_5H_{10}

(c) Source – Ethane. Process – Cracking

(d)(i) R – Ethanol; S – Butane. (ii) It is used as fuel.

2015 G.C.E OUESTION PAPER 3

89.(a) A – Cracking; S – Hydration (b) B – Ethene

(c) Cracking help to produce short – chain compounds with (C=C) double bond which makes it to be very reactive.

90.(a) Sodium, Aluminium, Zinc, Copper (b)(i) Copper. (ii) Copper pyrites.

(c) Because when aluminium exposes to air it reacts with oxygen in air forming a layer called aluminium oxide which protect aluminium further to react with oxygen in air.

91.(a) This is an acid which ionises partially in water.

(b) $2CH_3CH_2COOH + K_2CO_3 \rightarrow 2CH_3CH_2COOK + CO_2 + H_2O$ **(c)** Mass = ***2g***

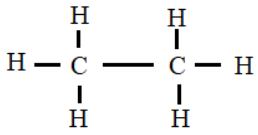
(d)(i) COOH – Carboxylic group; OH – Alcohol group

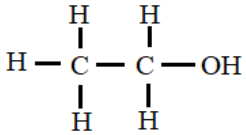
(ii) Condensation polymerisation

(e) – Throwing them in cities causes litter problem; - burning them causes air pollution

2016 G.C.E QUESTION PAPER 3

92.(a)(i) Ethane (ii) (iii) Alcohol (iv)





(v) carboxylic acid (vi) Ethanoic acid

93.(a) Cu2S (g) + O2 (g) → Cu (s) + SO2 (g)

(b) because it produces a poisonous gas called sulphur dioxide which chock the throat.

93.(e)(i) Native means unreactive (ii) Brass/bronze

94.(a) $(NH_4)_2SO_4$ (aq) + 2NaOH (aq) → Na_2SO_4 (aq) + $2H_2O$ (l) + $2NH_3$ (g)

(b) You place a red litmus paper to the vapour and if it turns blue meaning ammonia is produced

(c)(i) Hydrogen gas and Nitrogen gas (ii) $3H_2$ (g) + N_2 (g) → $2NH_3$ (g)

(iii) – Catalyst (iron); Temperature between 250 ^oC - 350 ^oC; Pressure about 450 atm

95.(a) catalyst phosphoric acid (b) Addition reaction (c)(i) no answer - error (ii) no answer – error

(iii) m = 2.1 x 60 ÷ 42 = ***3g*** (d) C_nH_{2n}

2016 QUESTION PAPER 2

96.(a)(i) Oxygen gas (ii) it relight the burning splint

(b)(i) to speed up the chemical reaction (ii) because it is sparingly soluble in water

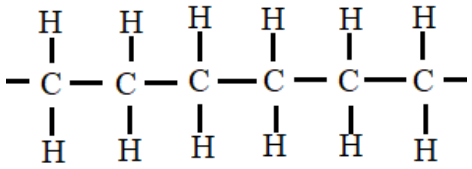
97.(a) A – Synthetic polymer. B- Terylene C – Synthetic polymer.

(b) the best way is to recycle them (c) Amino acid

98.(a)(i) Fermentation (ii) Zymase (iii) carbon dioxide

(b) Process – Addition polymerisation Catalyst – mixture of titanium and aluminium mixture

(c)



(d) Carbon dioxide and water

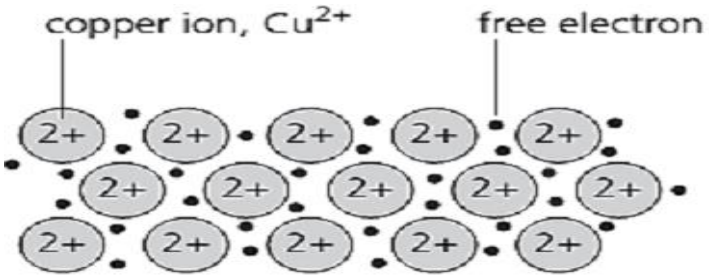
(e) Non- biodegradable means it means it cannot be decomposed by bacteria in the ground.

Effects: They causes land pollution/litter problem

99.(a)(i) To remove heavy impurities (ii) To remove iron oxide and some sulphur dioxide gas.

(b) $2CuFeS_2\ (s) + 4O_2\ (g) \rightarrow Cu_2S\ (s) + FeO\ (g) + 3SO_2(g)$

(c)(i)



(ii) This is because the positive ions are able to slide over each other without breaking the bond.

(d) To make electrical wire

100.(a) Ostwald process (b) Platinum – Rhodium alloy

(c) $2NH_3\ (g) + O_2\ (g) \rightarrow 2NO\ (g) + 3H_2O\ (g)$

$4NH_3\ (g) + 5\ O_2\ (g) \rightarrow 4NO\ (g) + 6H_2O\ (g)$

(d) $2NO\ (g) + O_2\ (g) \rightarrow 2NO_2\ (g)$ (e) It is dissolved in water.

(f) Because it is miscible with water in all its proportions, forming hydrates at low temperature.

(g) Because sulphuric acid acts as dehydrating agent.

(h) It is used to make explosive e.g Trinitrotoluene (TNT)

- It is used to make fertilizer, dyes, etc.

101. (a) coke and limestone (b) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

(C) Presence of water vapour; presence of Oxygen

102.(a) Exhaust from car engine; power station/ factories

(b)(i) It causes acid rain.

(ii) Acid rain reacts with lime forming salt hence lime will be washed away from the buildings.

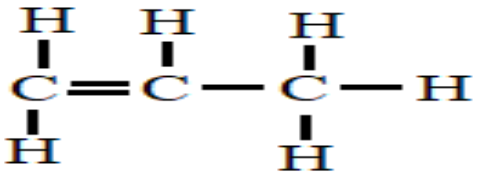
(c) Carbon

103.(a) Precipitation (b) Displacement (c) Neutralisation (d) Combustion (d) Synthesis

104.(a) Y, Z, X, W (b) Metal X (c) By reducing it with carbon or carbon monoxide

(d) ZCO_3 (s) $\rightarrow$ ZO (s) + CO2 (g) (e) Silver because it does not rust

105.(a)(i) Polypropene (ii) Name – propene. Structure –



(b)(i) Synthetic polymer (ii) To make plastics

(c) Cassava is crushed and pressure cooked to release starch (glucose). Then it will be mixed with water and yeast then allow it to react for few days in the absence of air (oxygen) at the temperature of 37°C. The enzyme in yeast (zymase) will help to convert starch to ethanol and carbon dioxide gas will be produced.

(d) C_2H_5OH (l) + $3O_2 \rightarrow 2CO_2$ (g) + $3H_2O$ (l)

106.(a)(i) The catalystic converter has two compartments marked A and B. In compartment A, nitrogen monoxide is converted into nitrogen gas and oxygen gas. In compartment B, carbon monoxide is combined with oxygen from compartment A forming a harmless gas carbon dioxide.

(ii) 2NO (g) $\rightarrow N_2$ (g) + O_2 (g) OR 2CO (g) + O_2 (g) $\rightarrow 2CO_2$ (g)

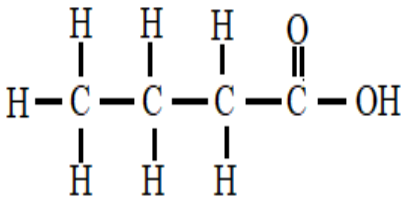
(b) Carbon monoxide, it reacts with the haemoglobin in blood, and prevents it from carrying oxygen around the body – so you die from oxygen starvation.

(c) By using more air (oxygen) during combustion

- Design car engine which run at lower temperature.

2017 QUESTION PAPER 2

107.(a) Ethene – C_2H_4 (b)



(C) $CH_3CH_2CO_2H$ and CH_3COCH_2OH

(d) CH_3OH and HCO_2H

108(a)(i) Ethanol

(ii) Cassava/maize is crushed and pressure cooked to release starch (glucose). Then it will be mixed with water and yeast then allow it to react for few days in the absence of air (oxygen) at the temperature of 37°C. The enzyme in yeast (zymase) will help to convert starch to ethanol and carbon dioxide gas will be produced.

(iii)

$$C_6H_{12}O_6\,(l) \xrightarrow[\text{Zymase}]{\text{Yeast}} 2C_2H_5OH\,(g) + 2CO_2\,(g)$$

(b) Use – alcoholic drinking Bad effect – damages the liver

2019 .G.C.E QUESTION PAPER 2

109 (a) Corrosion is the chemical wearing of metal resulting from attack by atmospheric oxygen in the presence of moisture.

(b) Potassium (c) presence of moisture (water vapour); Presence of oxygen

(d) Greasing and oiling; galvanizing; Painting

110. (a) Allotropes are elements in different physical forms but in the same state.

(b) Graphite and Diamond

(c) Graphite is used as lubricant, this is because the layer of carbon atoms slide over each other easily.

111.(a)(i) Metal X, because its metal nitrate decomposes into metal nitrites and oxygen only.

(ii) Metal W, because its metal nitrate decomposes into nitrogen dioxide and oxygen gas

(b) Metal X

(c)(i) Metal X, because its metal nitrate decomposes into metal nitrites and oxygen only.

(ii) Metal W, because its metal nitrate decomposes into nitrogen dioxide and oxygen gas.

(d) Oxygen relight the growing splint; - used in disinfectant

112.(a) K – Hydration. L – Oxidation M – Esterification

(b) C_2H_4 (g) + H_2O (g) → C_2H_5OH (g)

(c) common name – Acetic acid. Chemical name – Ethanoic acid

(d) $C_nH_{2n+1}COOH$ or $C_nH_2O_2$ (e) – Used to make perfumes. - used in food flavouring